

Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

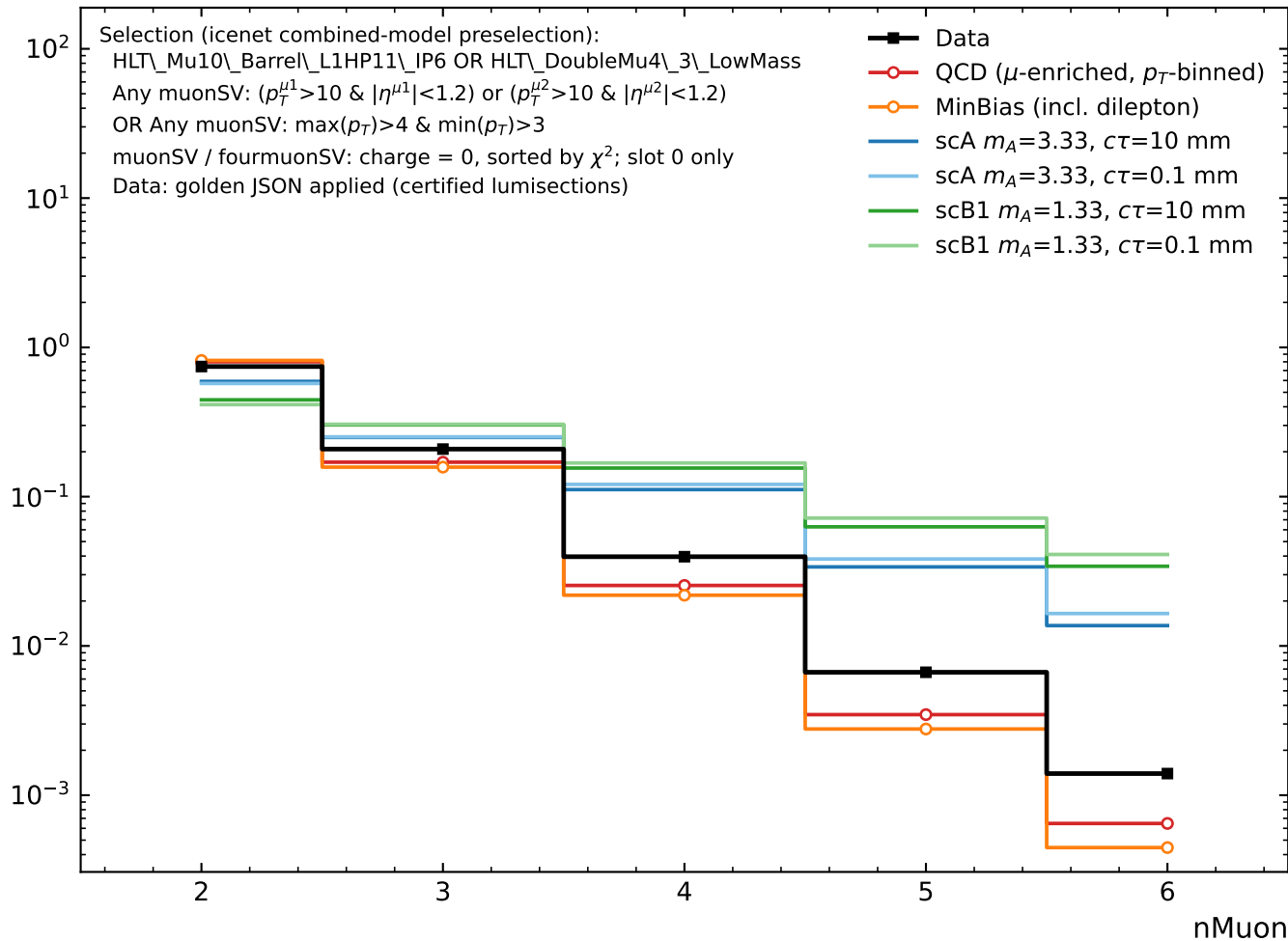
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



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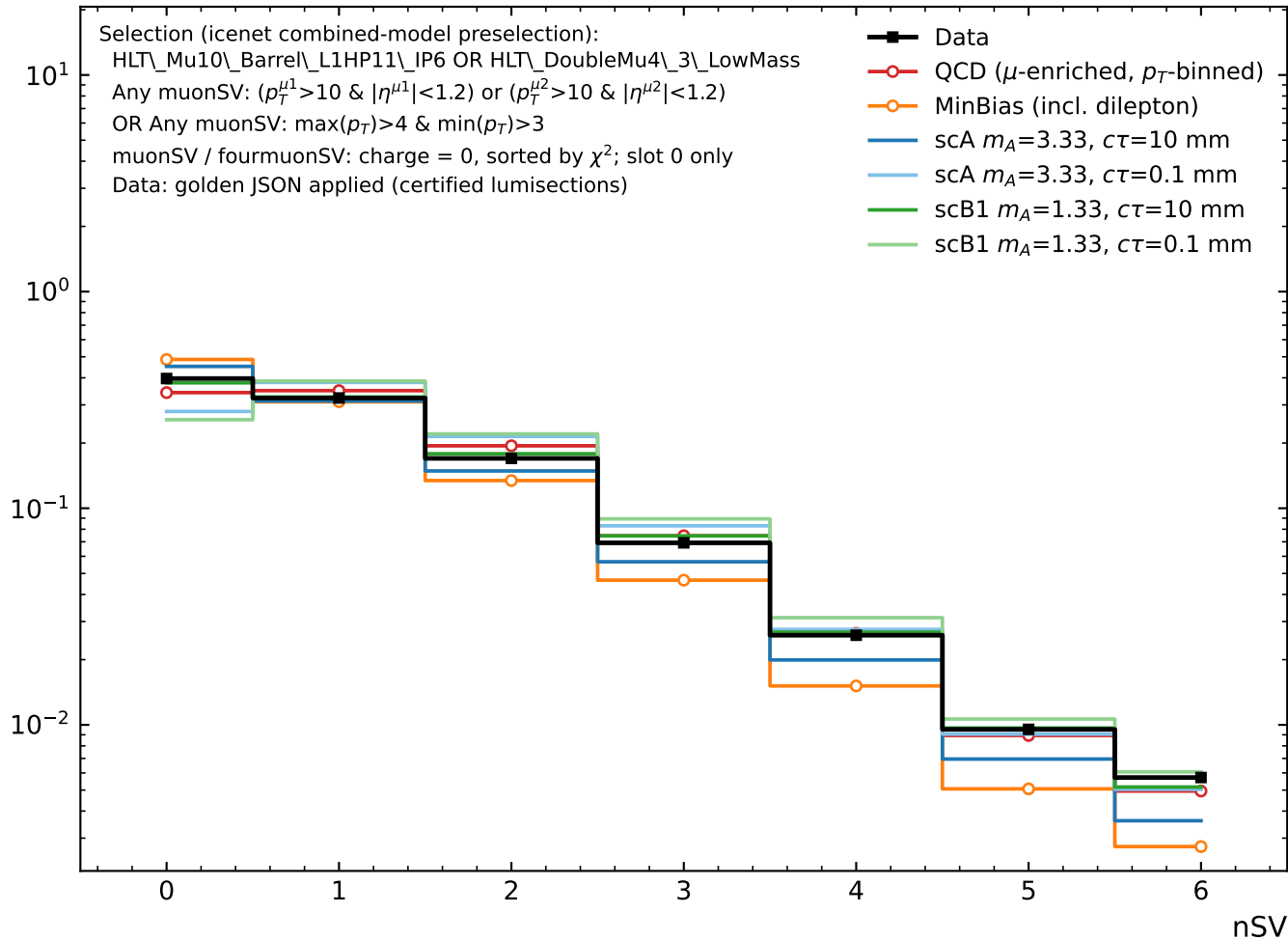
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- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A = 3.33$, $c\tau = 10$ mm
- scA $m_A = 3.33$, $c\tau = 0.1$ mm
- scB1 $m_A = 1.33$, $c\tau = 10$ mm
- scB1 $m_A = 1.33$, $c\tau = 0.1$ mm

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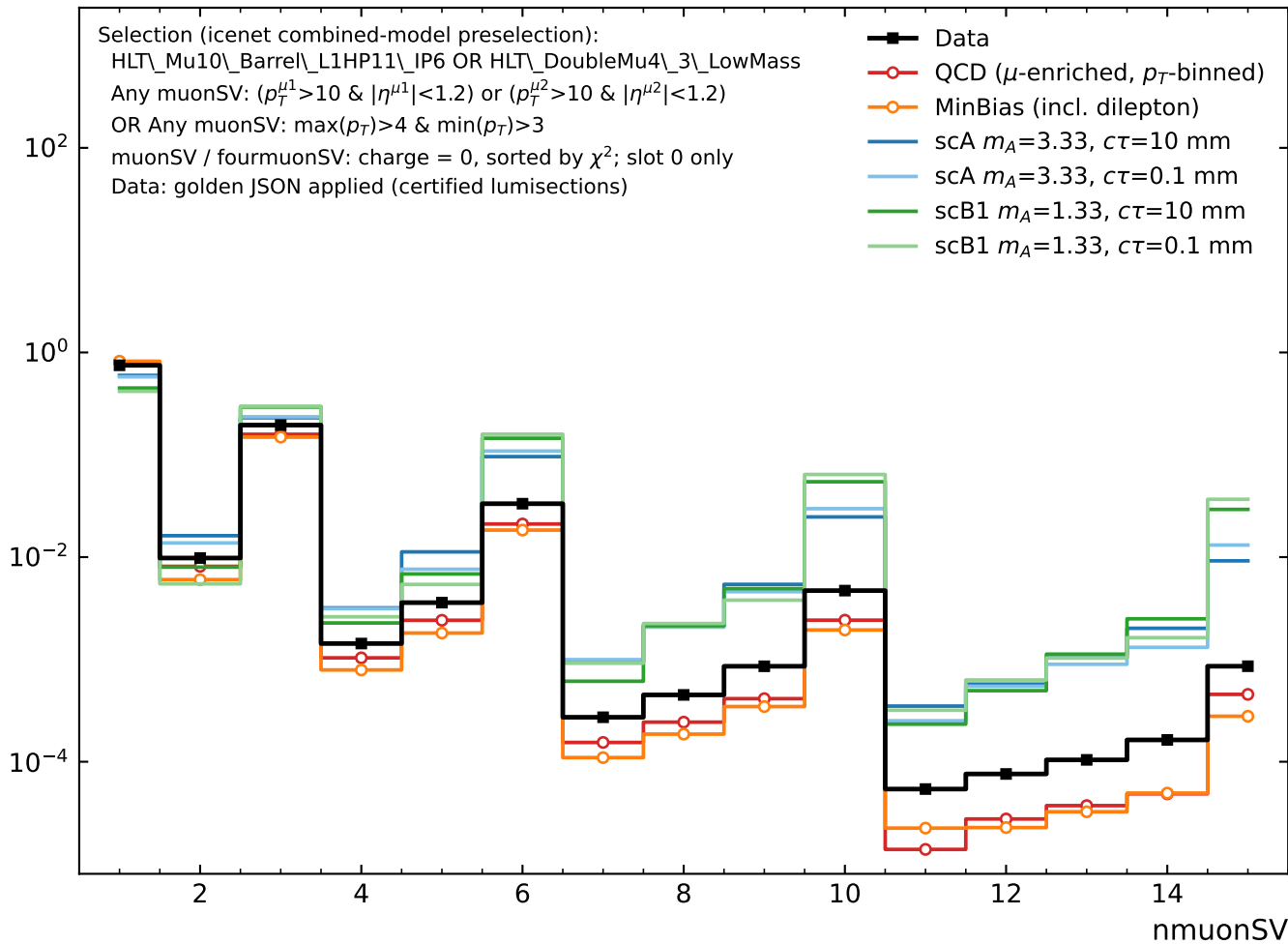
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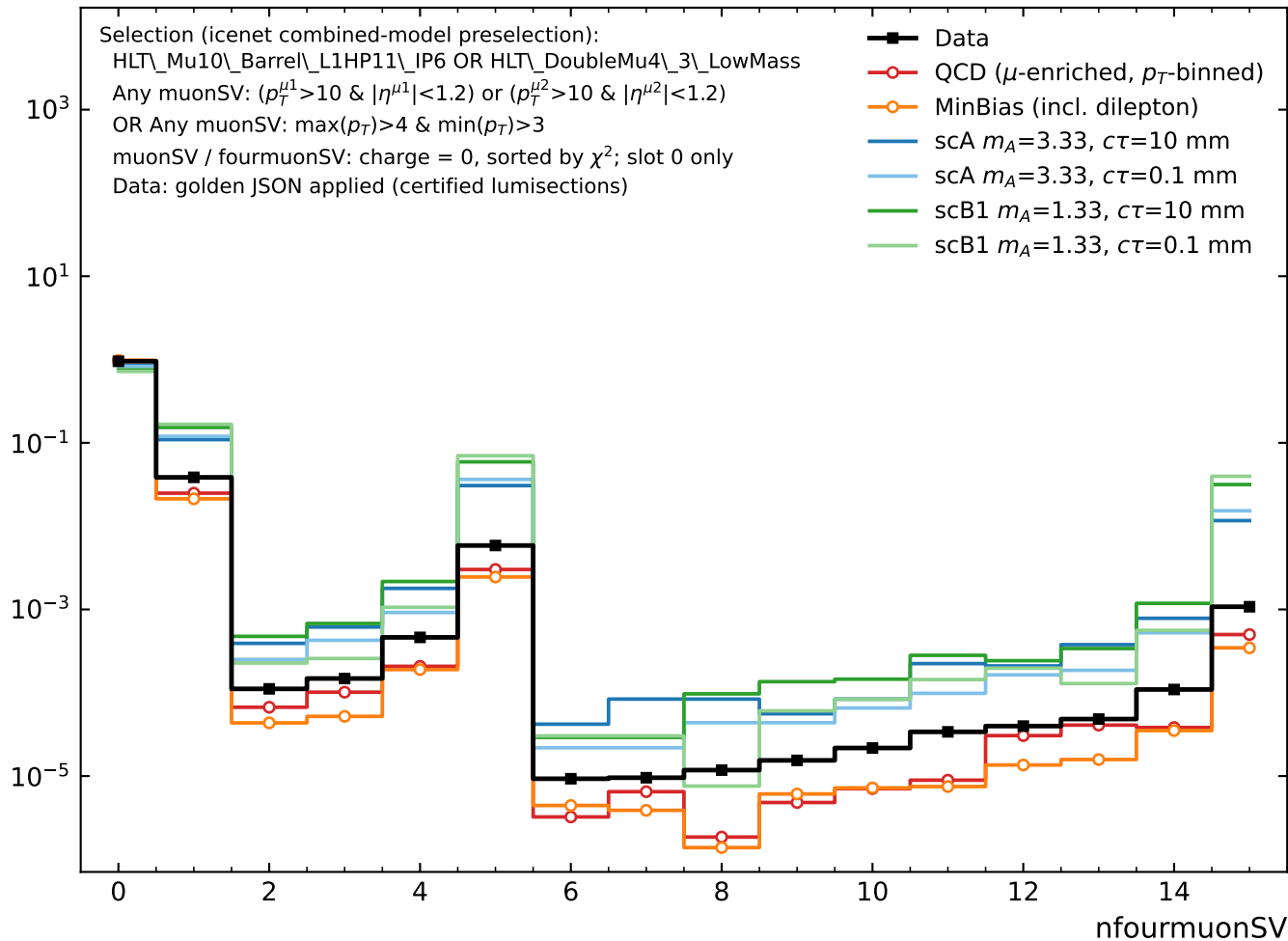
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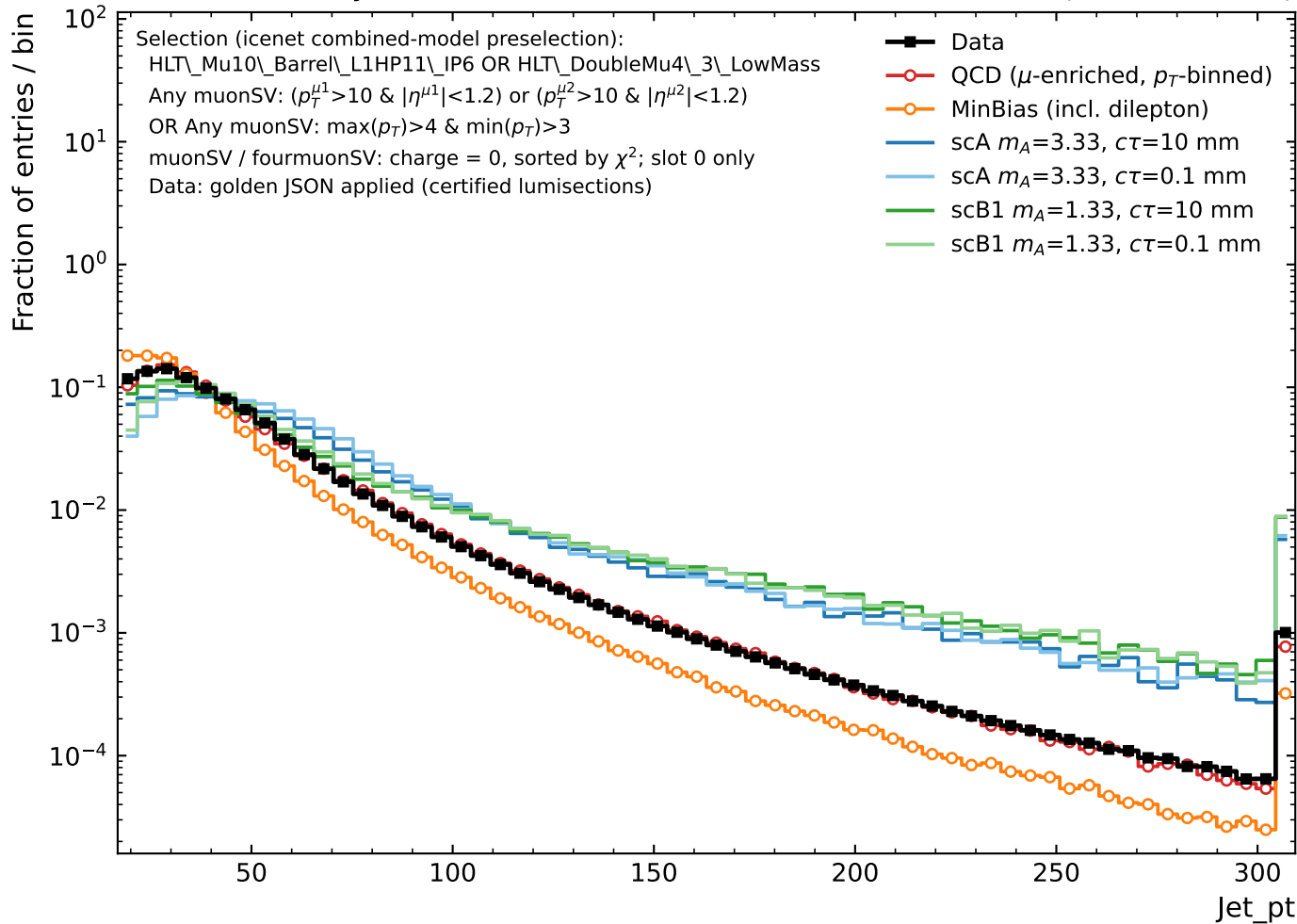
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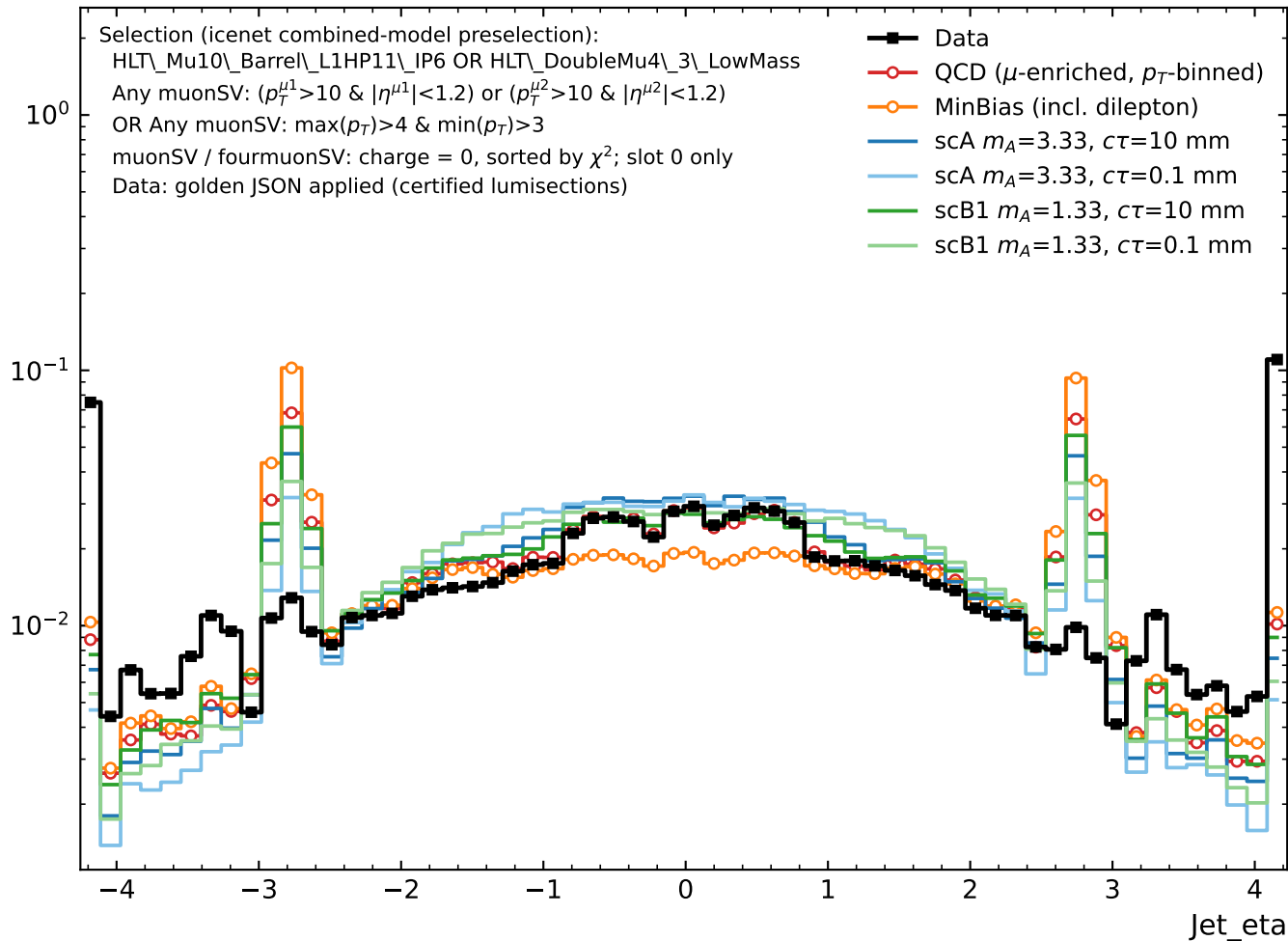
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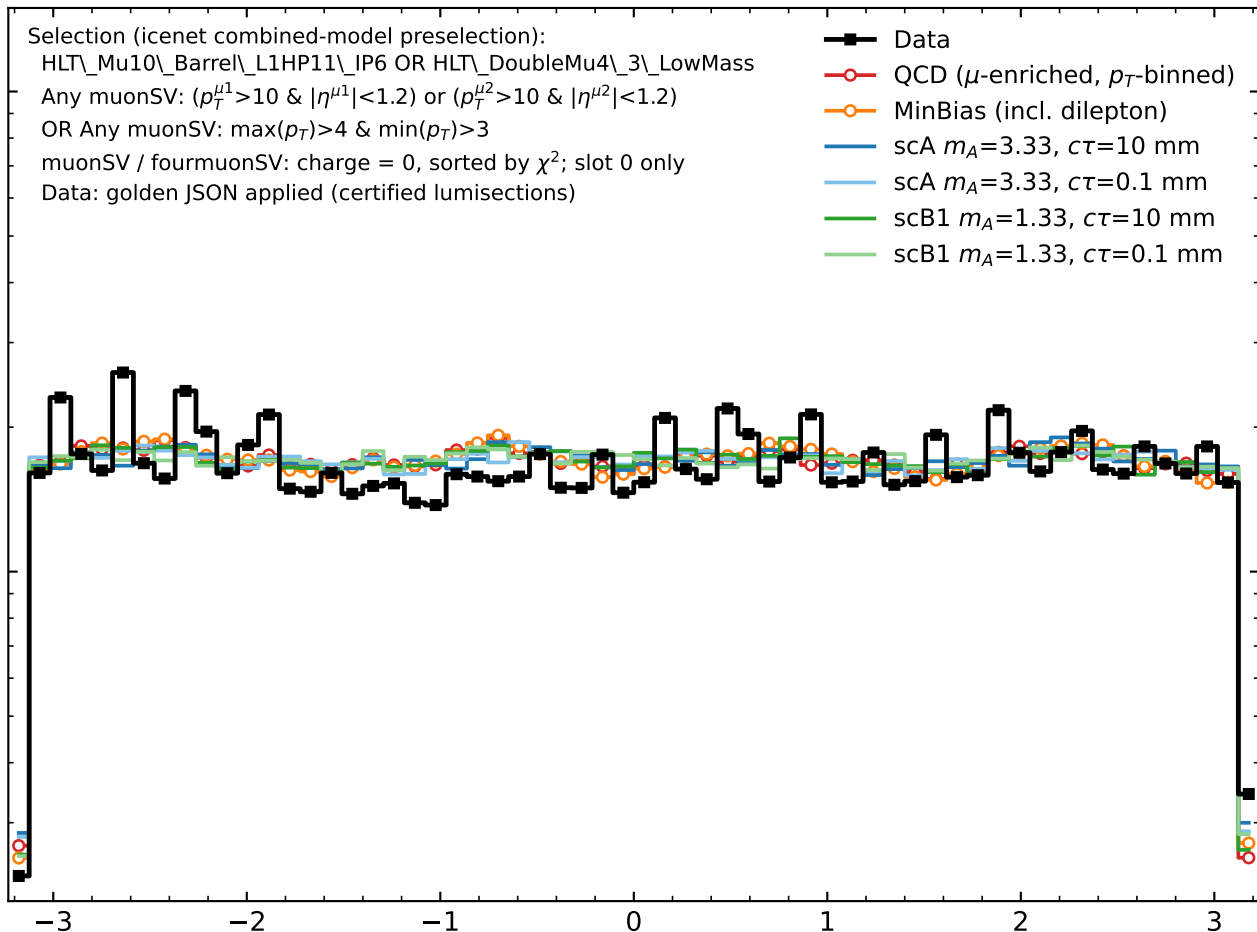
Fraction of entries / bin

10^{-2}

10^{-1}

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Jet_phi



Selection (icenet combined-model preselection):

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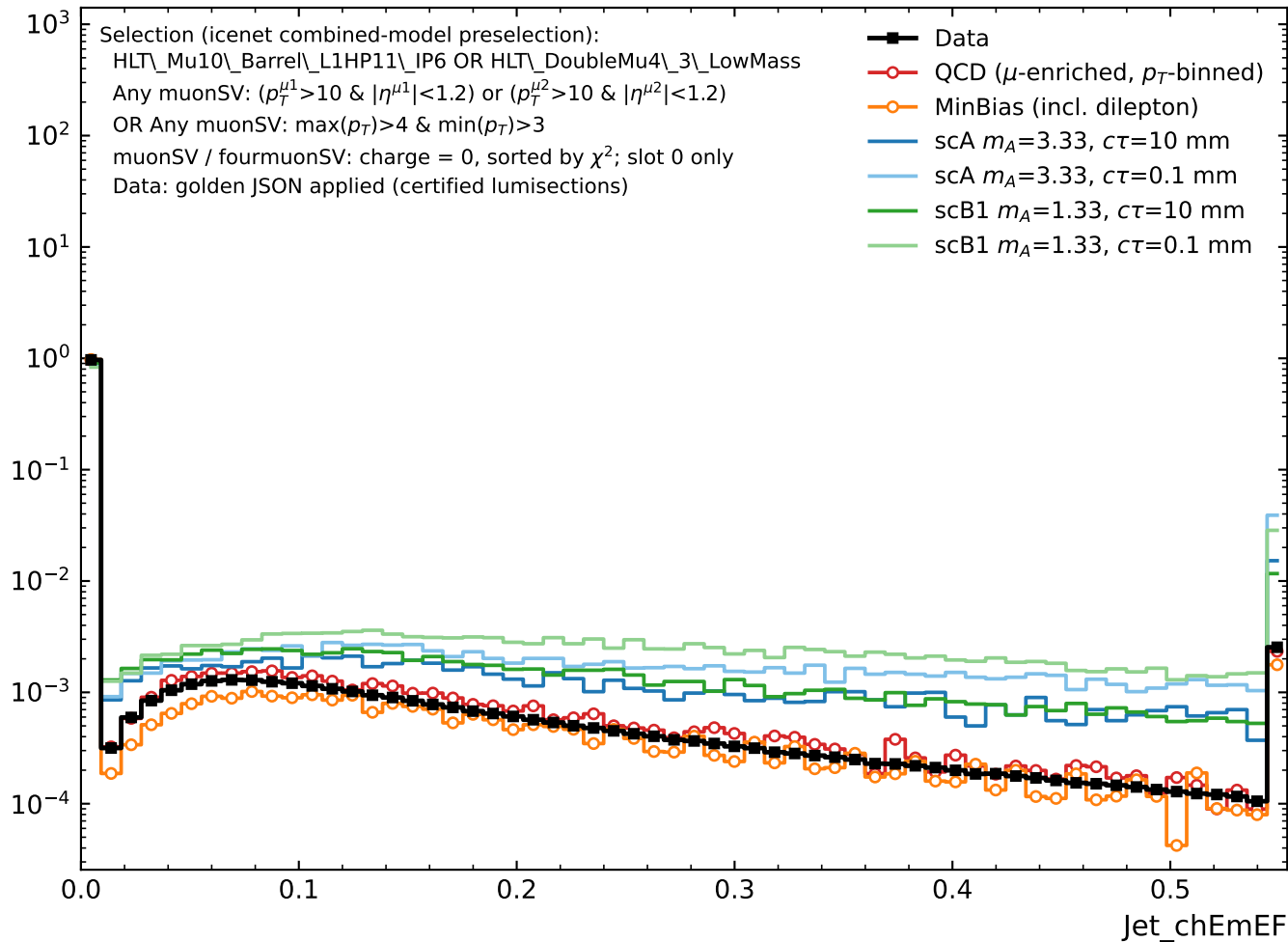
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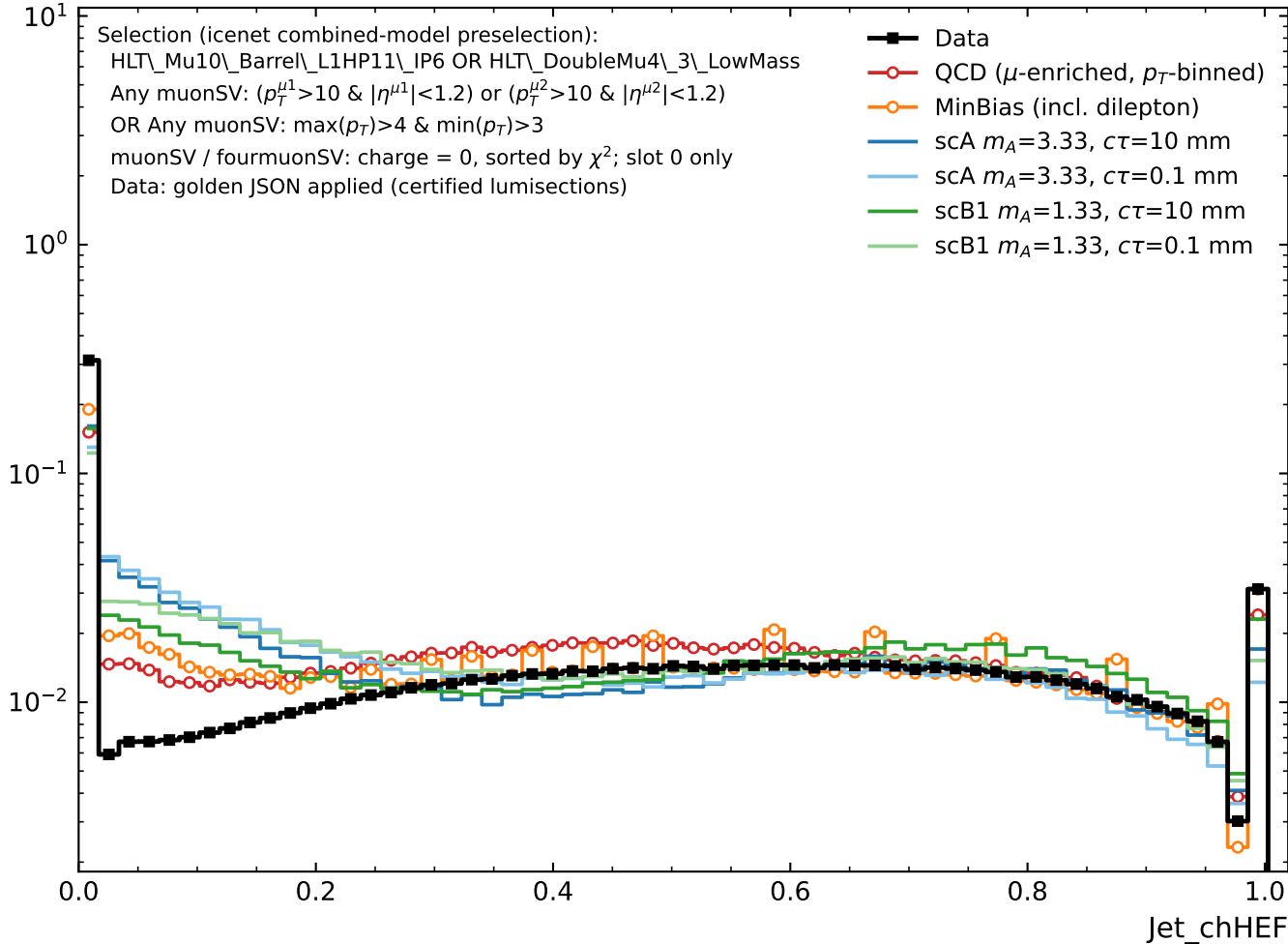
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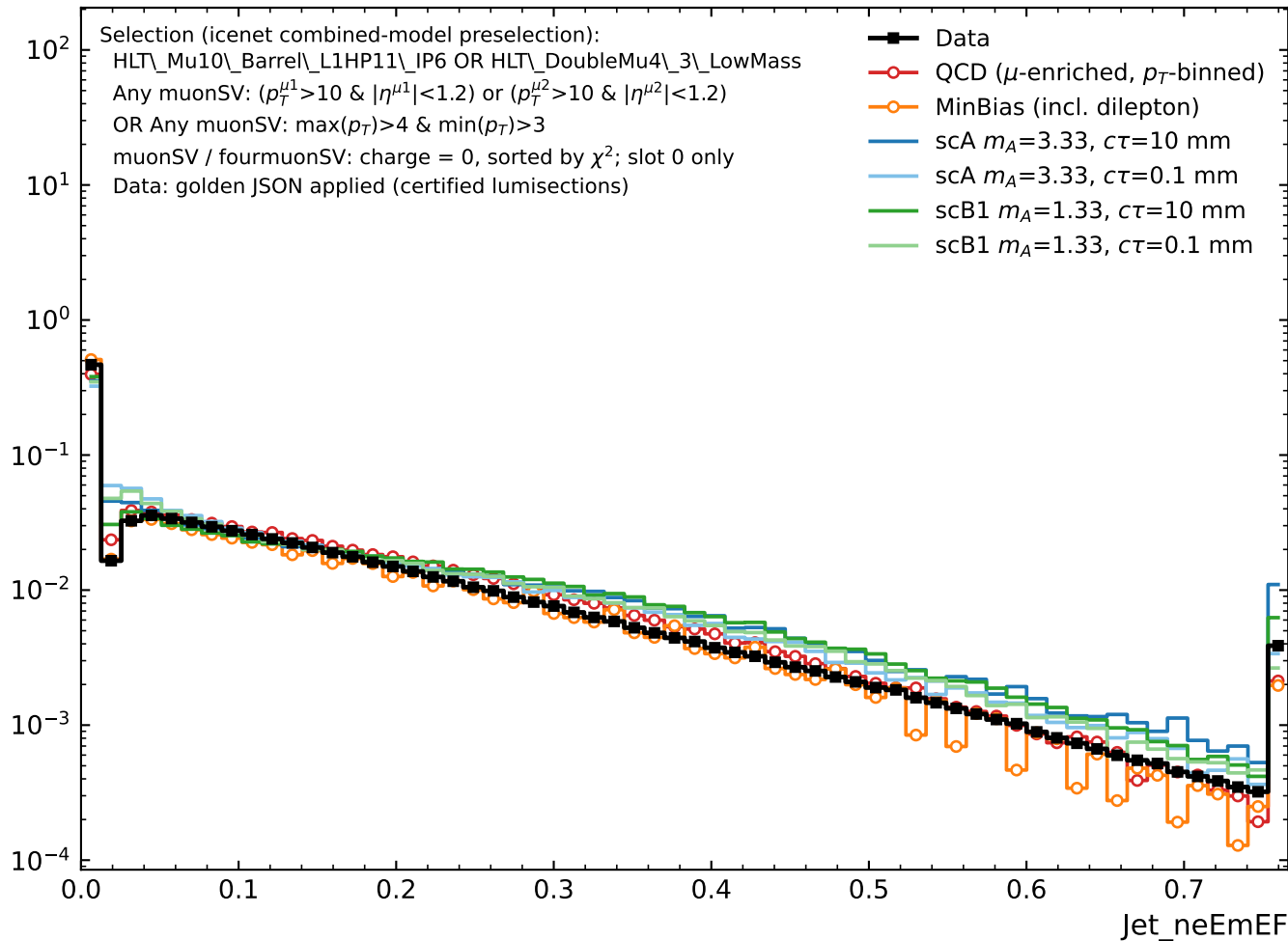
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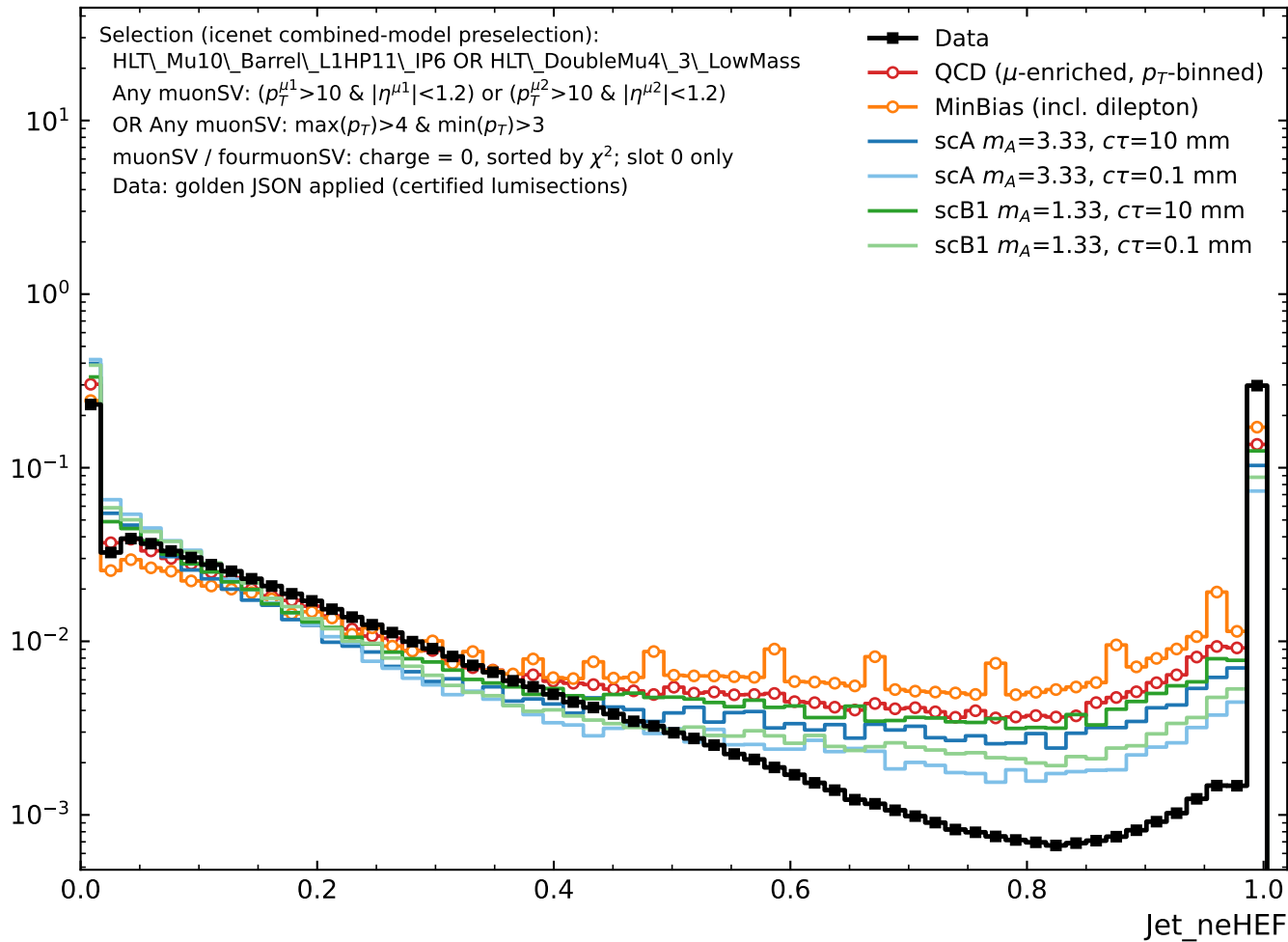
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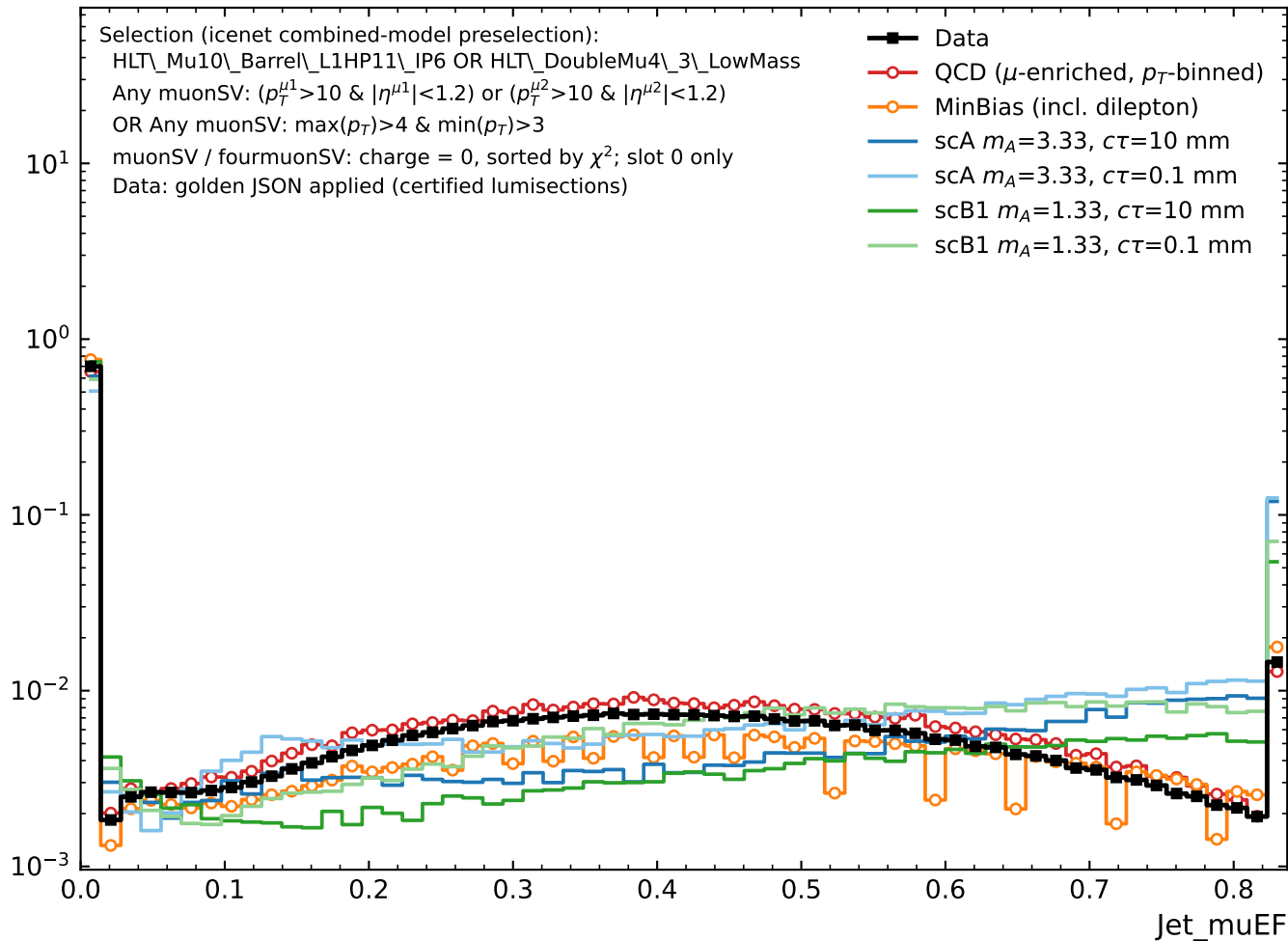
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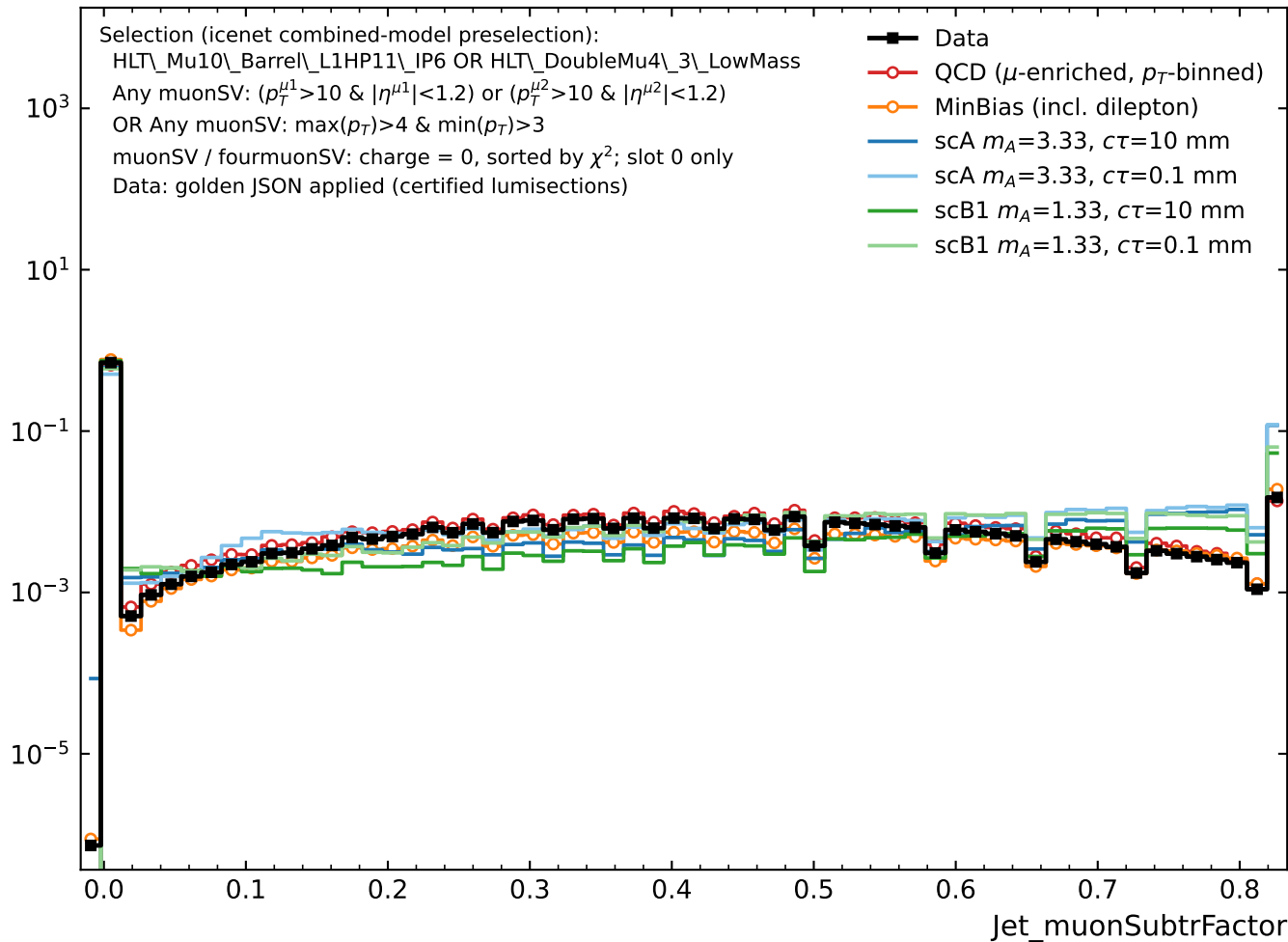
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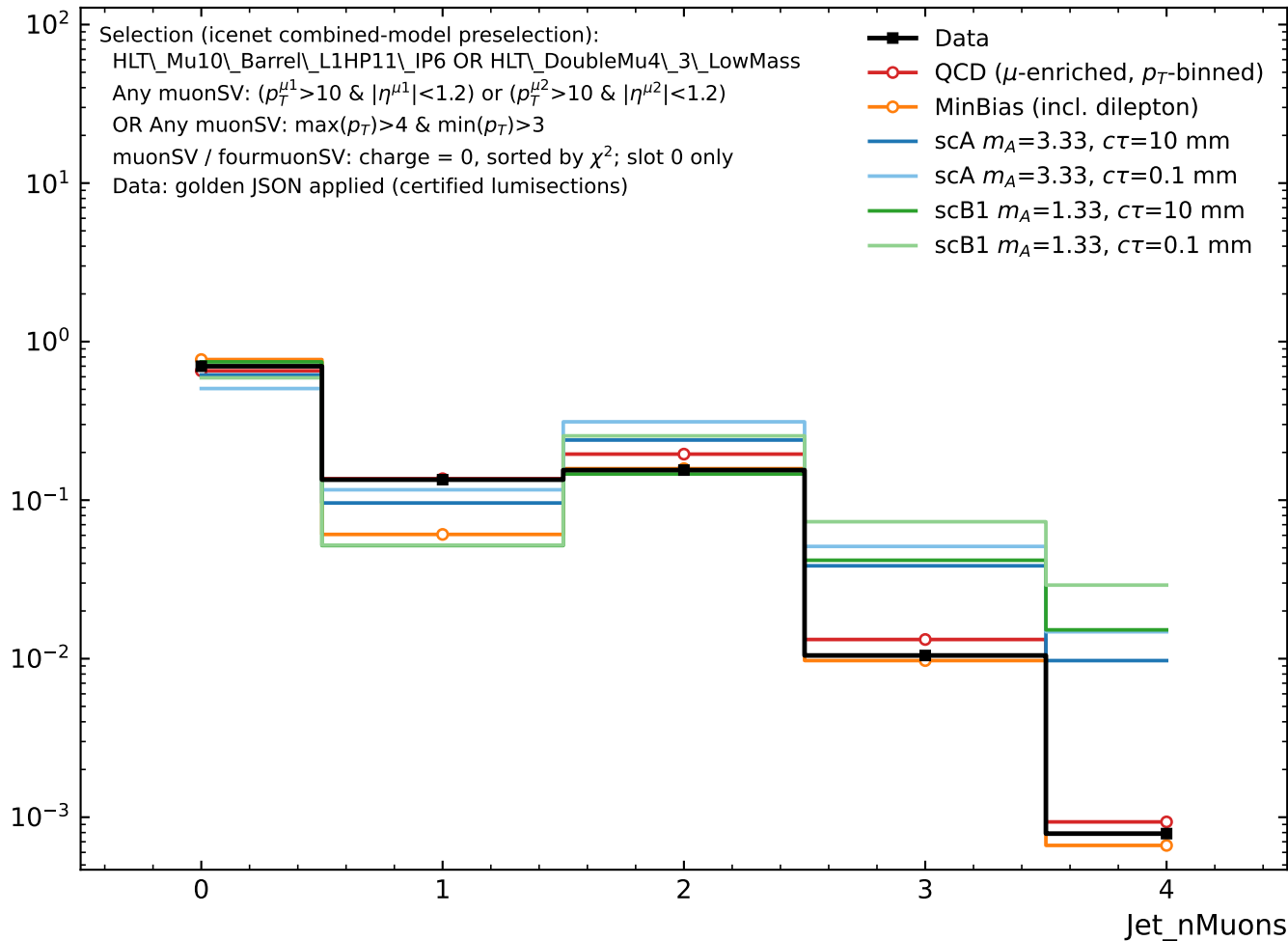
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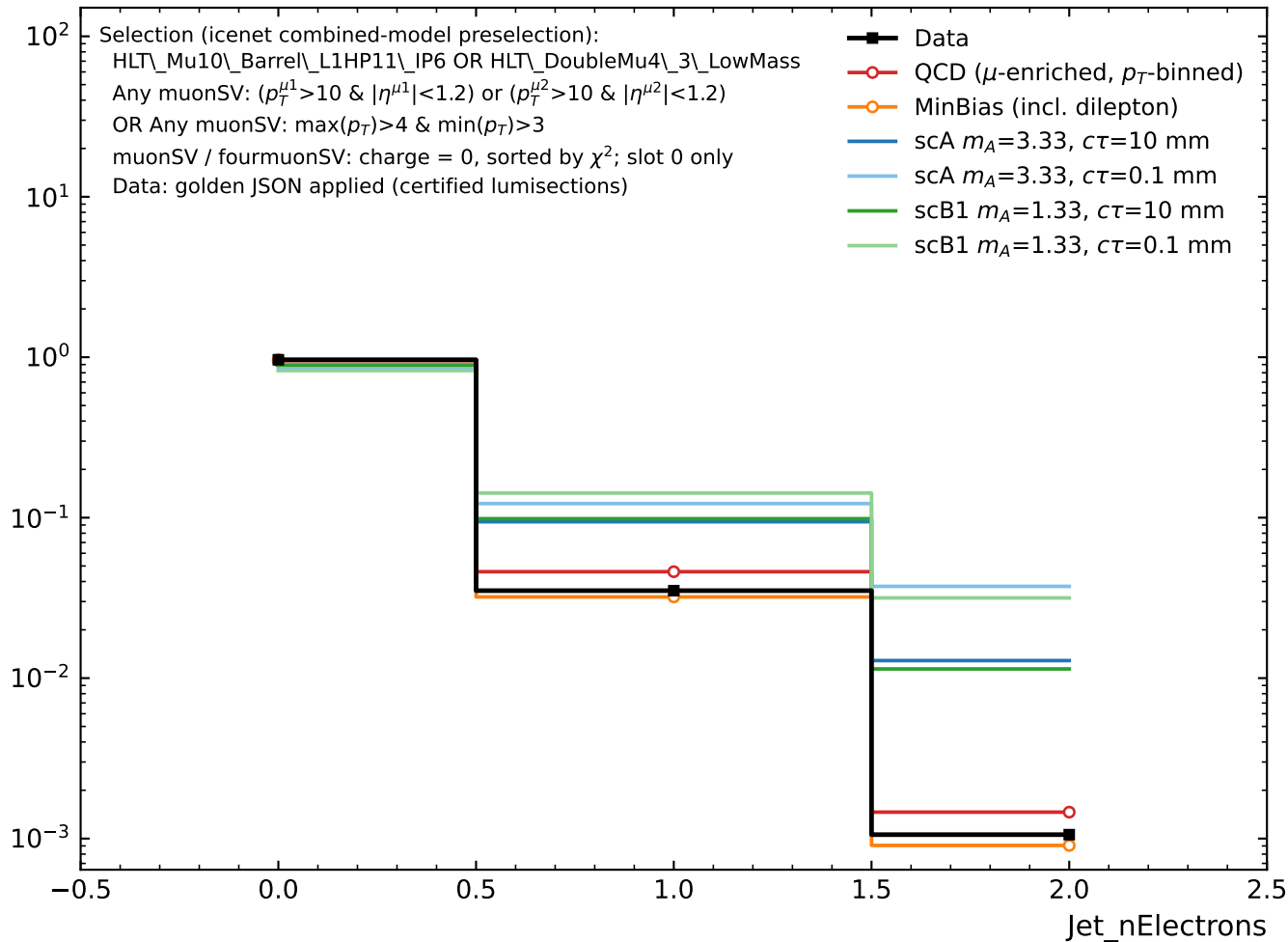
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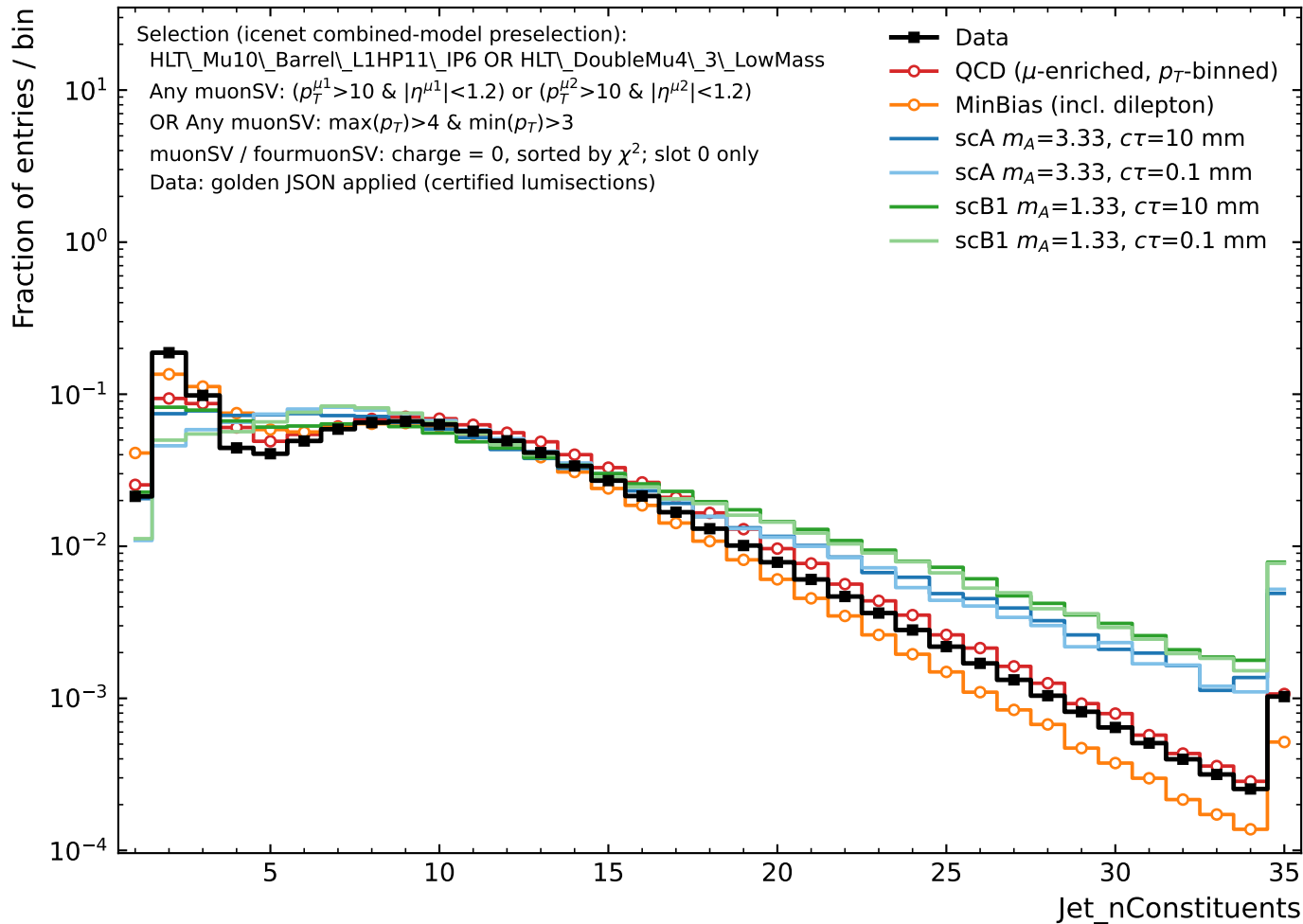
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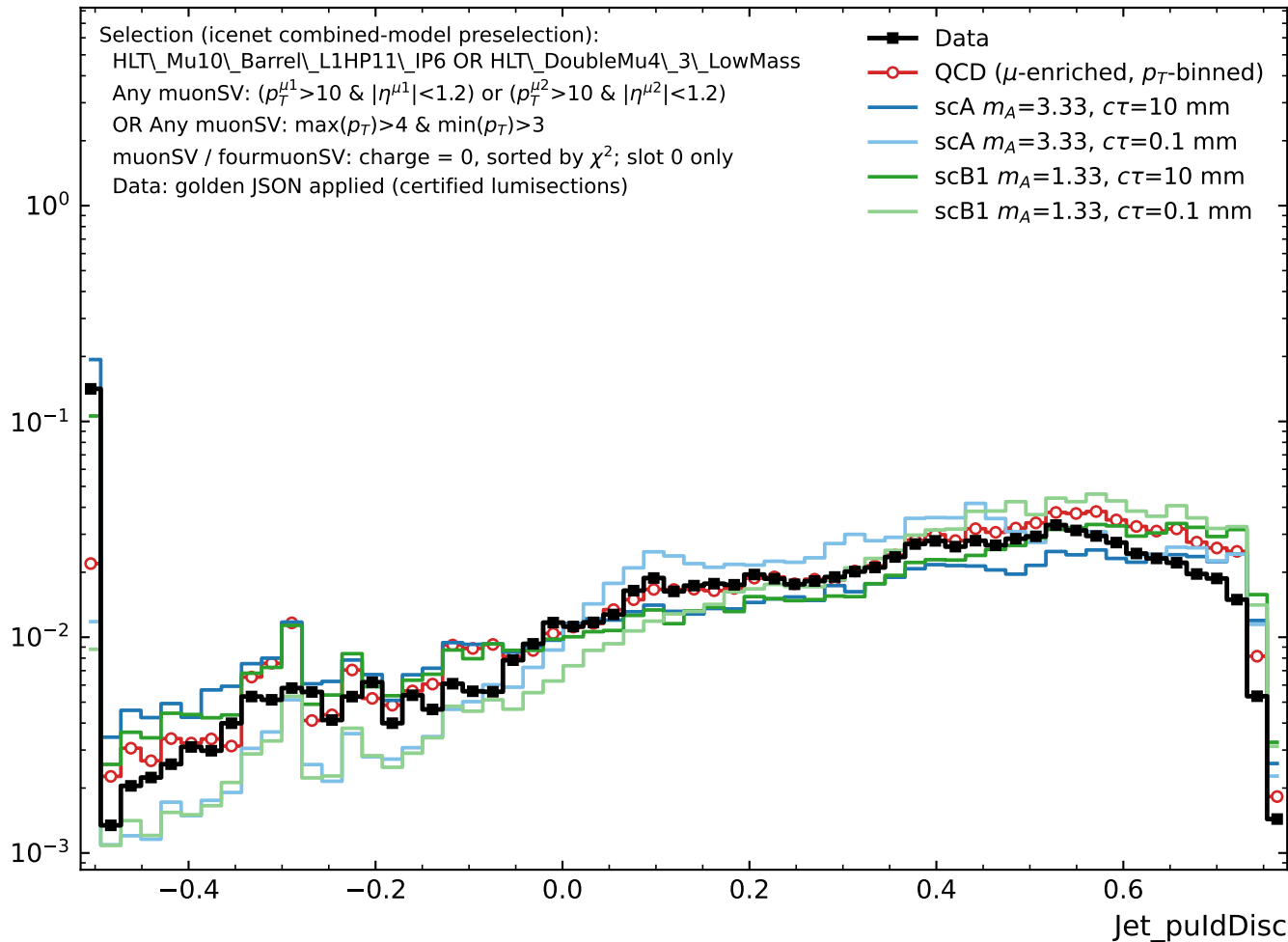
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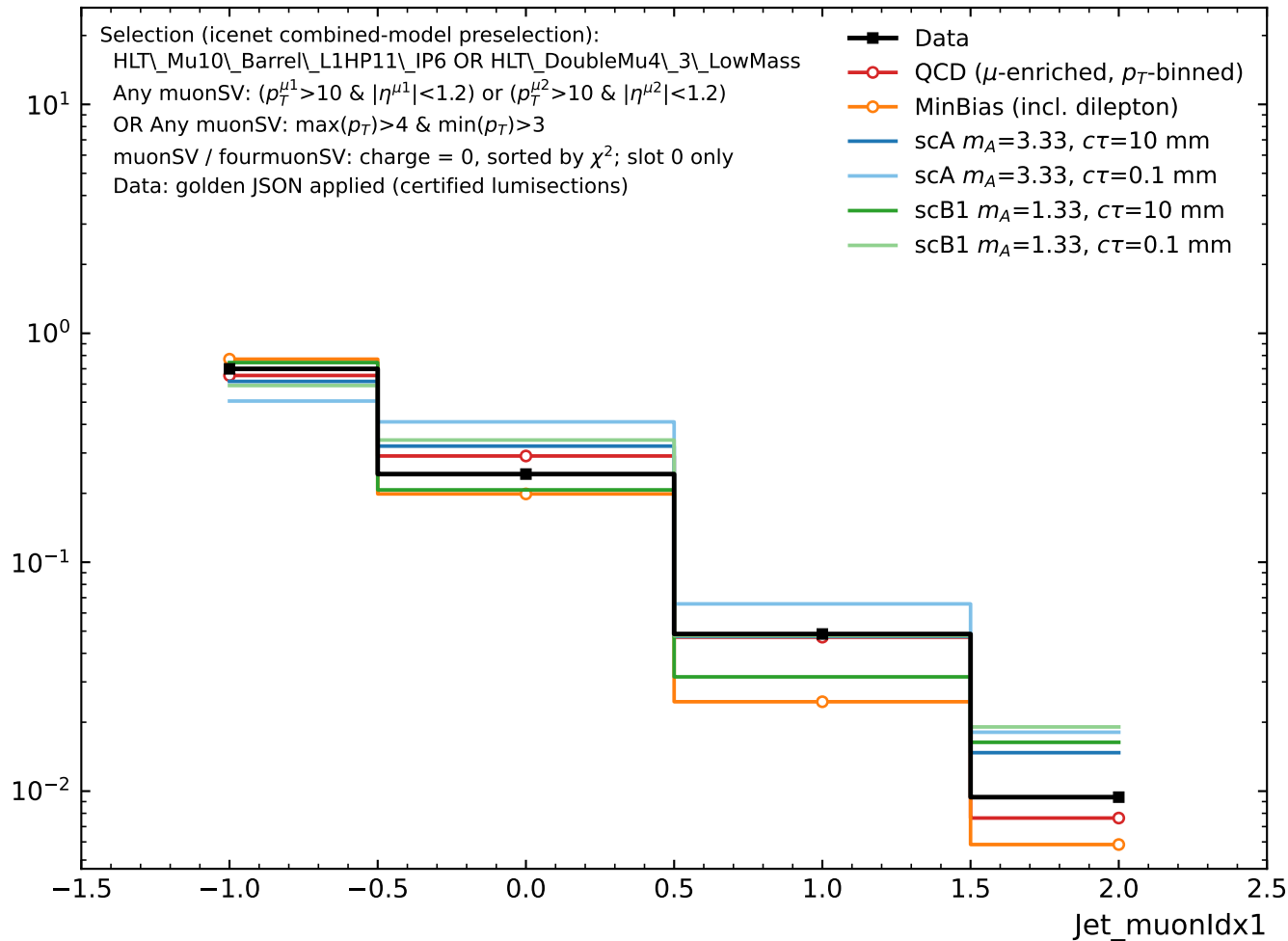
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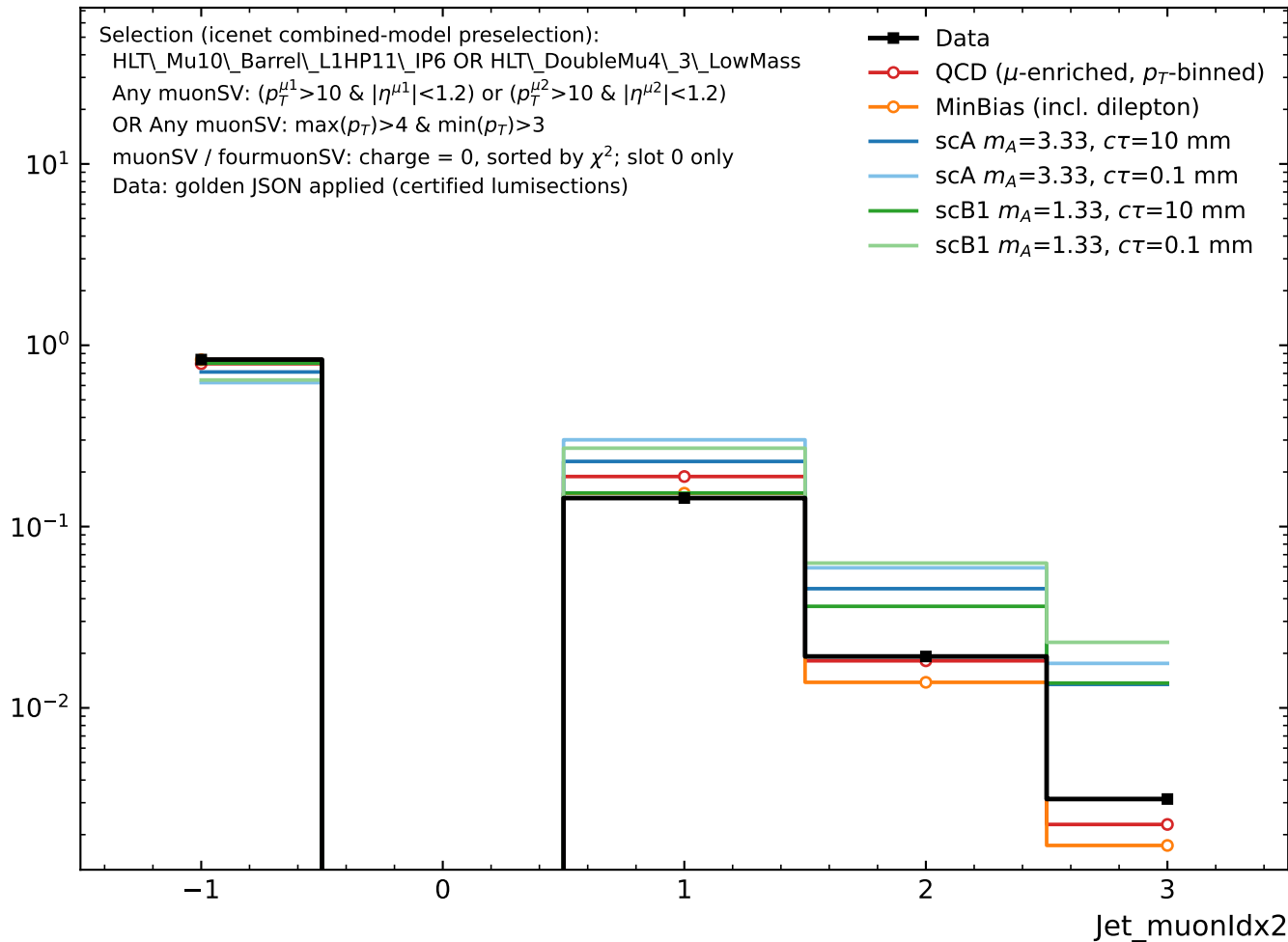
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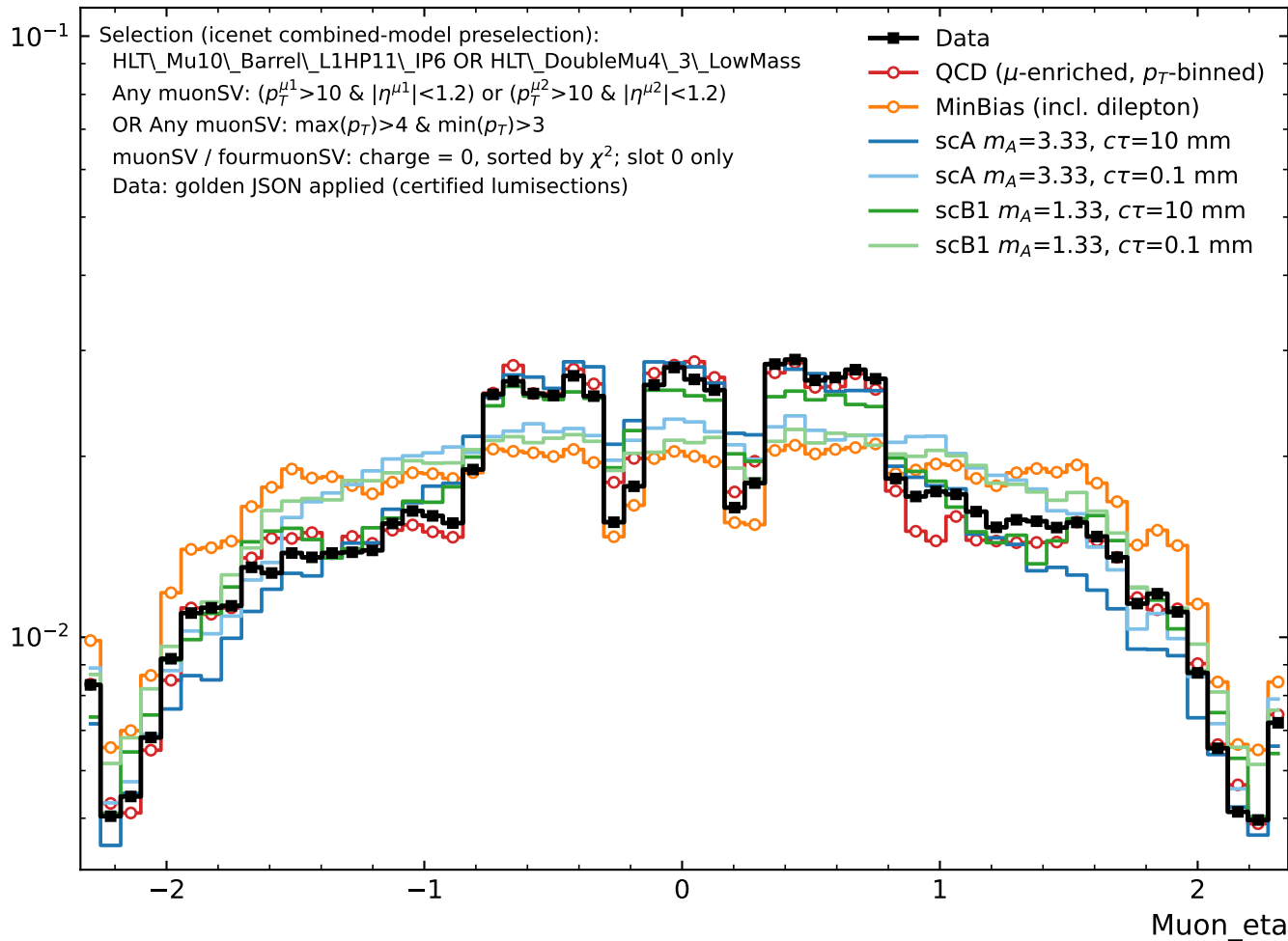
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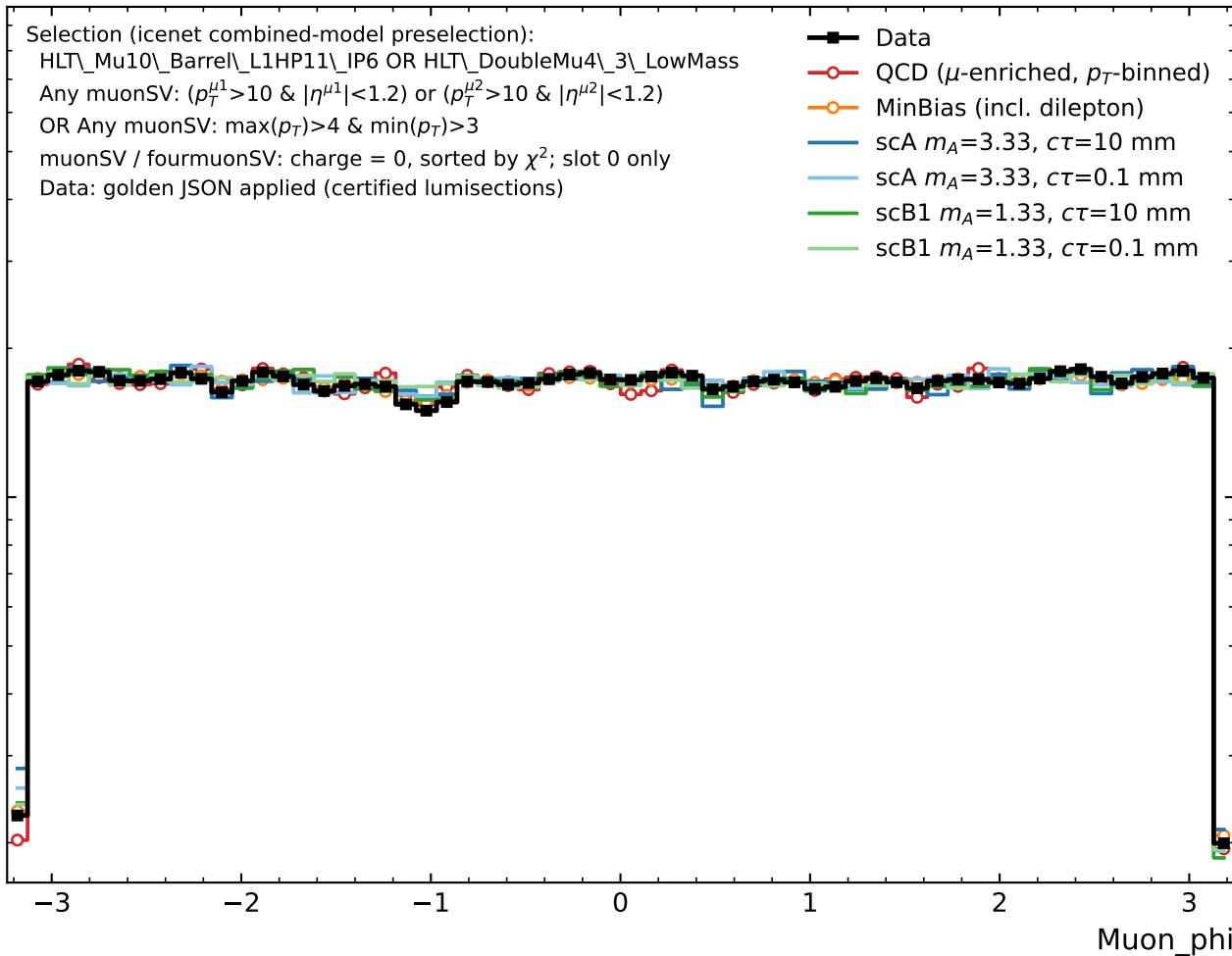
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Fraction of entries / bin

10^{-2}

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Muon_phi



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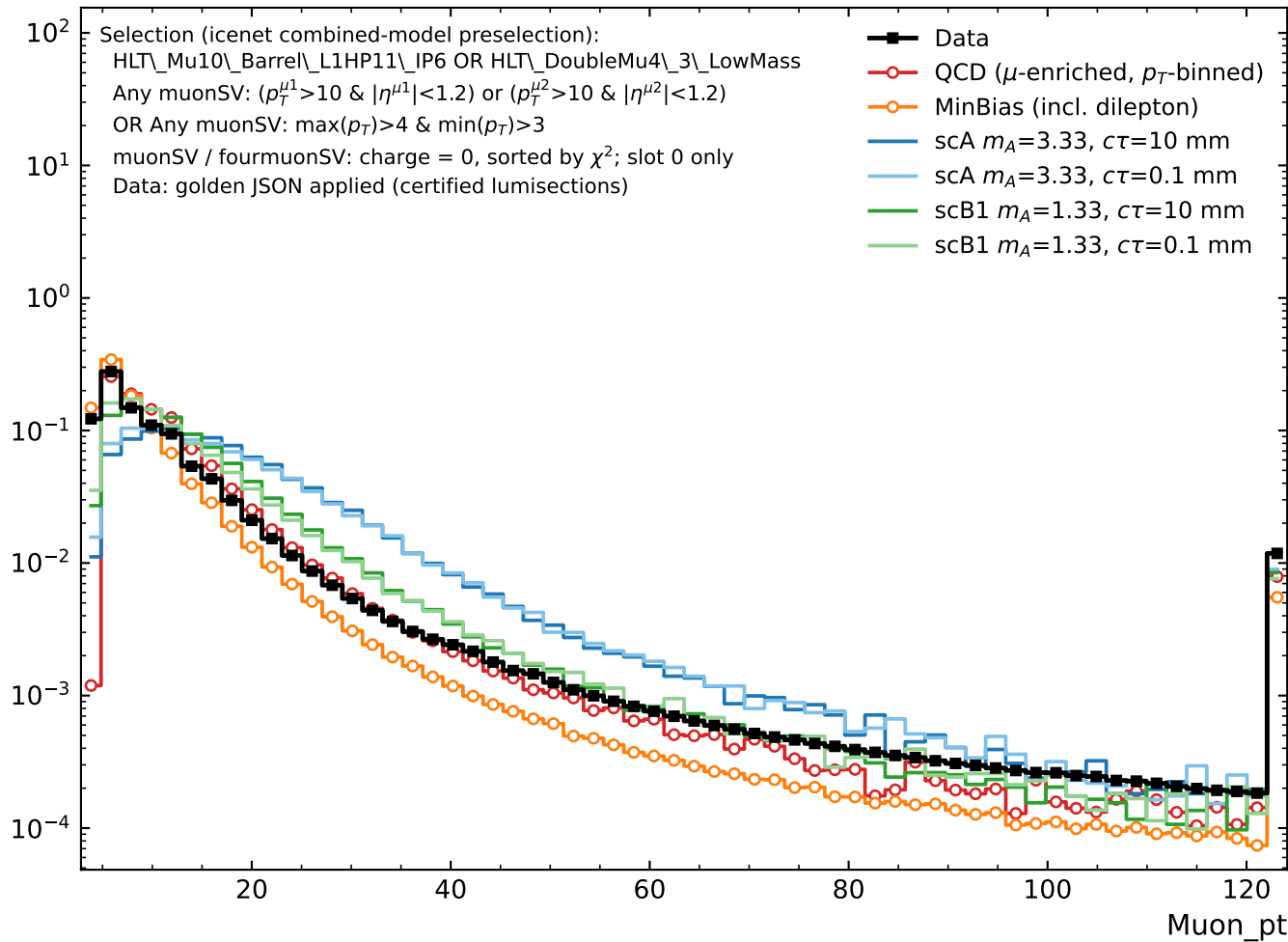
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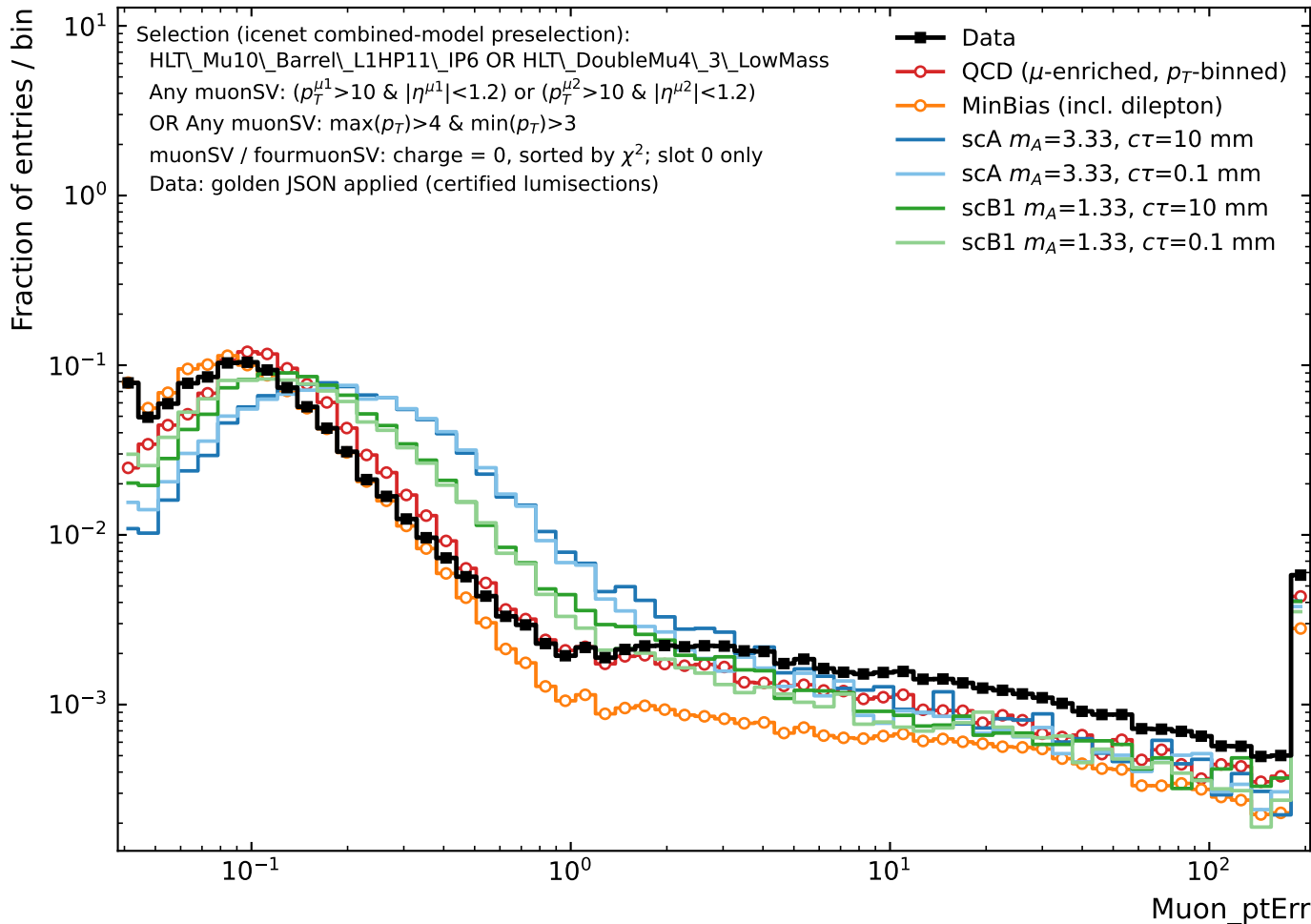
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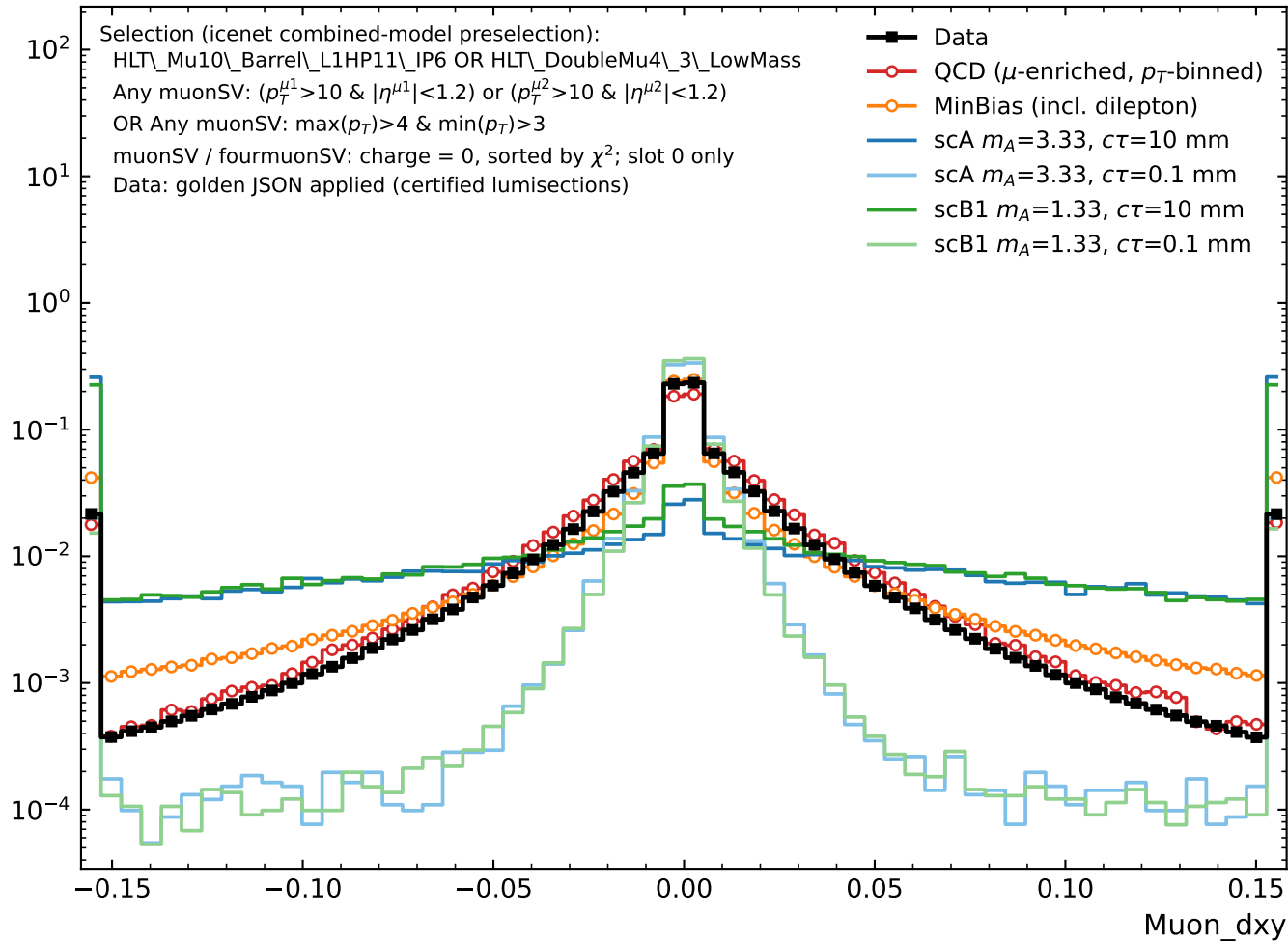
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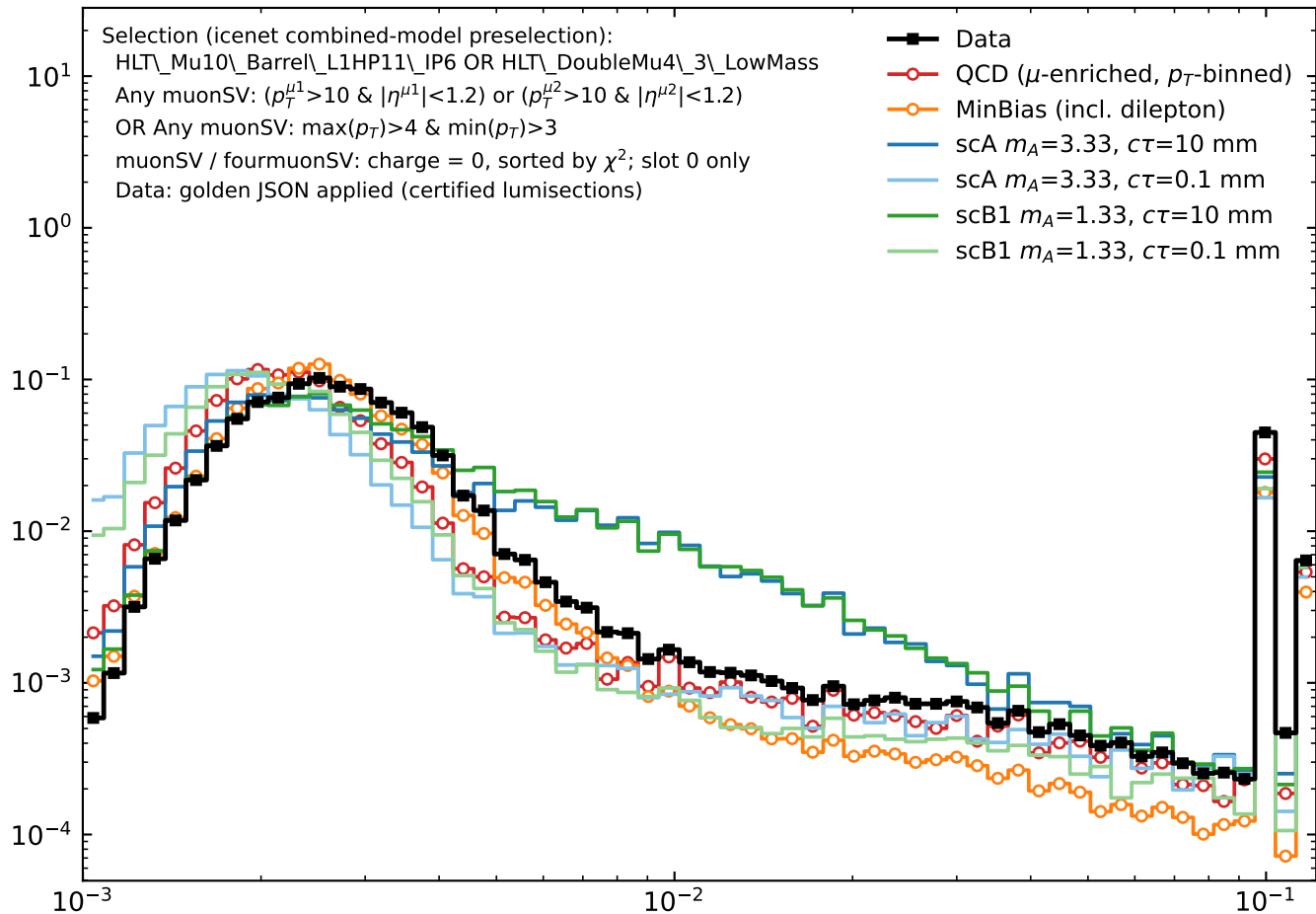
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Muon_dxyErr

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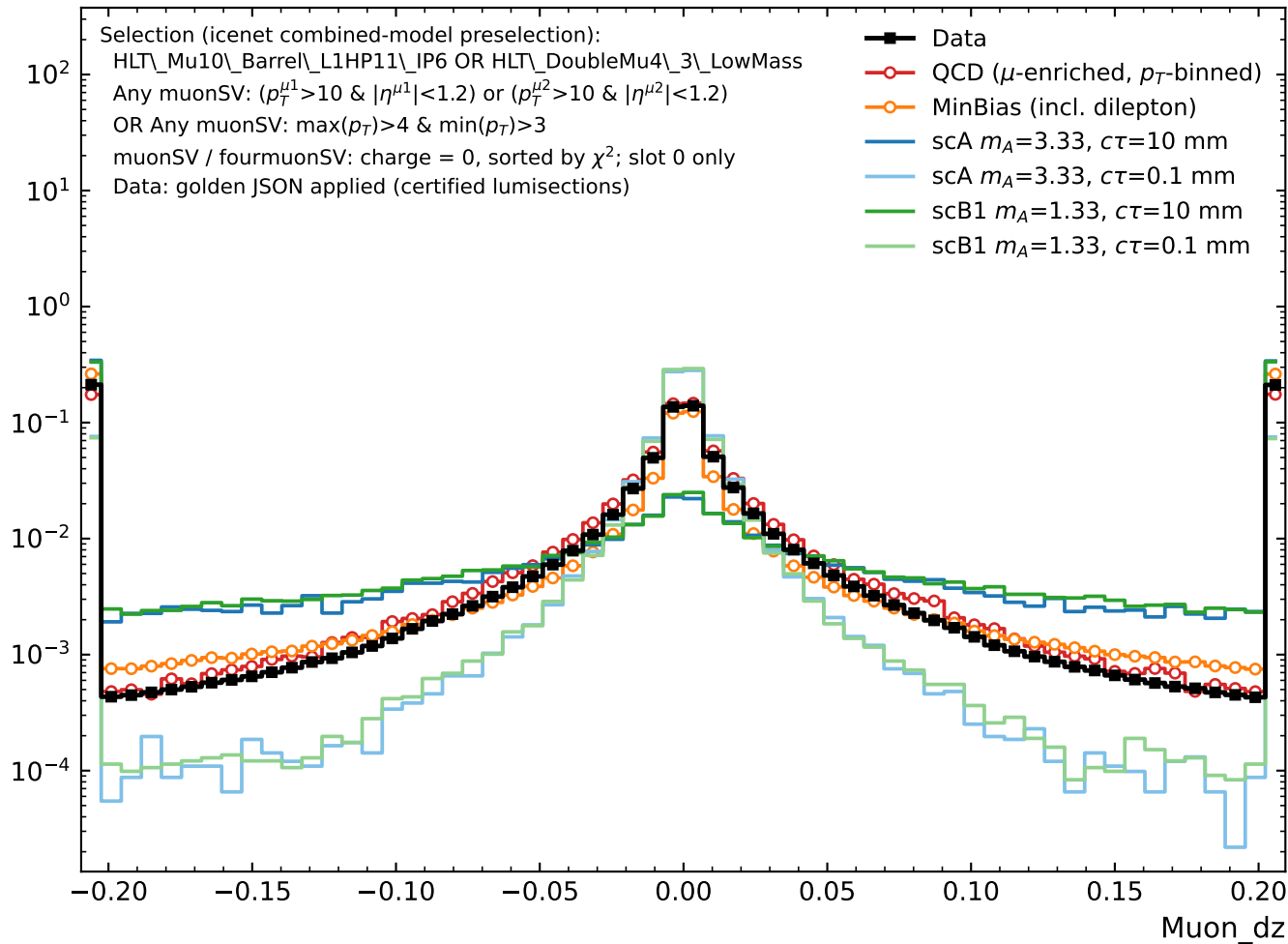
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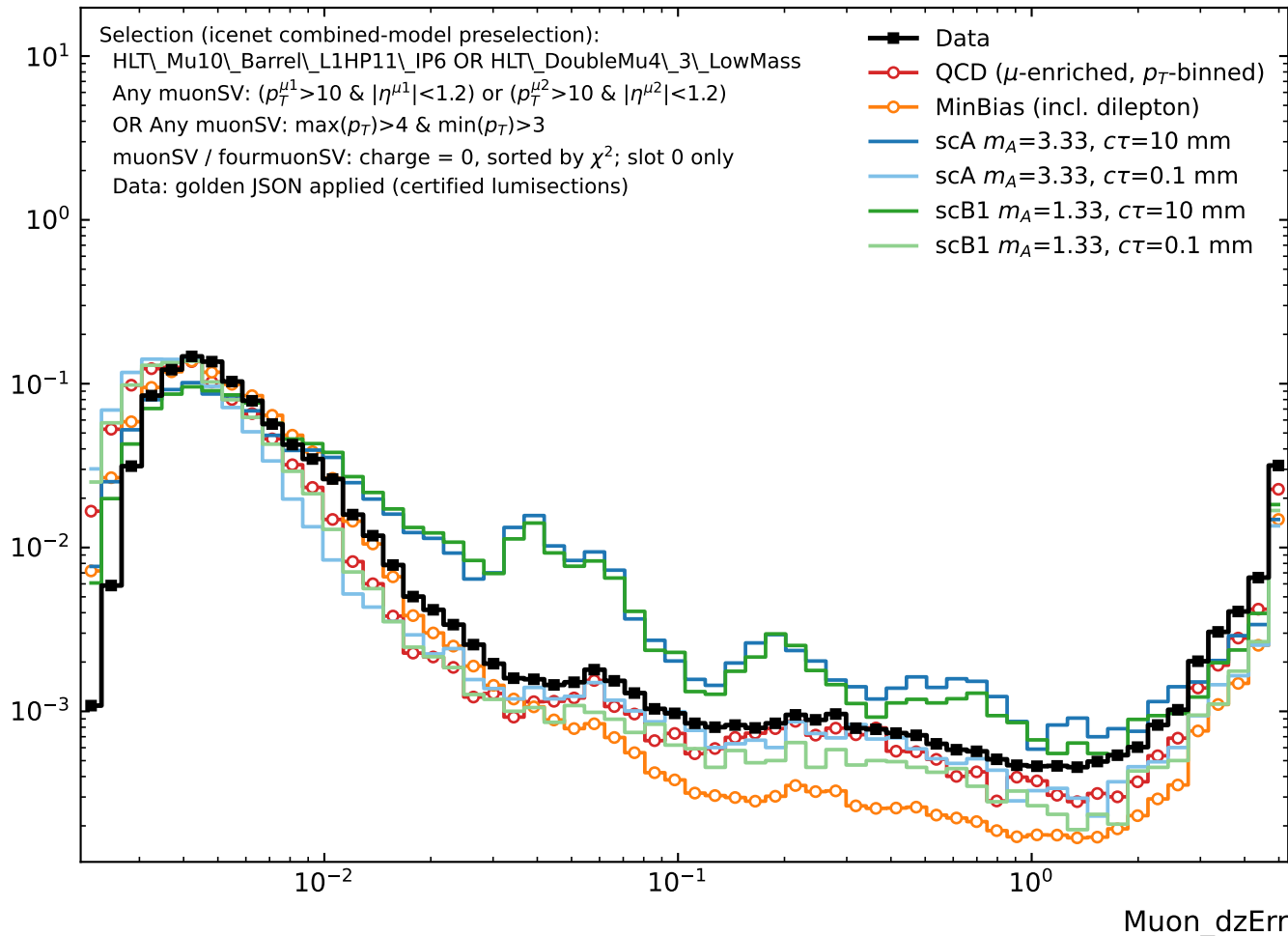
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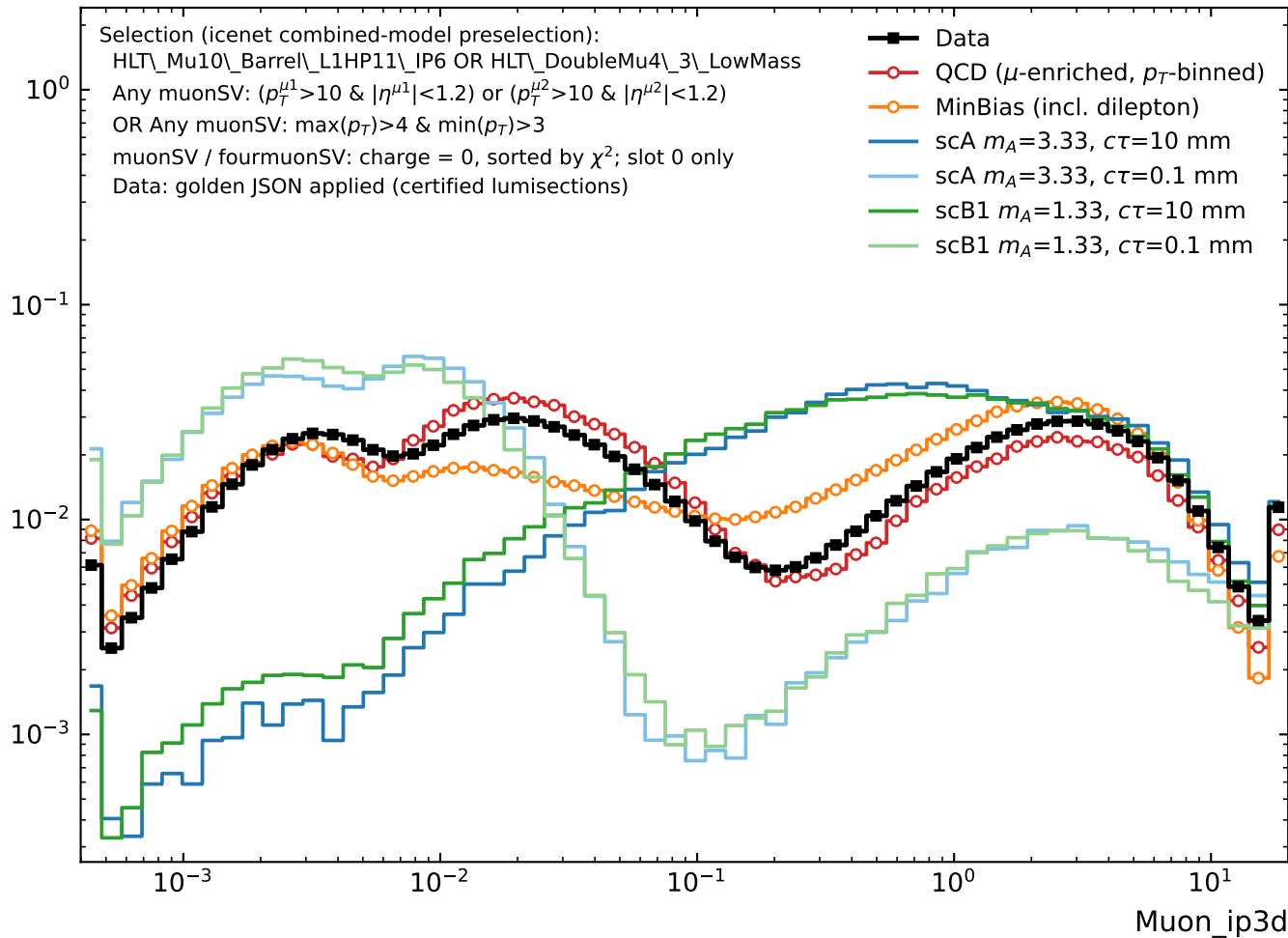
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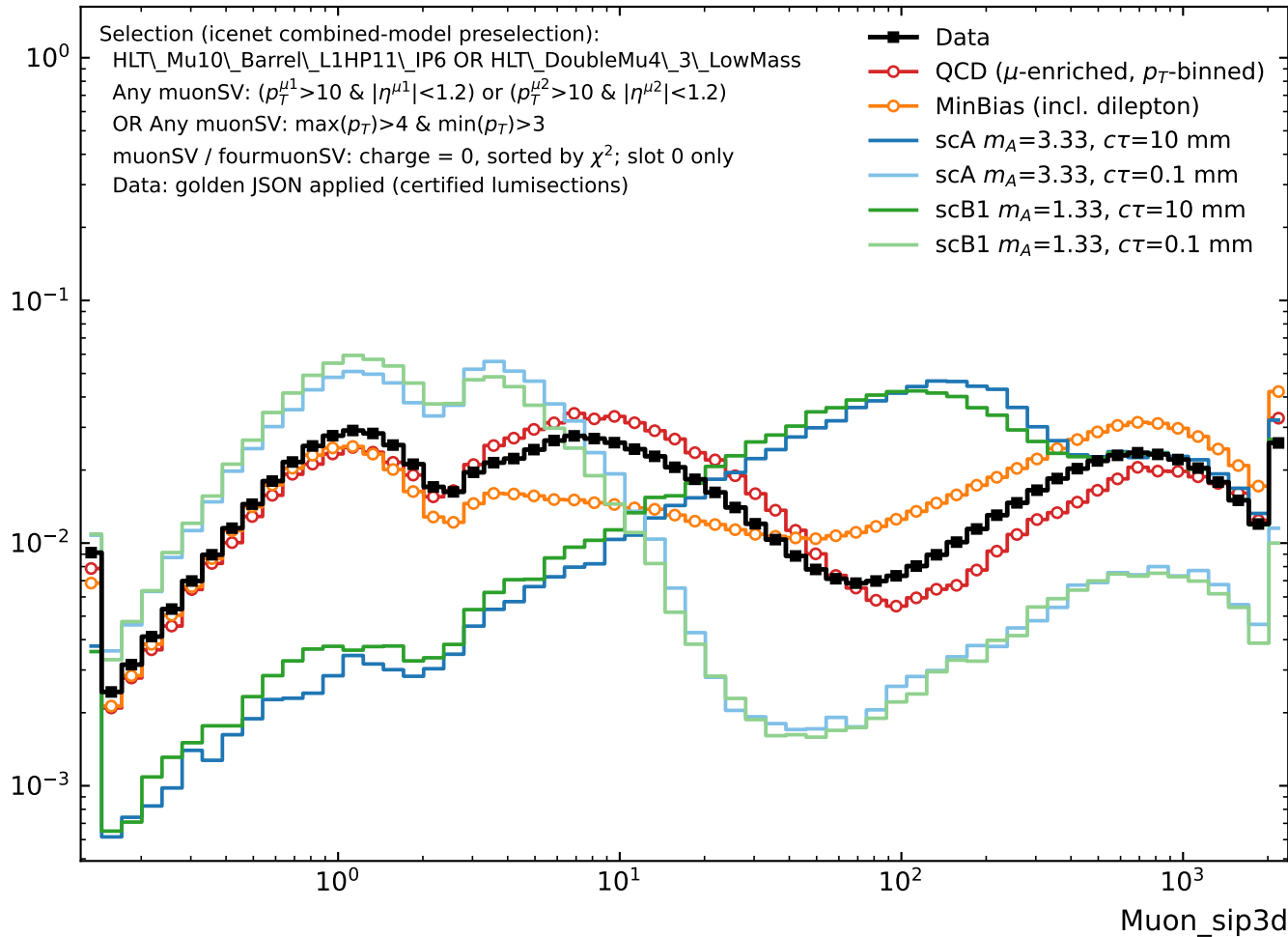
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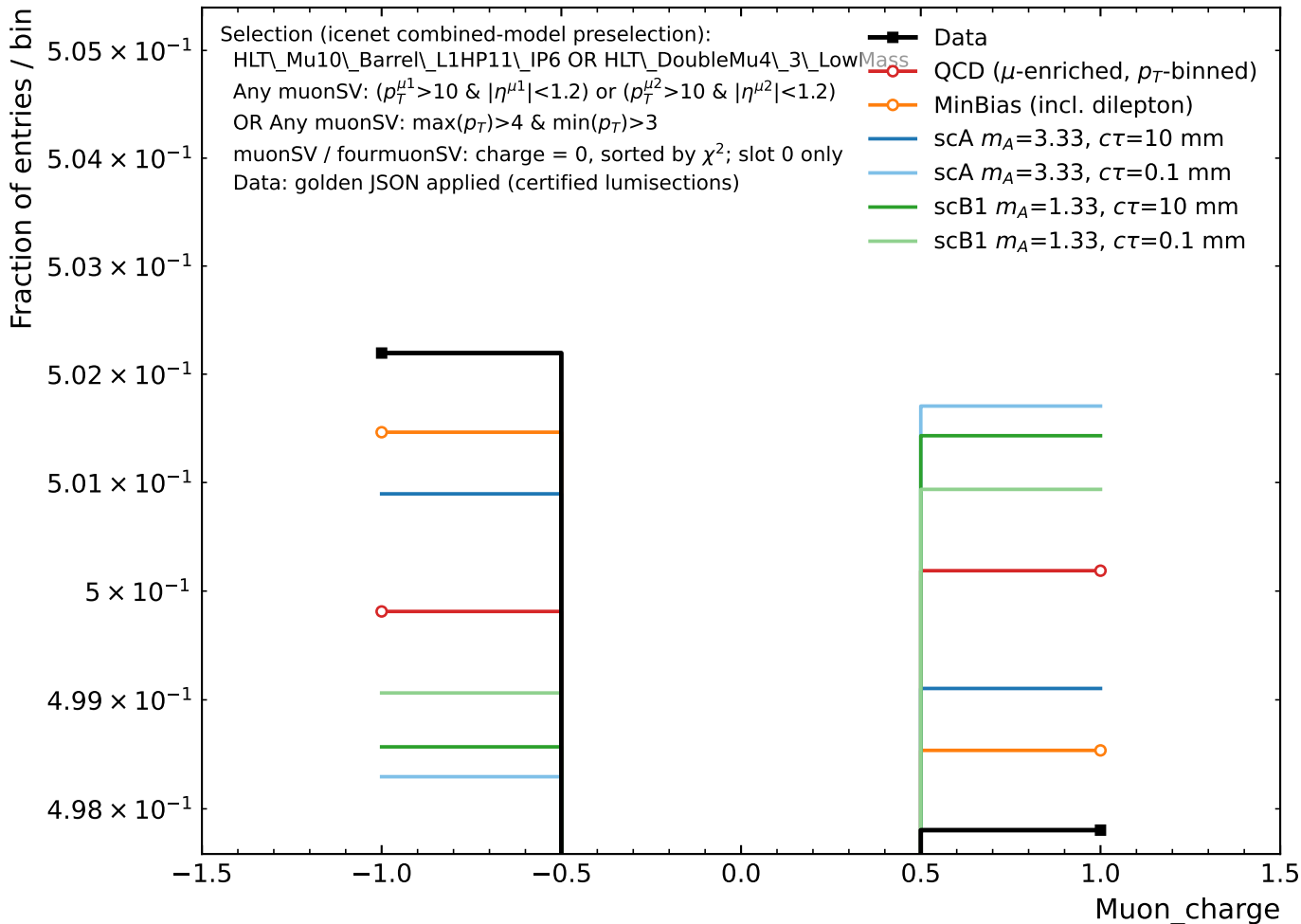
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Fraction of entries / bin





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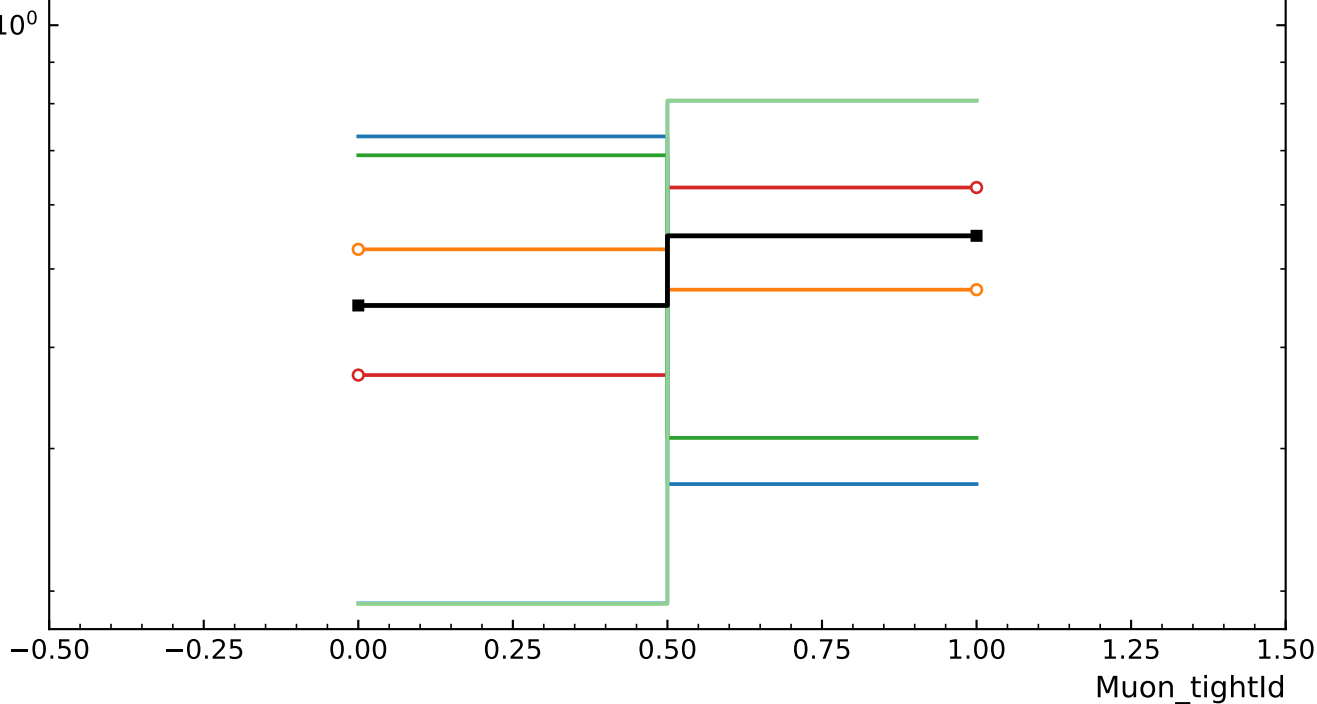
Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$ muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

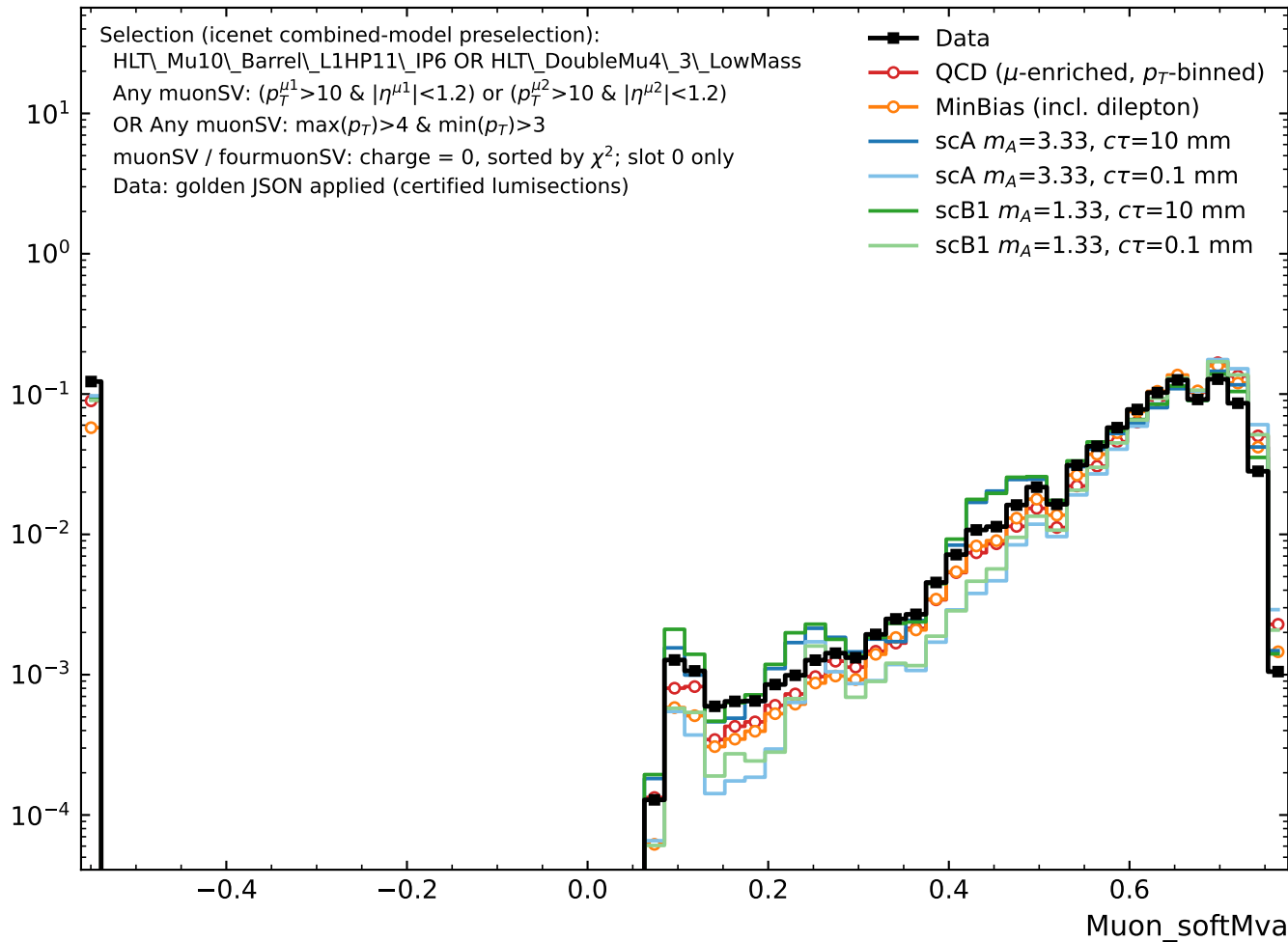
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

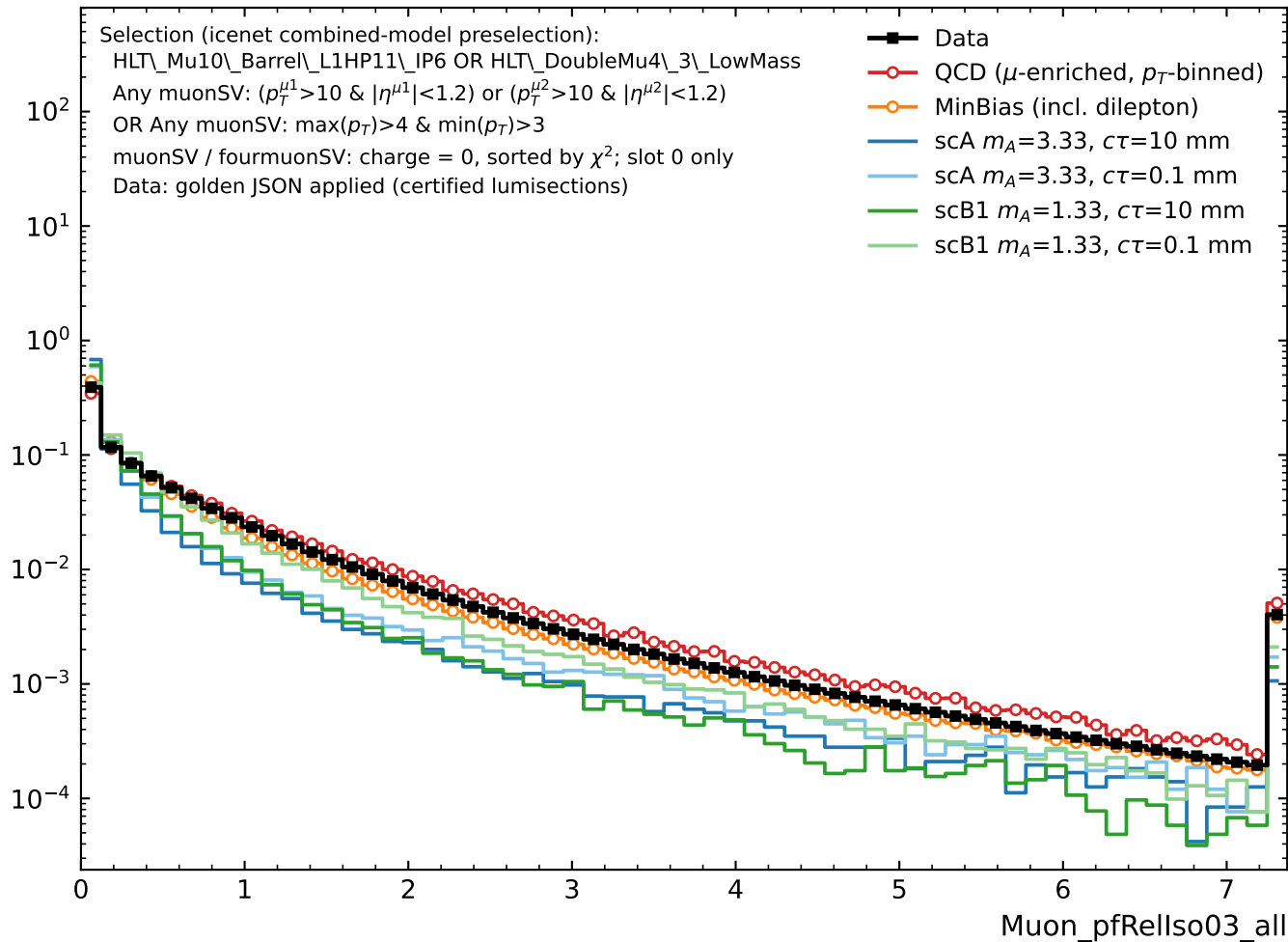
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

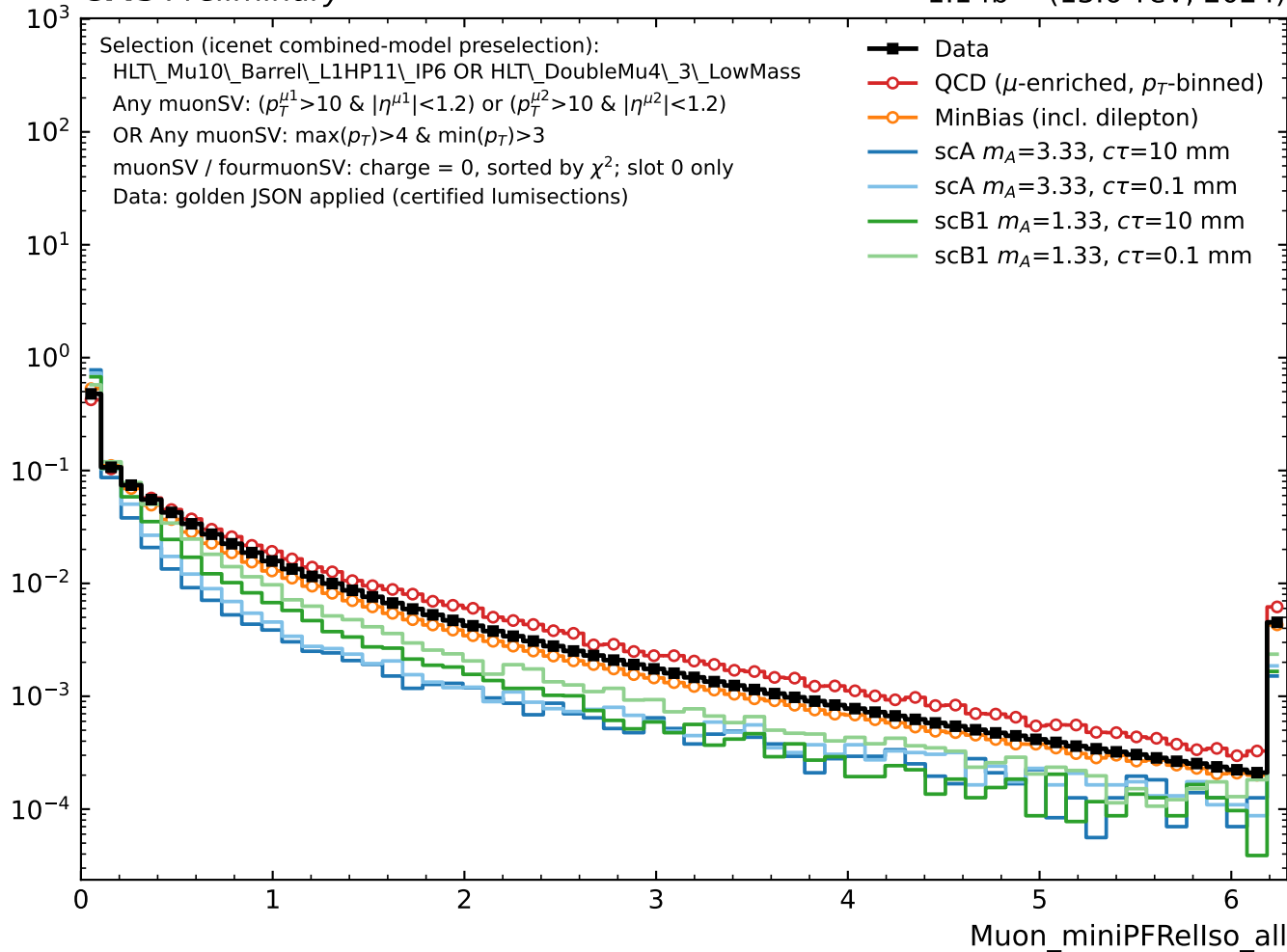
OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

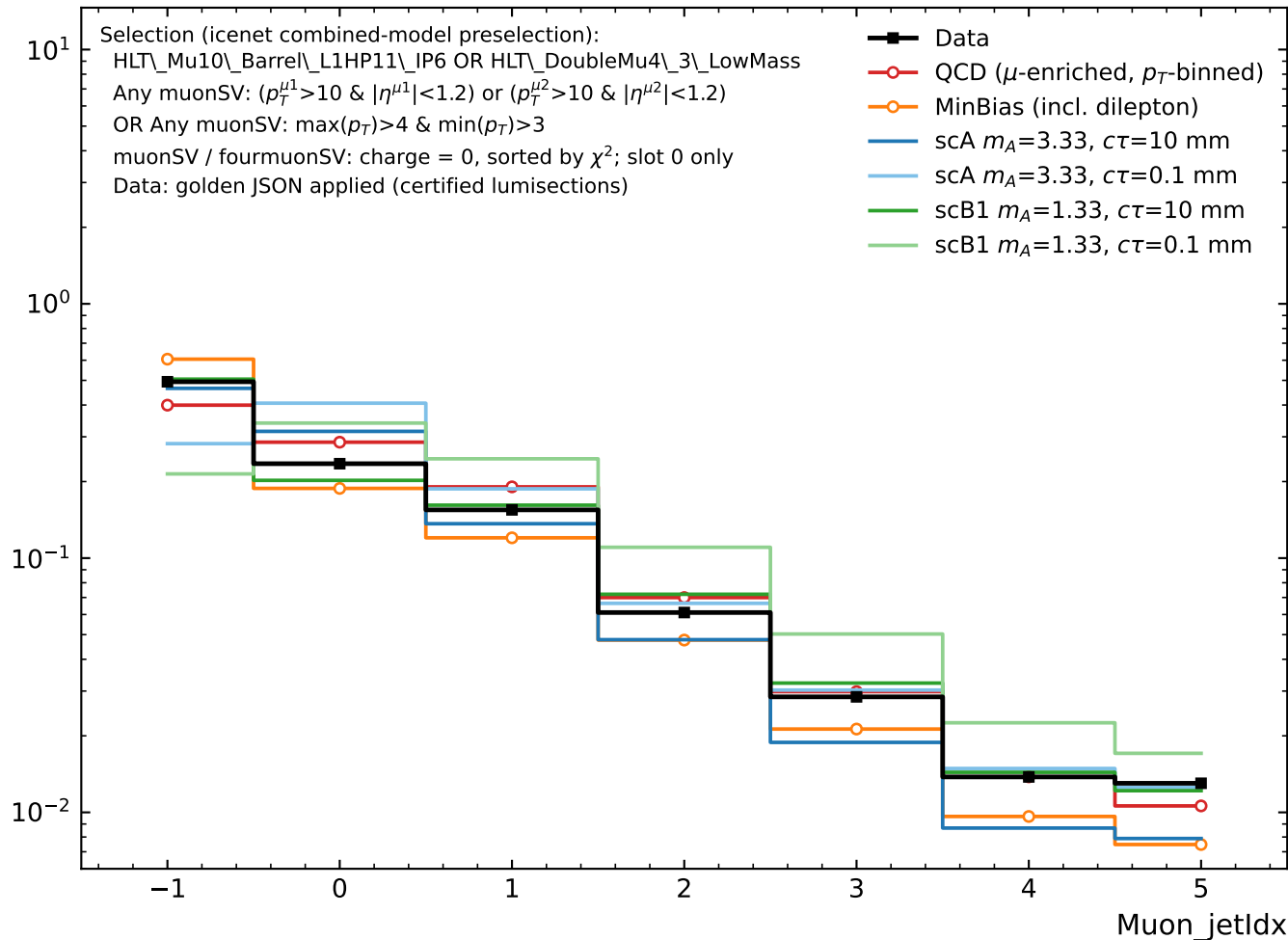
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

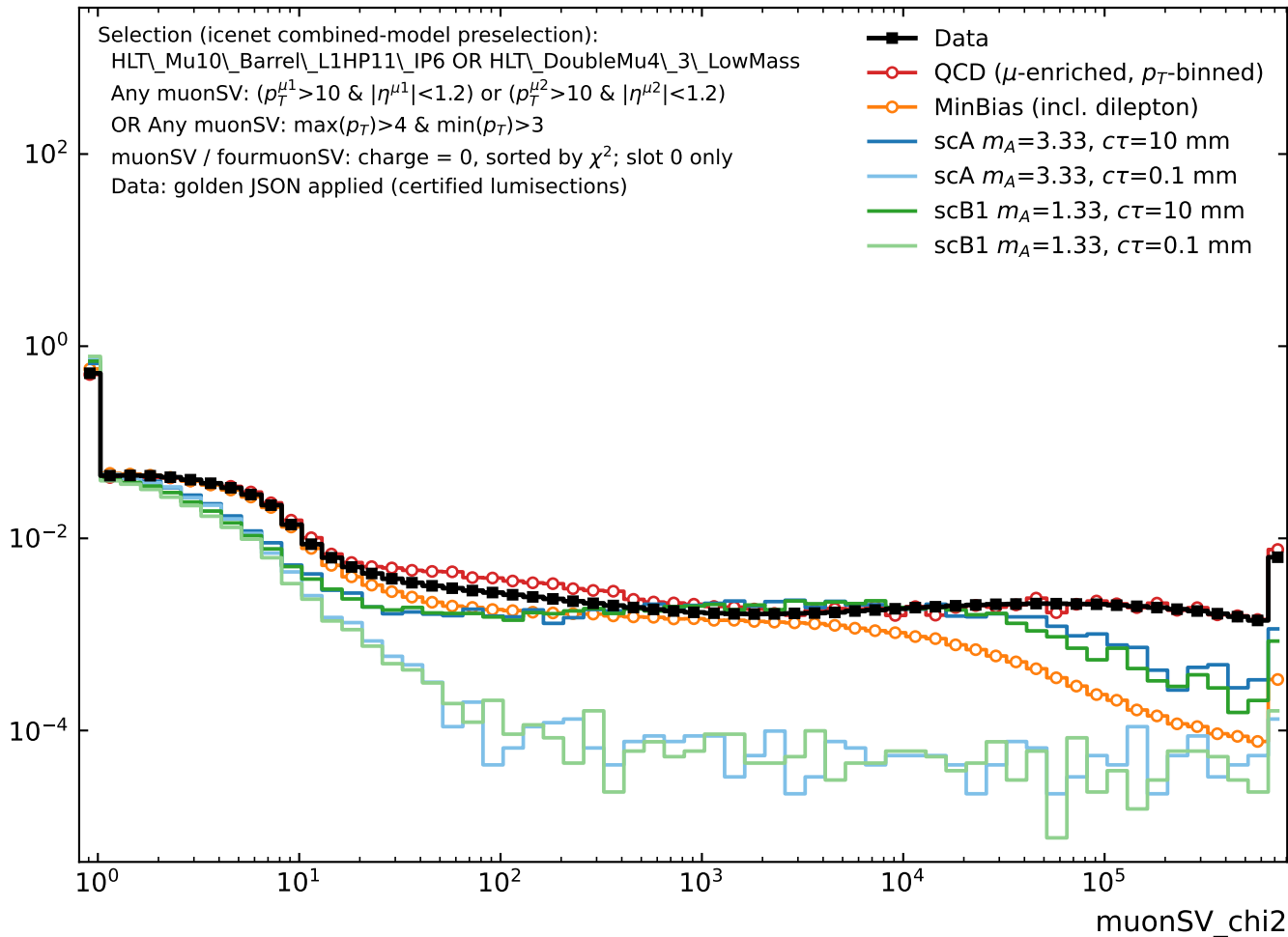
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

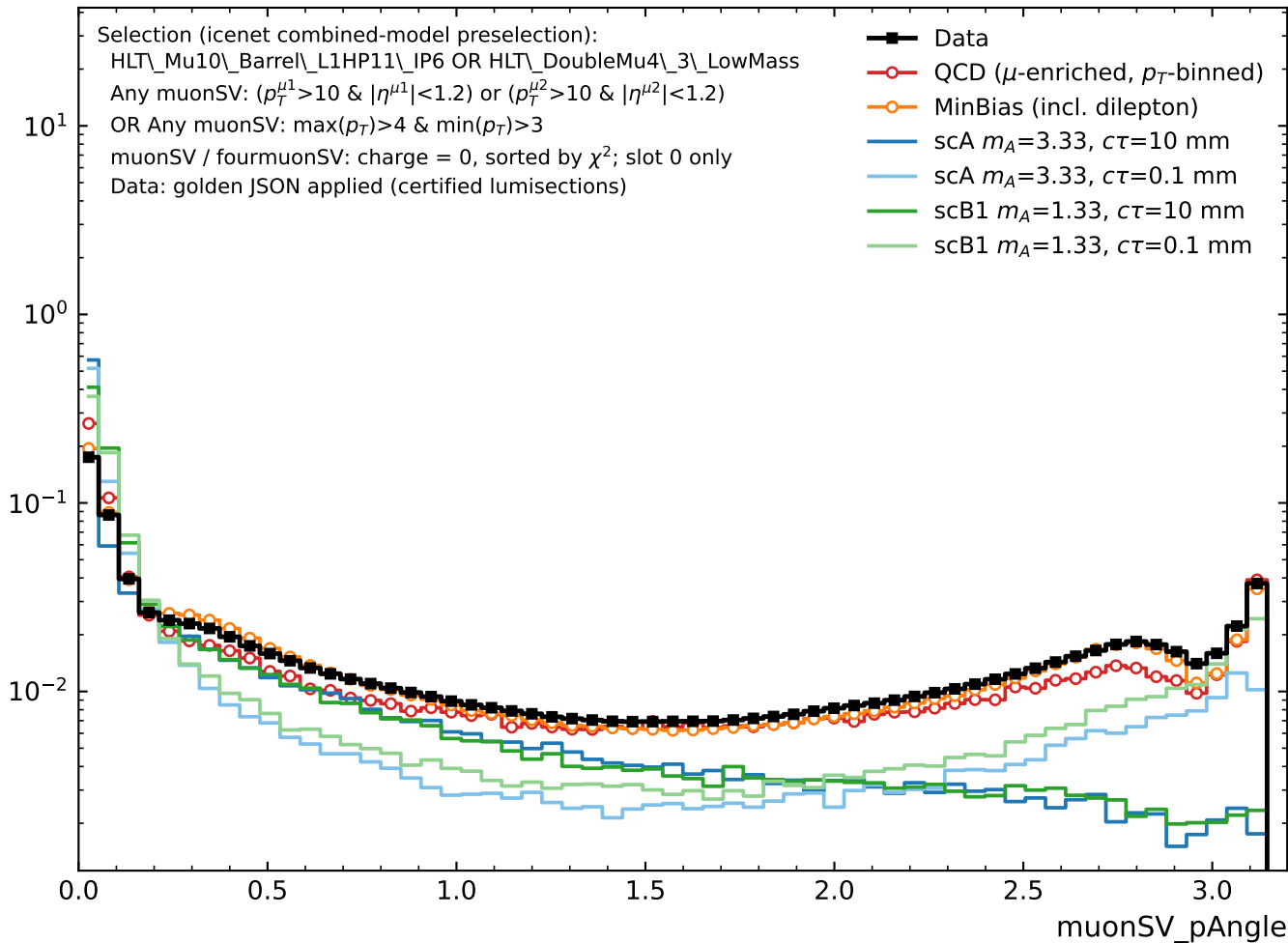
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

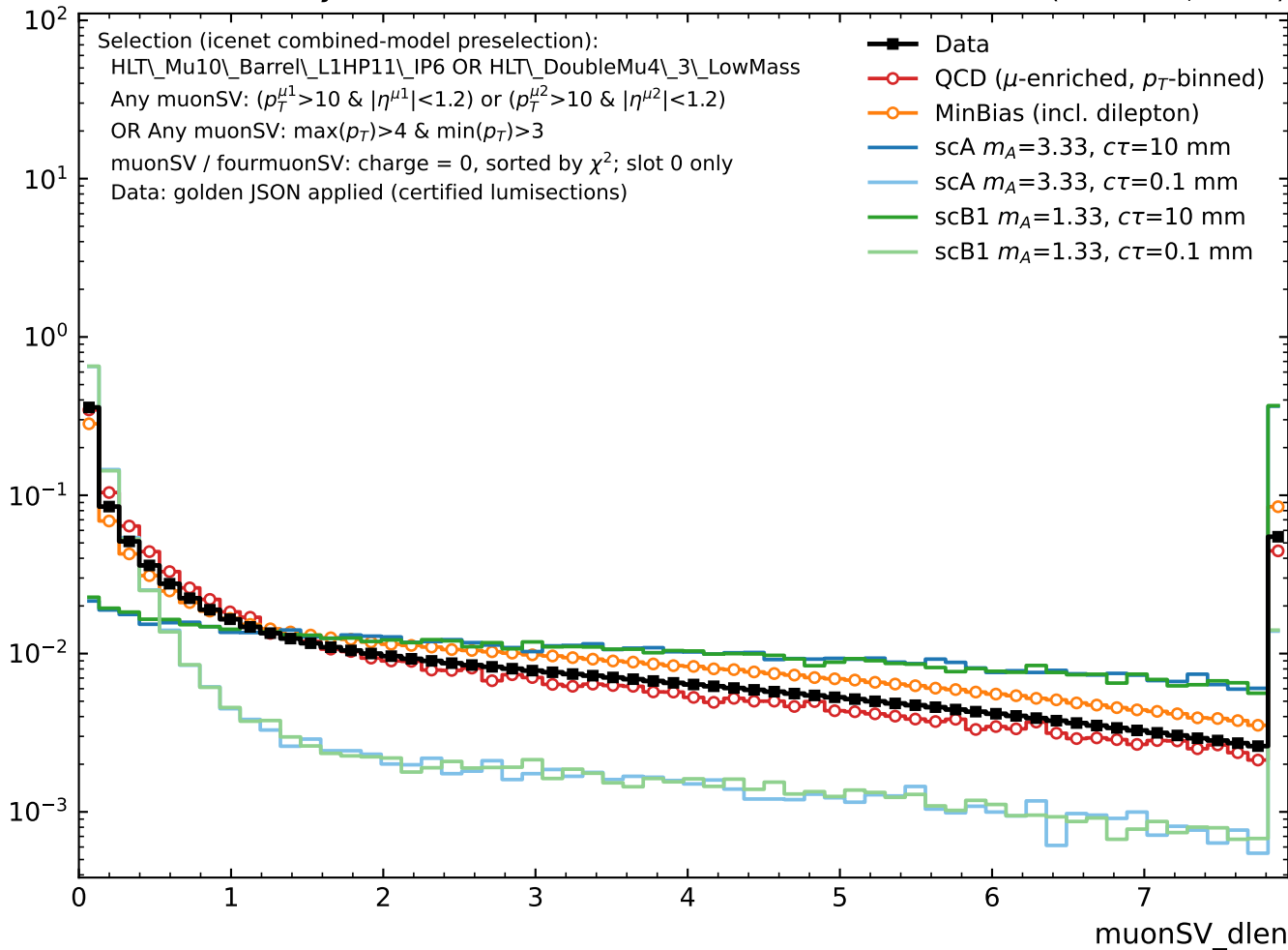
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

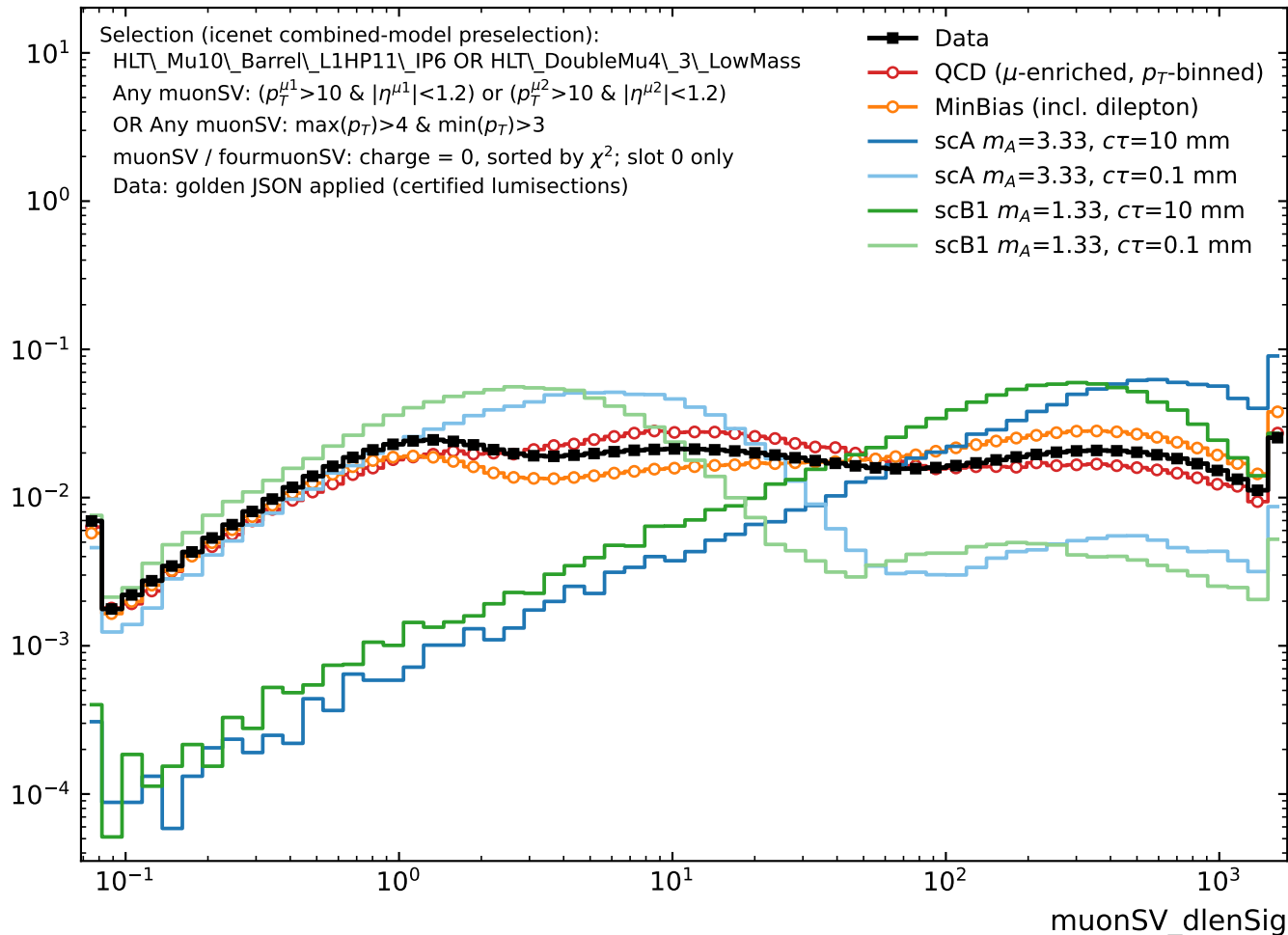
muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

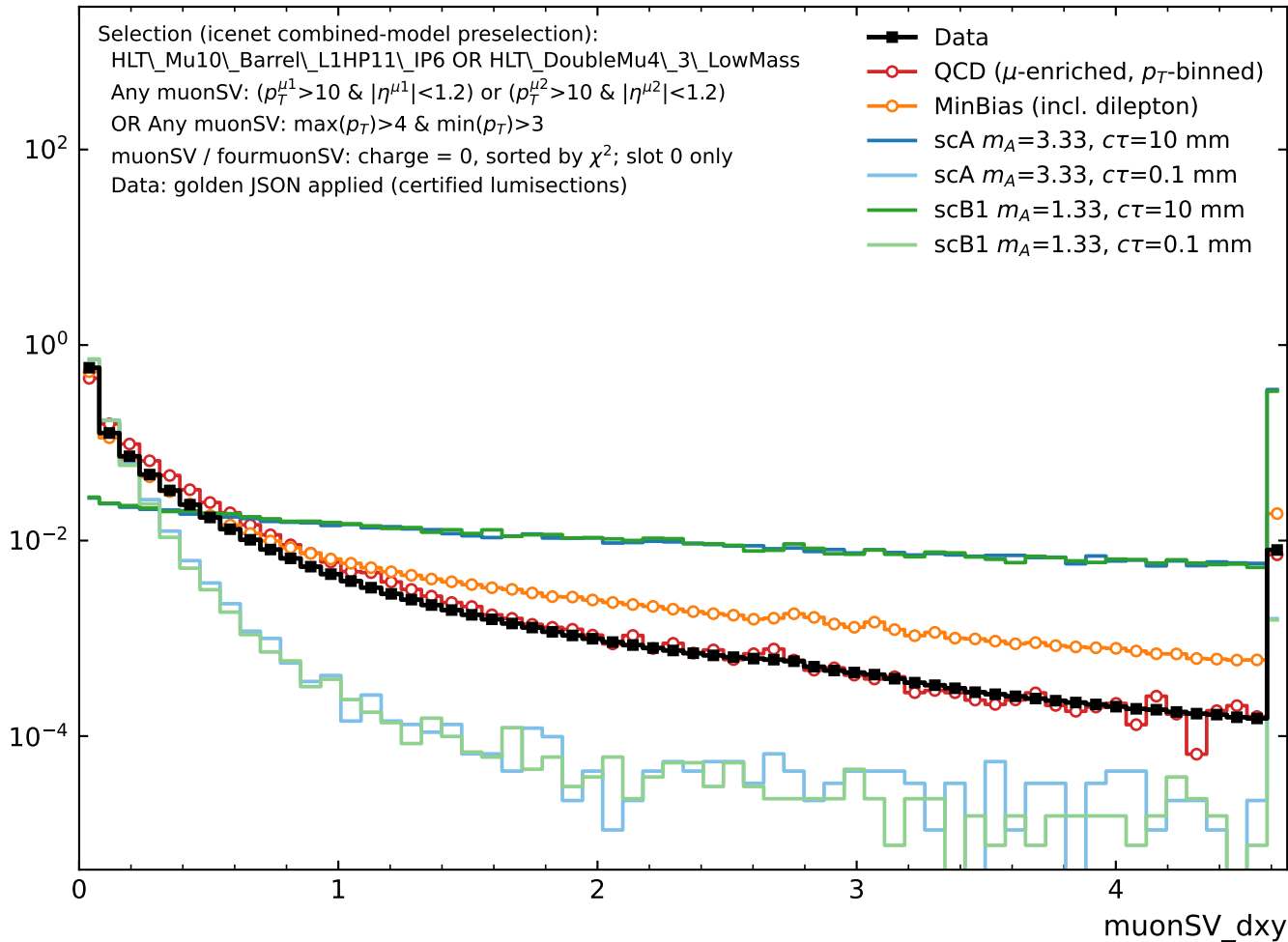
OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

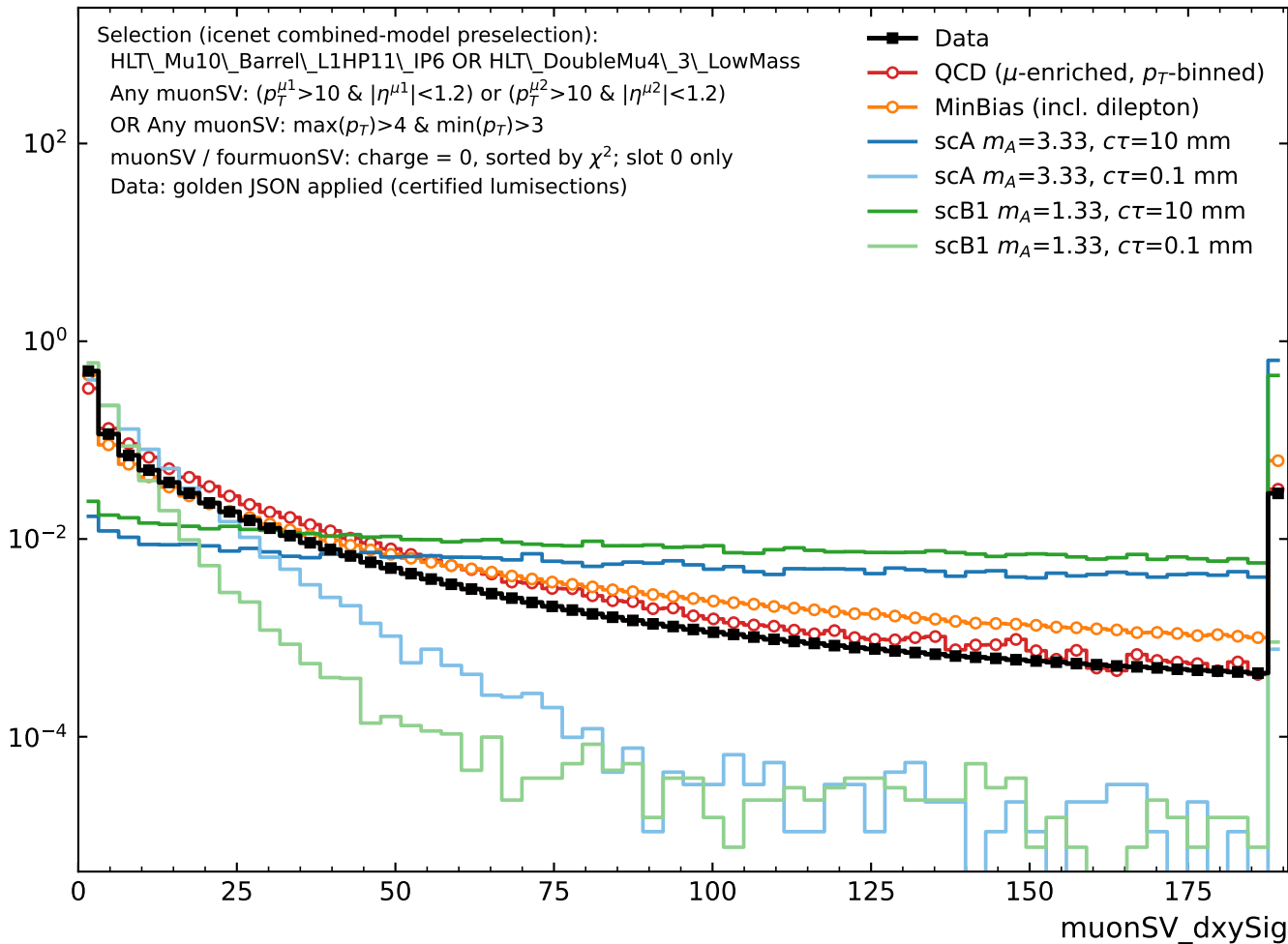
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

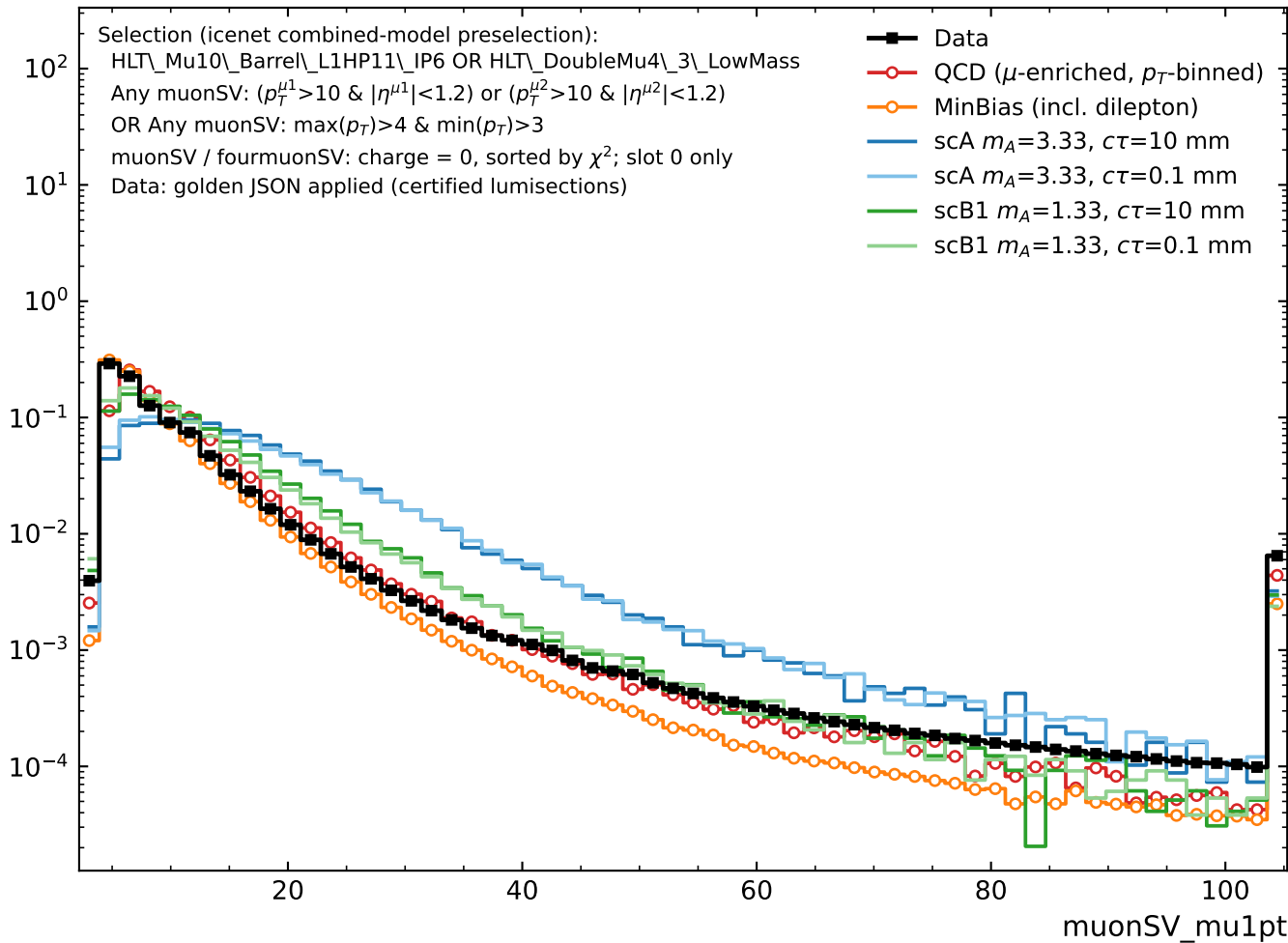
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

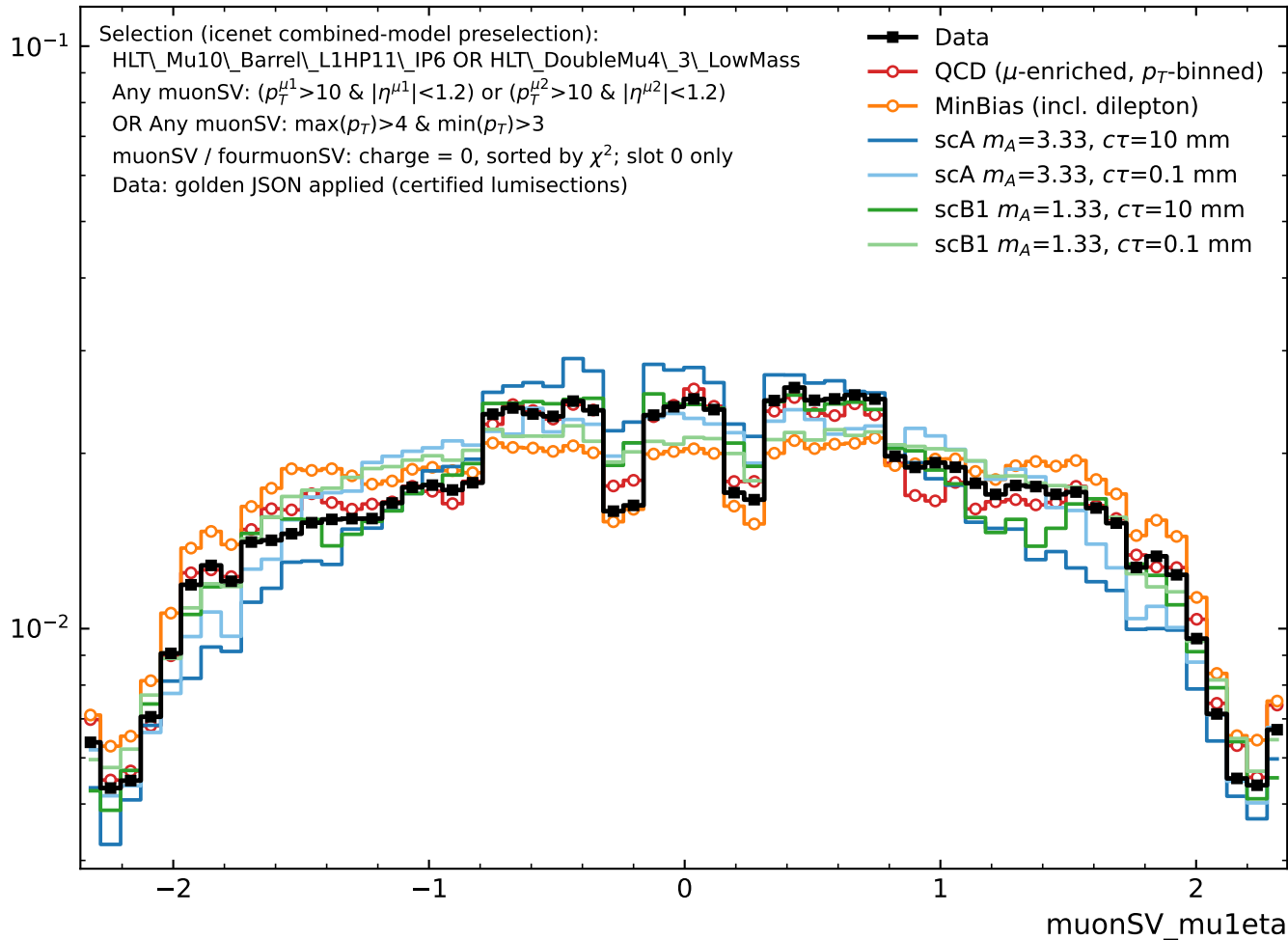
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



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HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

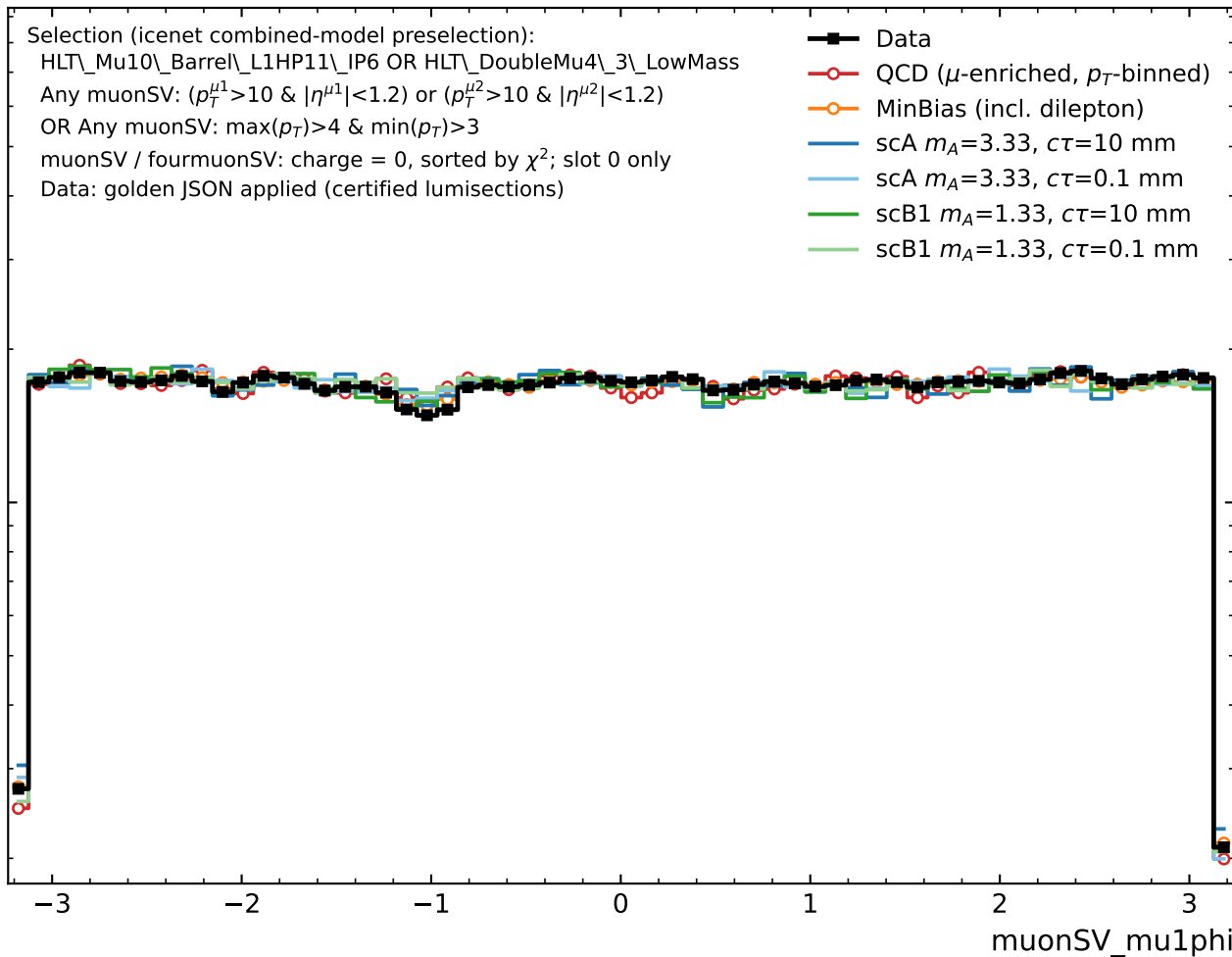
Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

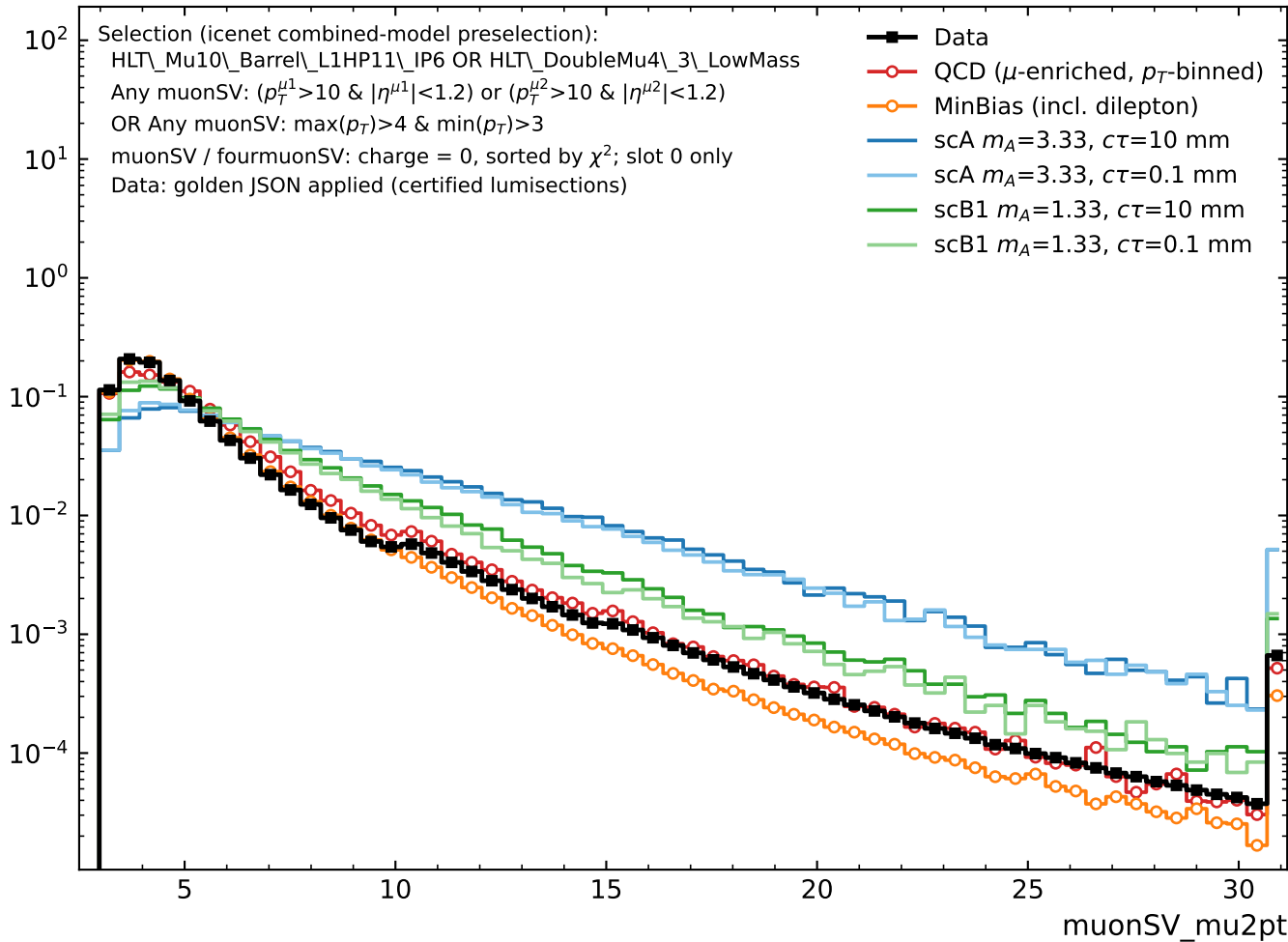
Fraction of entries / bin

10^{-2}

muonSV_mu1phi



Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

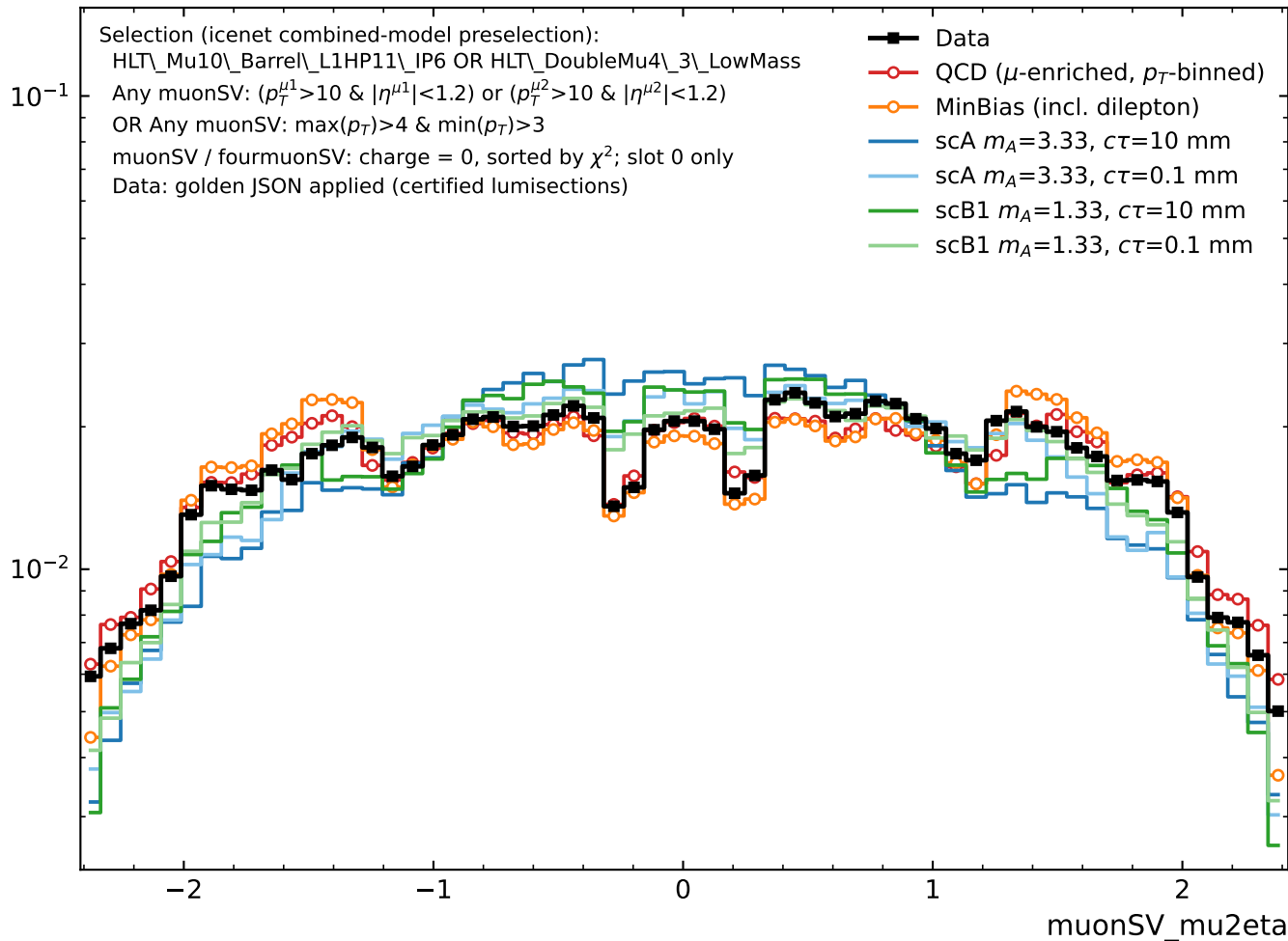
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin

10^{-2}

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

-3

-2

-1

0

1

2

3

muonSV_mu2phi

Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

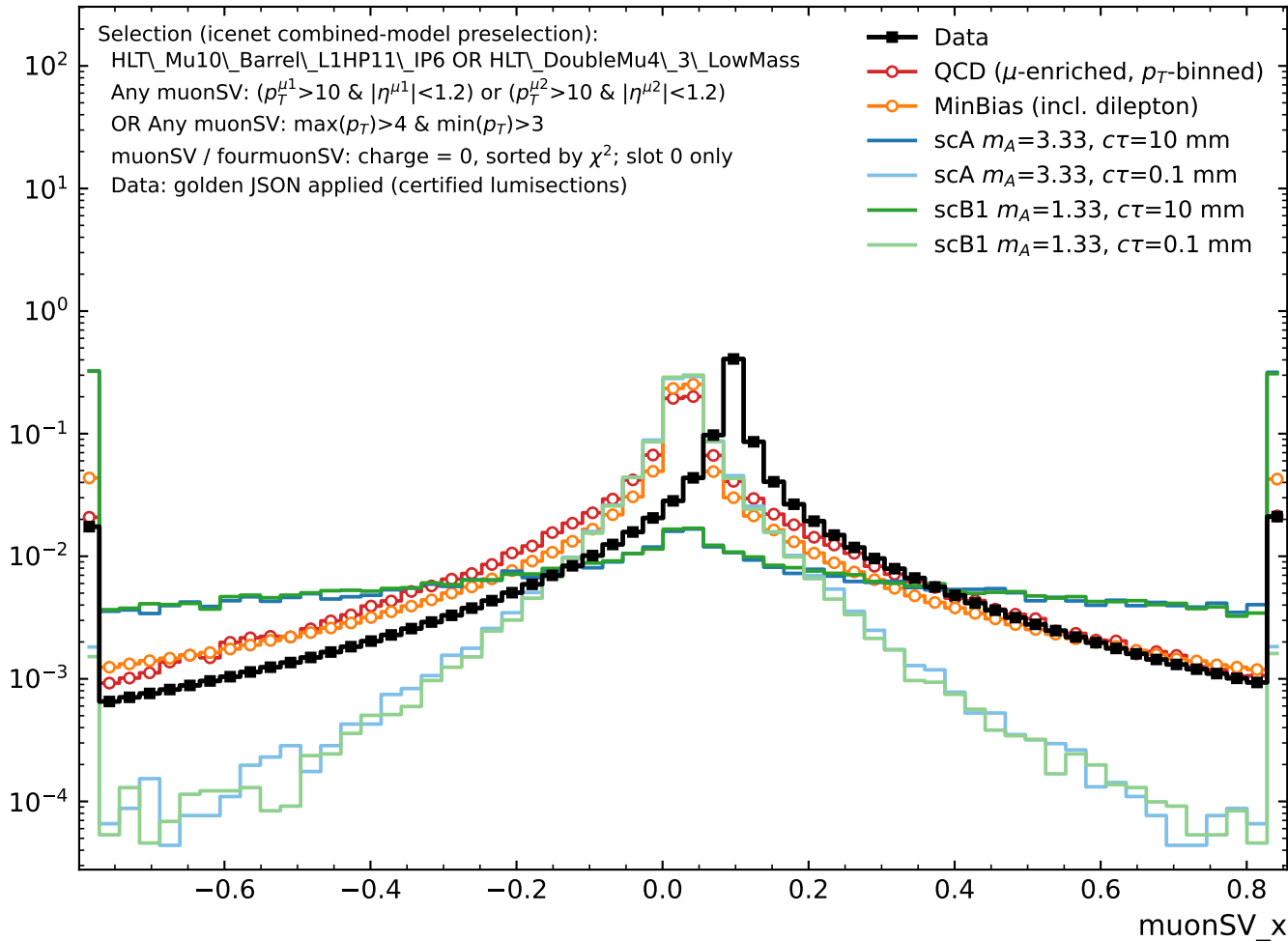
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

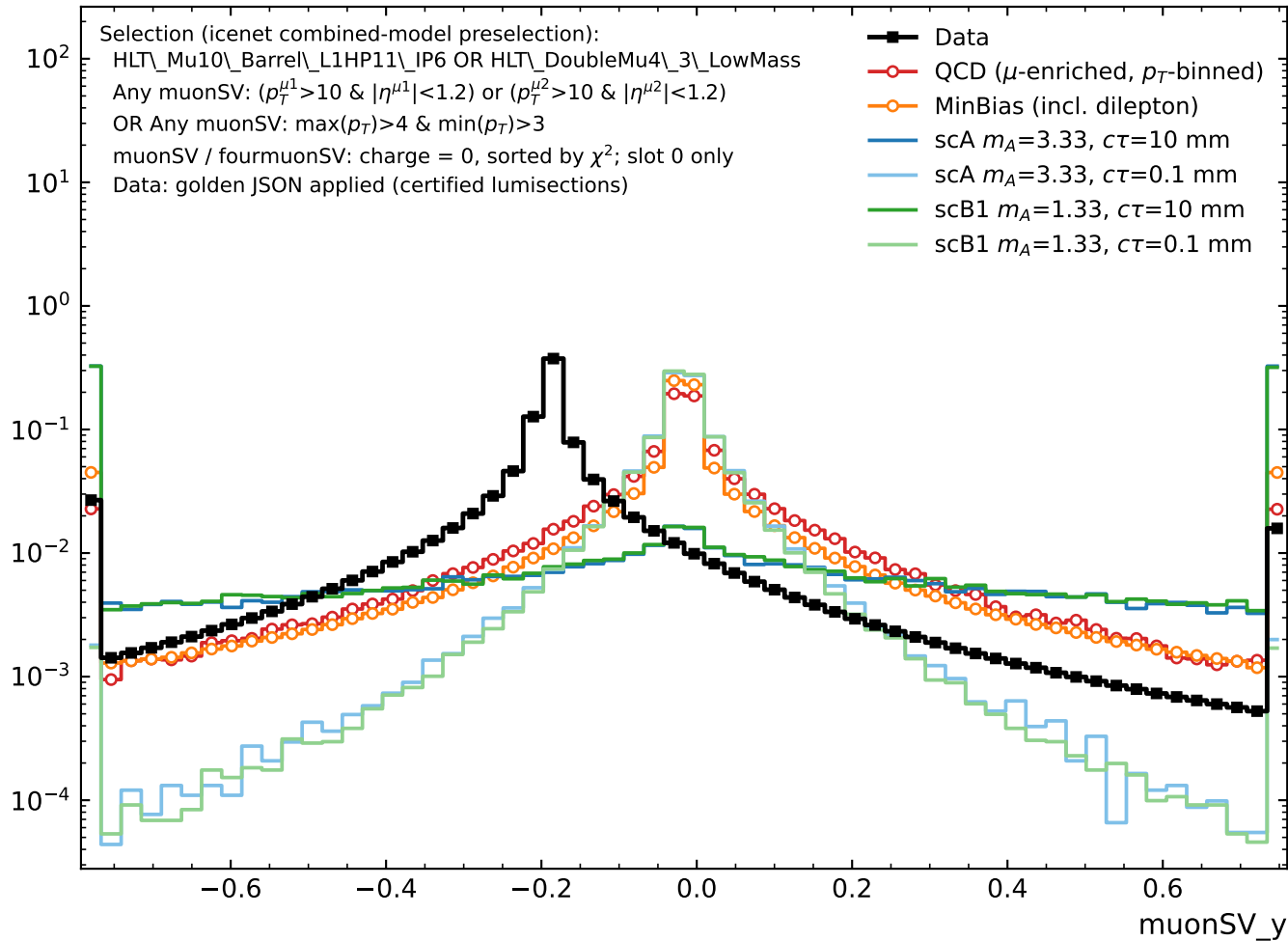
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

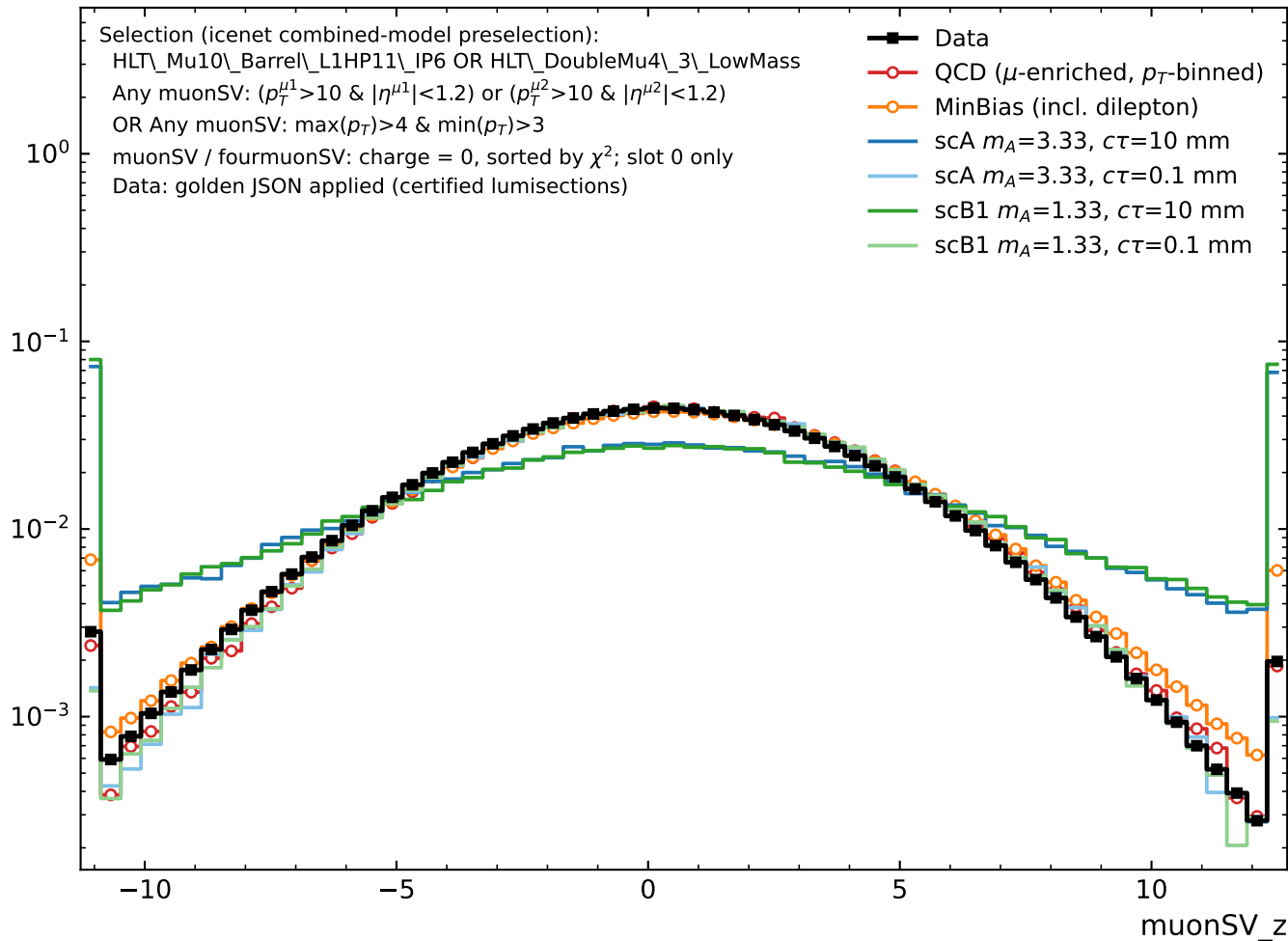
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

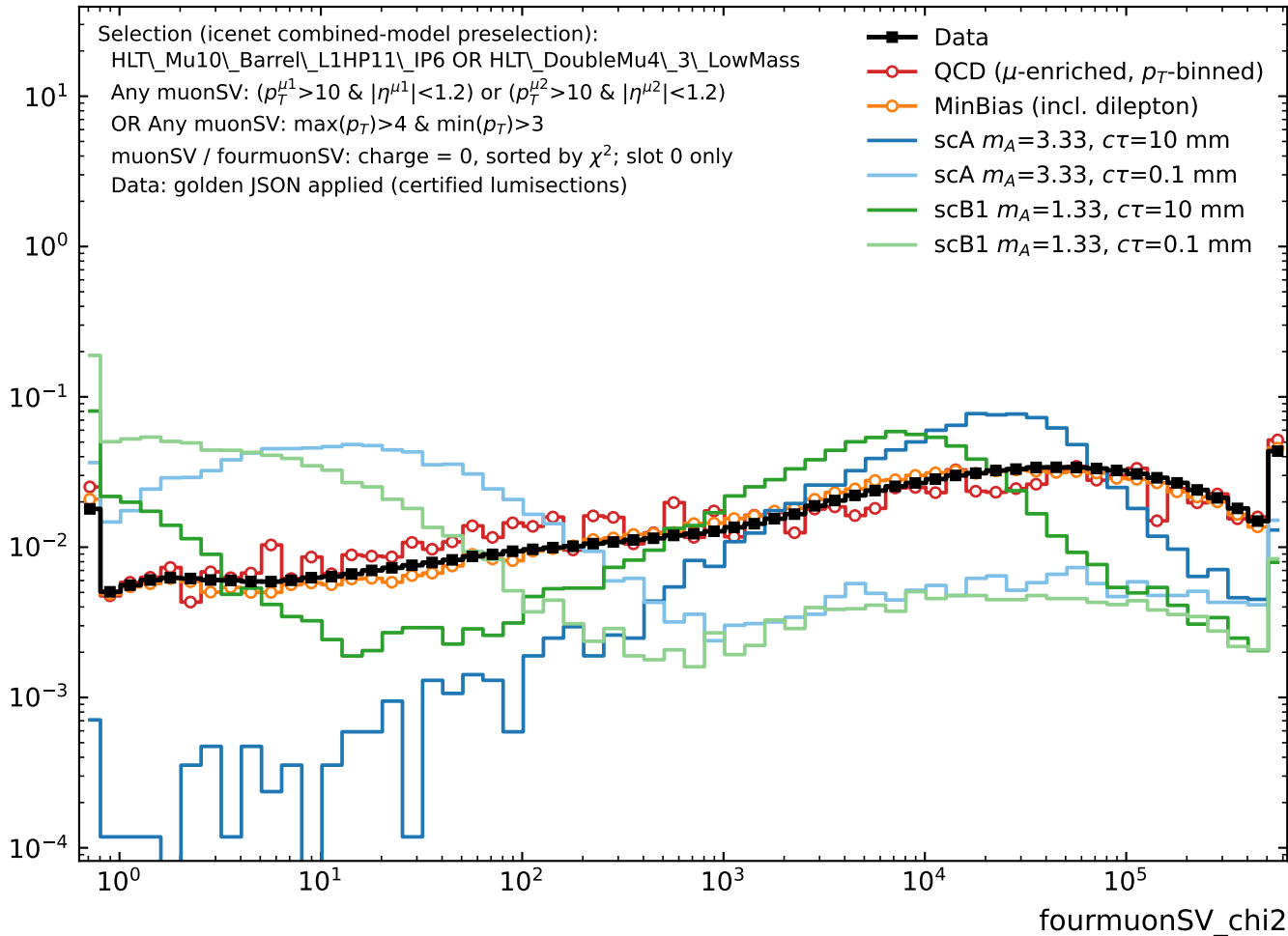
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

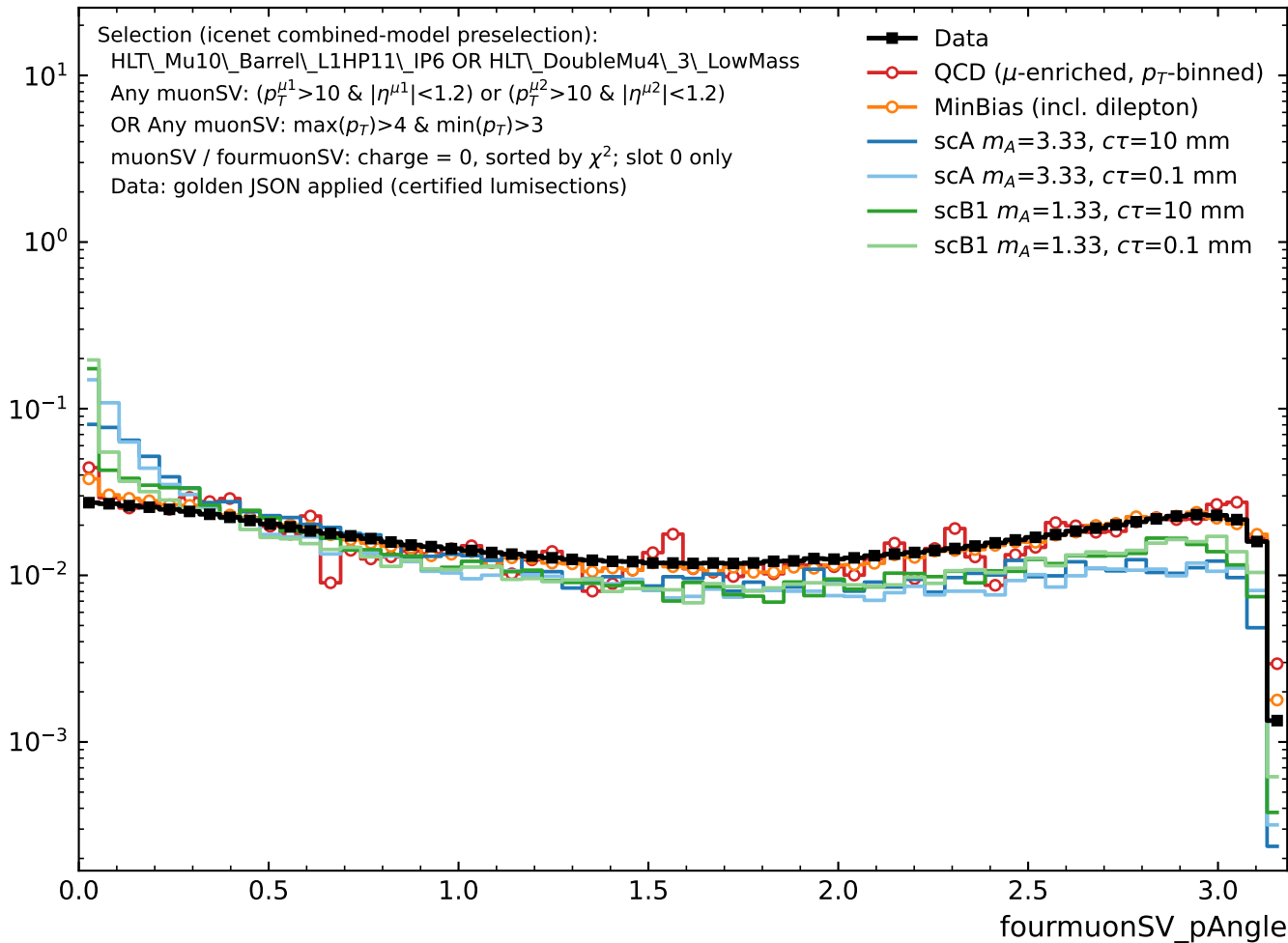
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

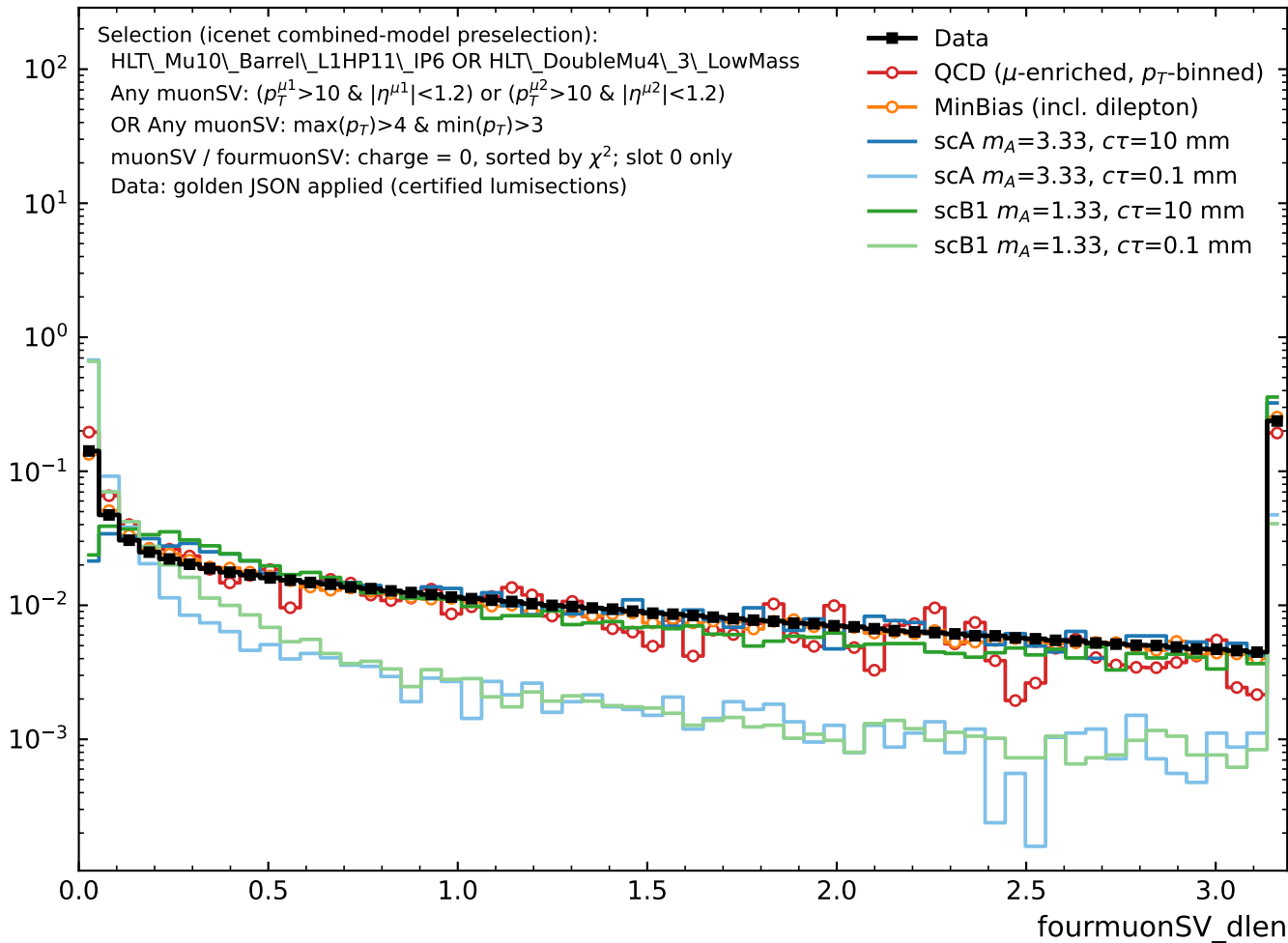
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

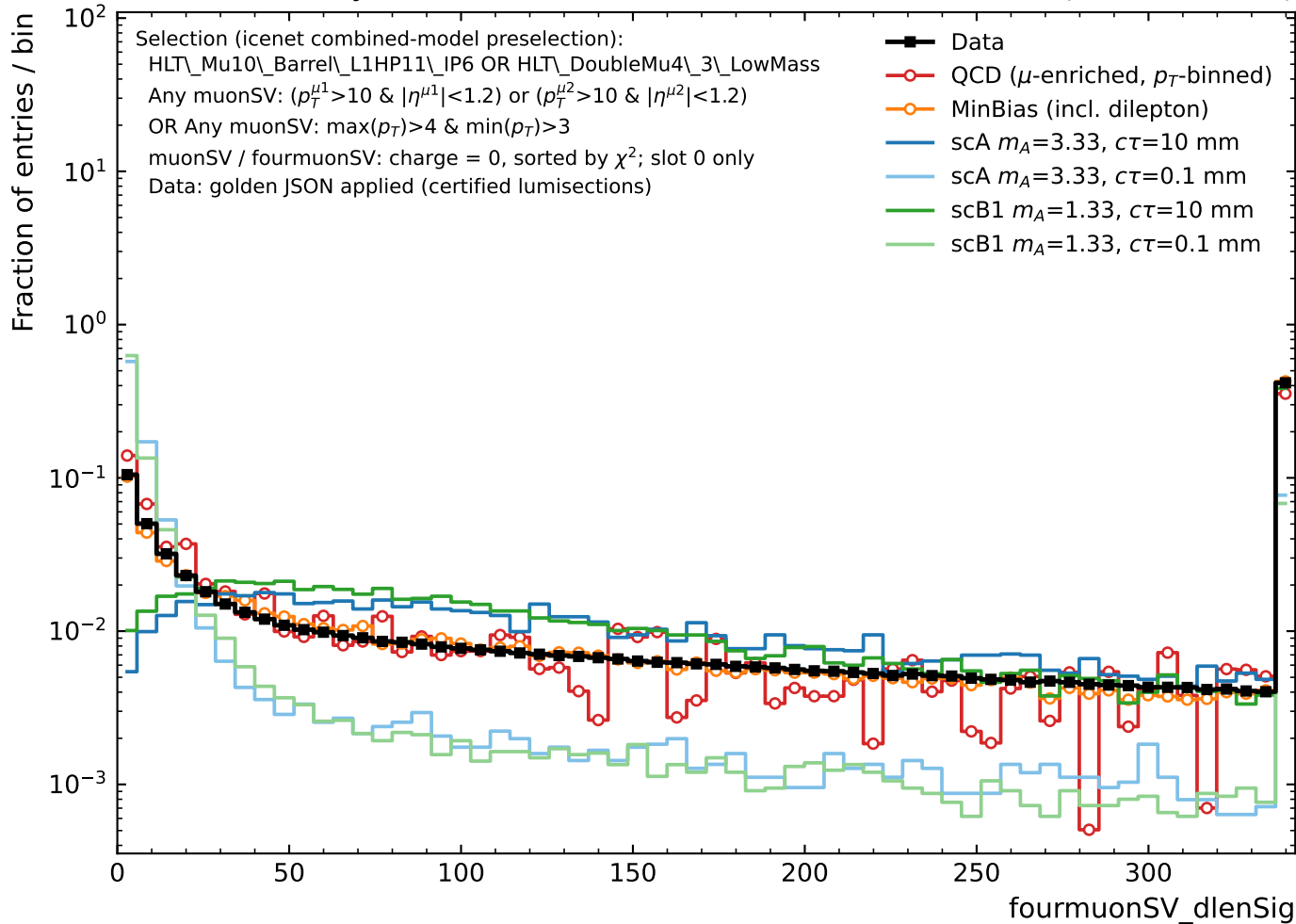
HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

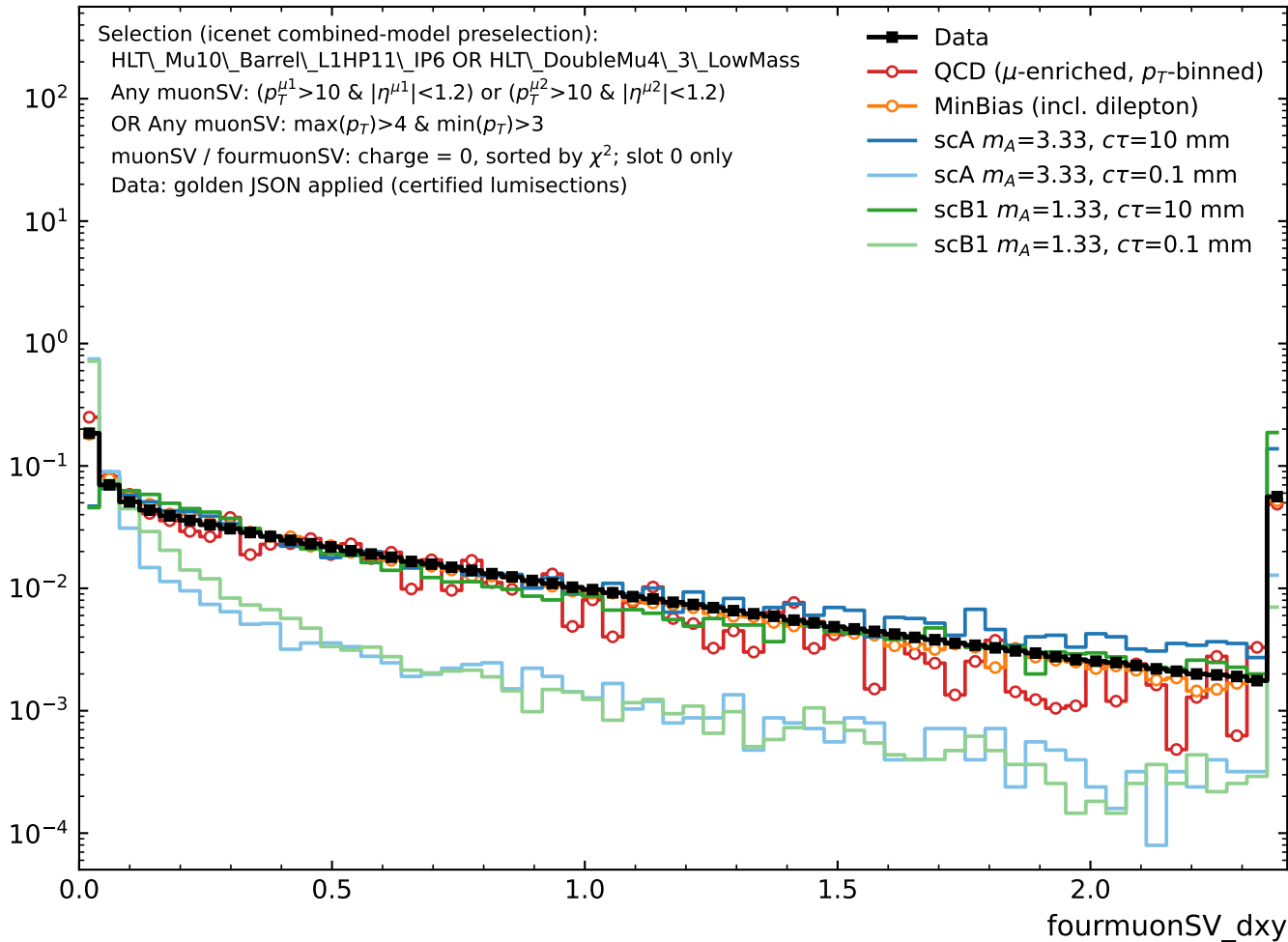
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

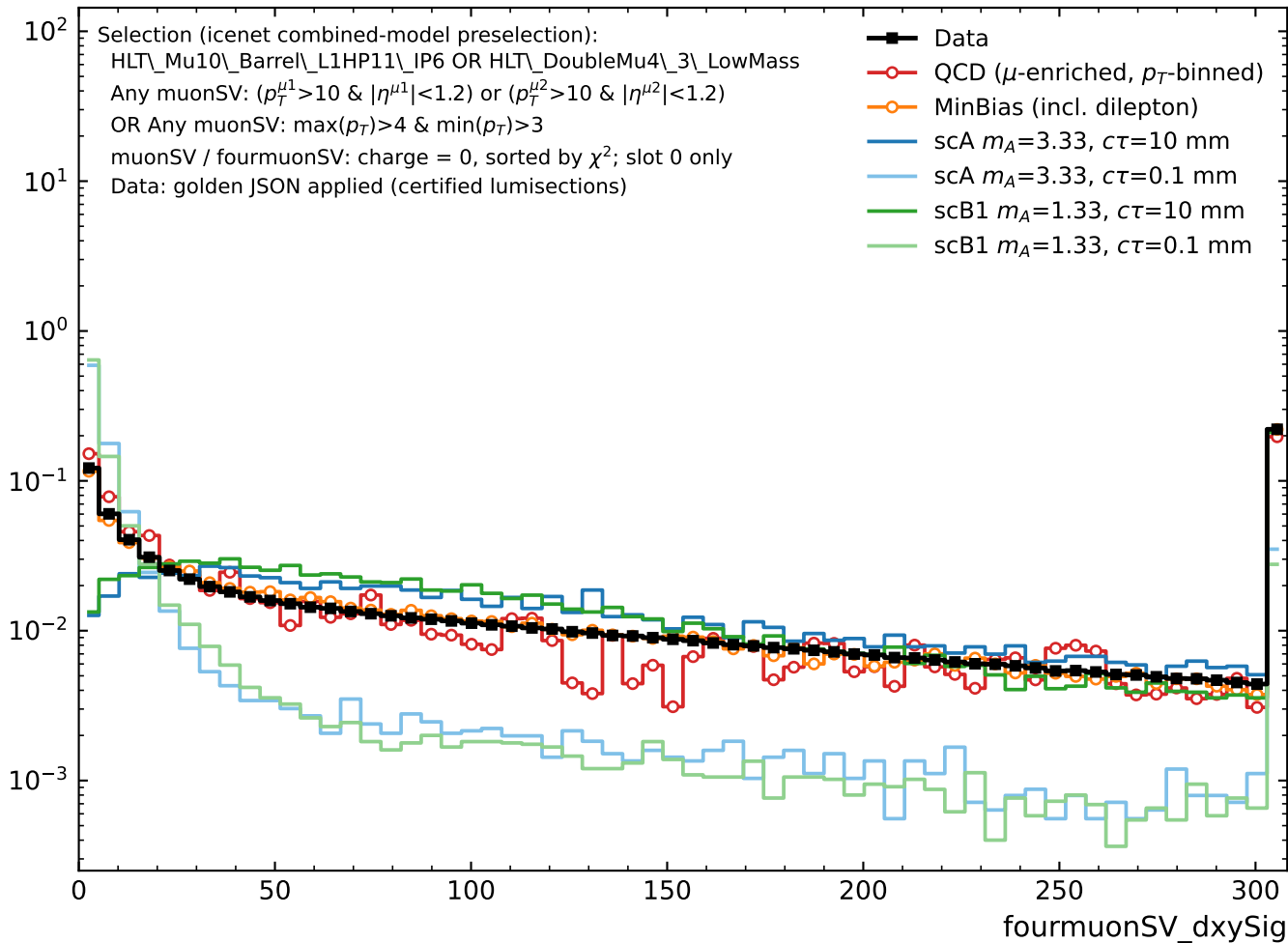
muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

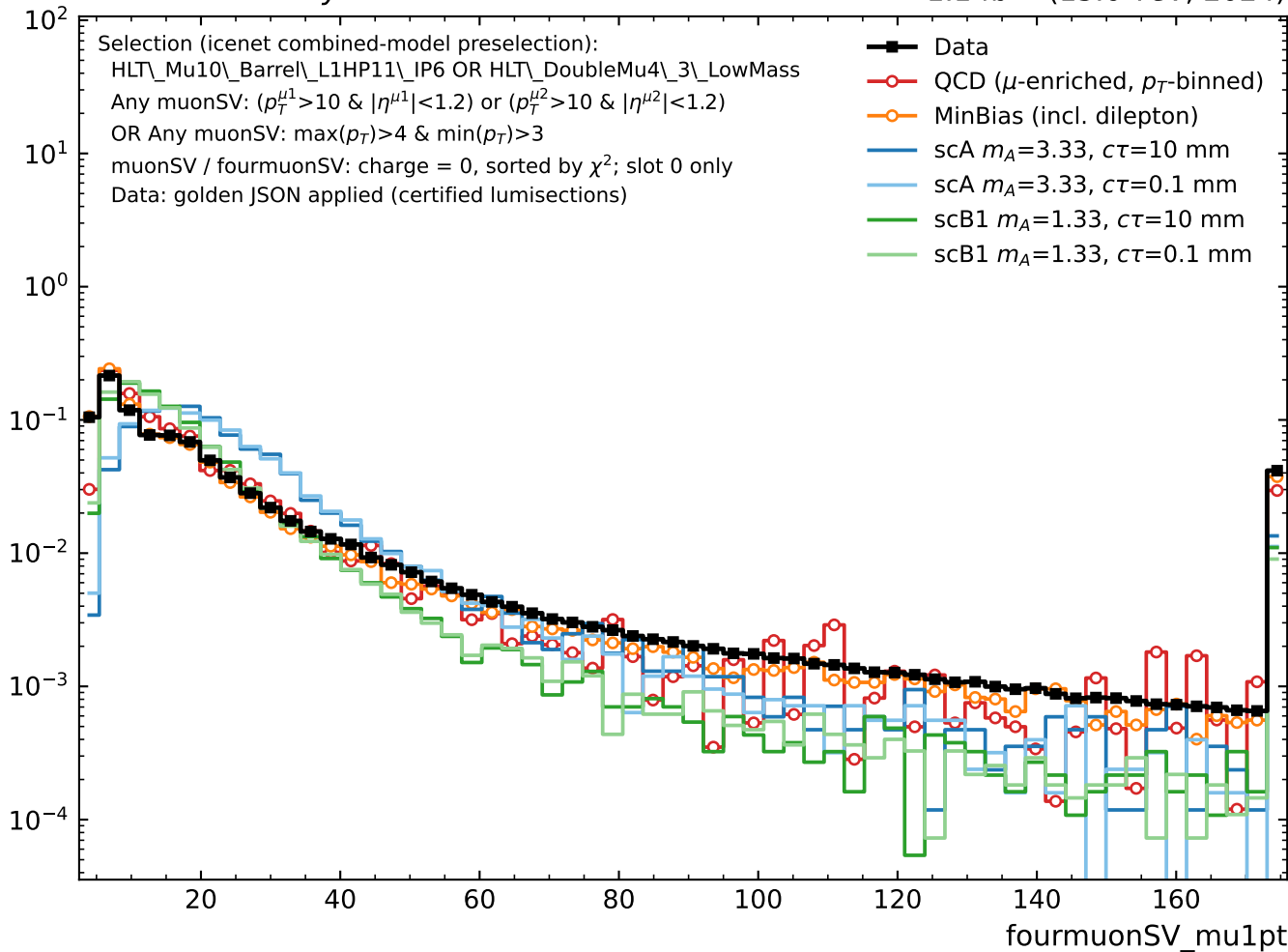
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



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HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

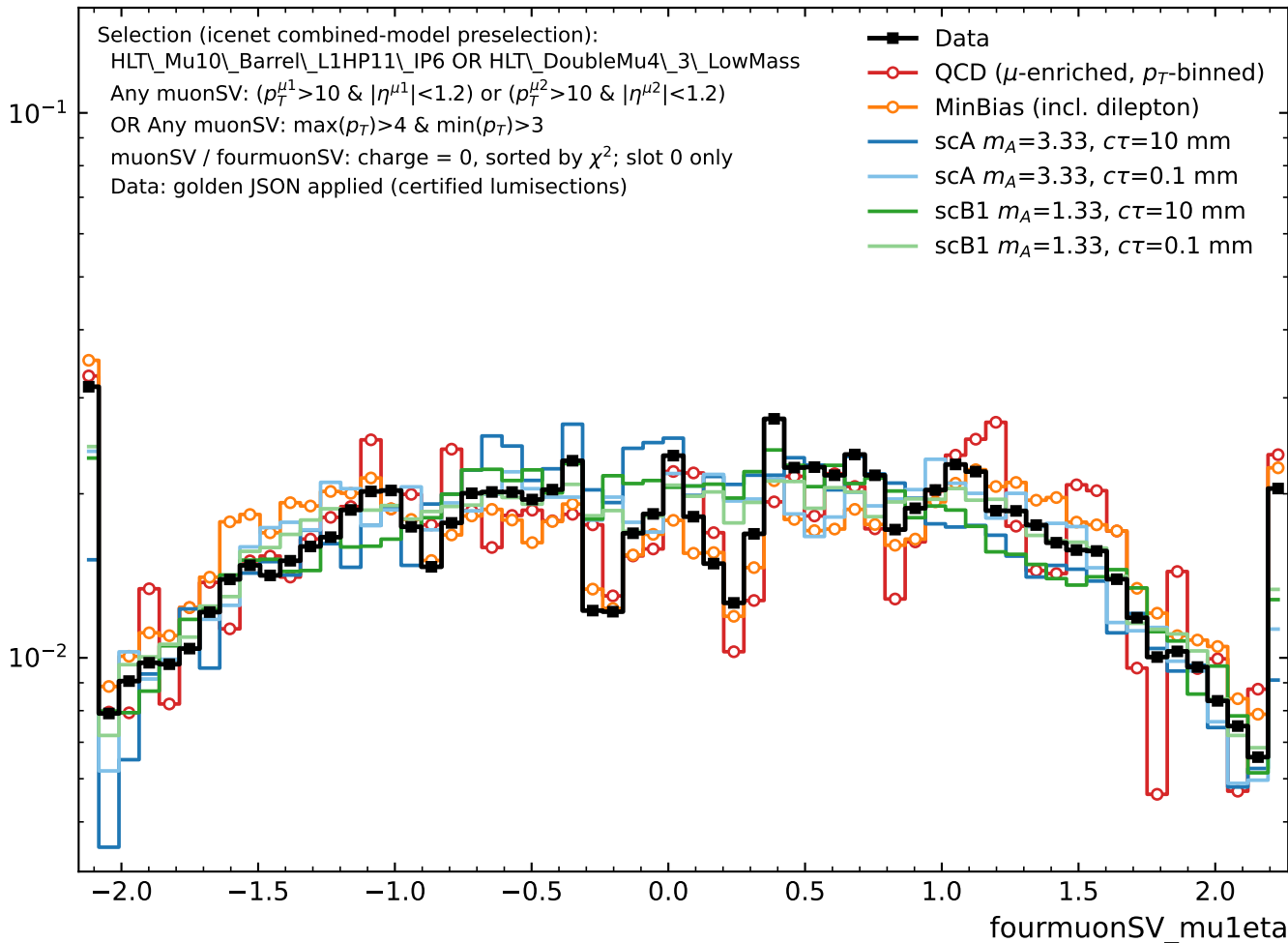
OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A = 3.33$, $c\tau = 10$ mm
- scA $m_A = 3.33$, $c\tau = 0.1$ mm
- scB1 $m_A = 1.33$, $c\tau = 10$ mm
- scB1 $m_A = 1.33$, $c\tau = 0.1$ mm

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

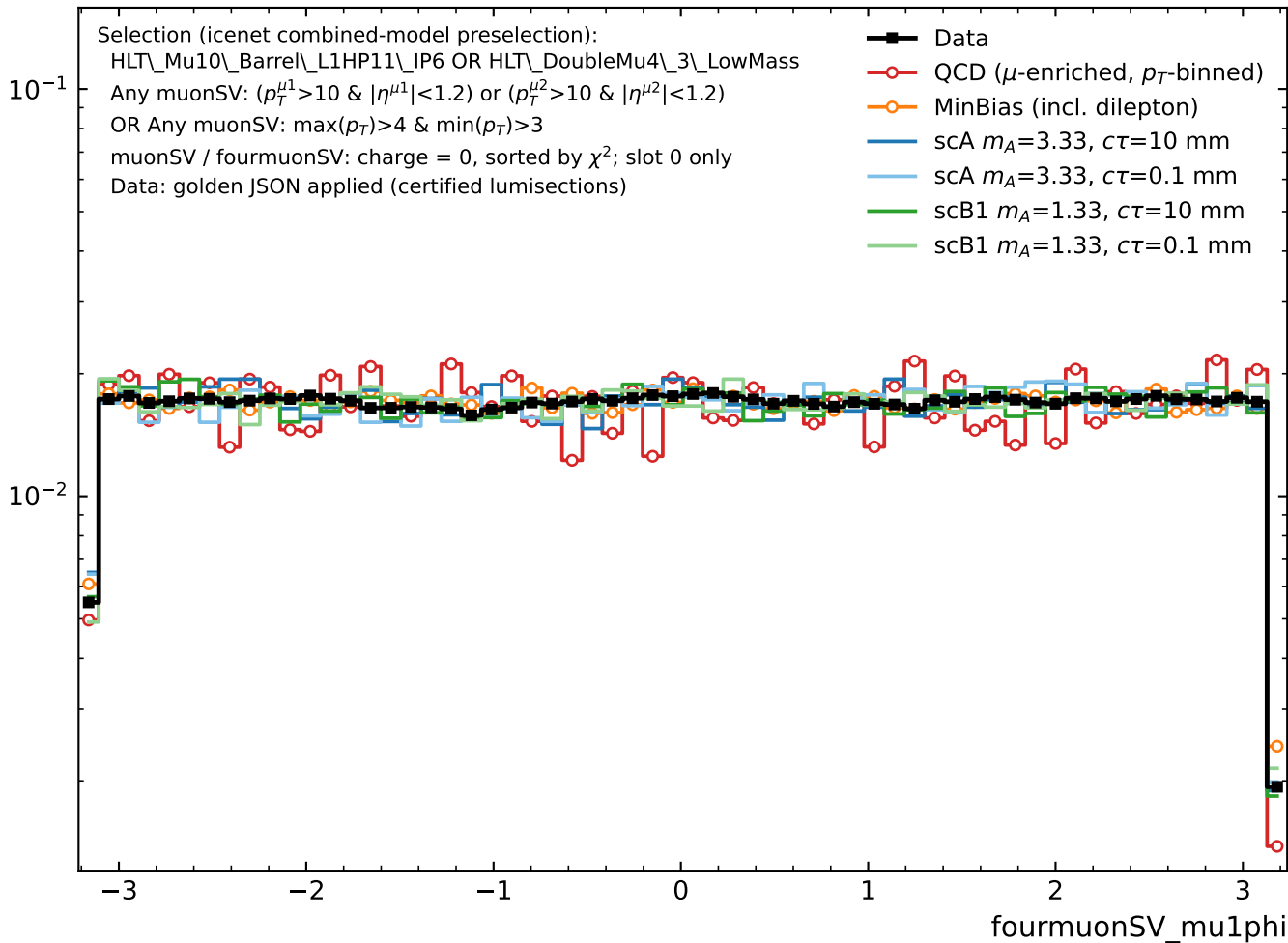
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

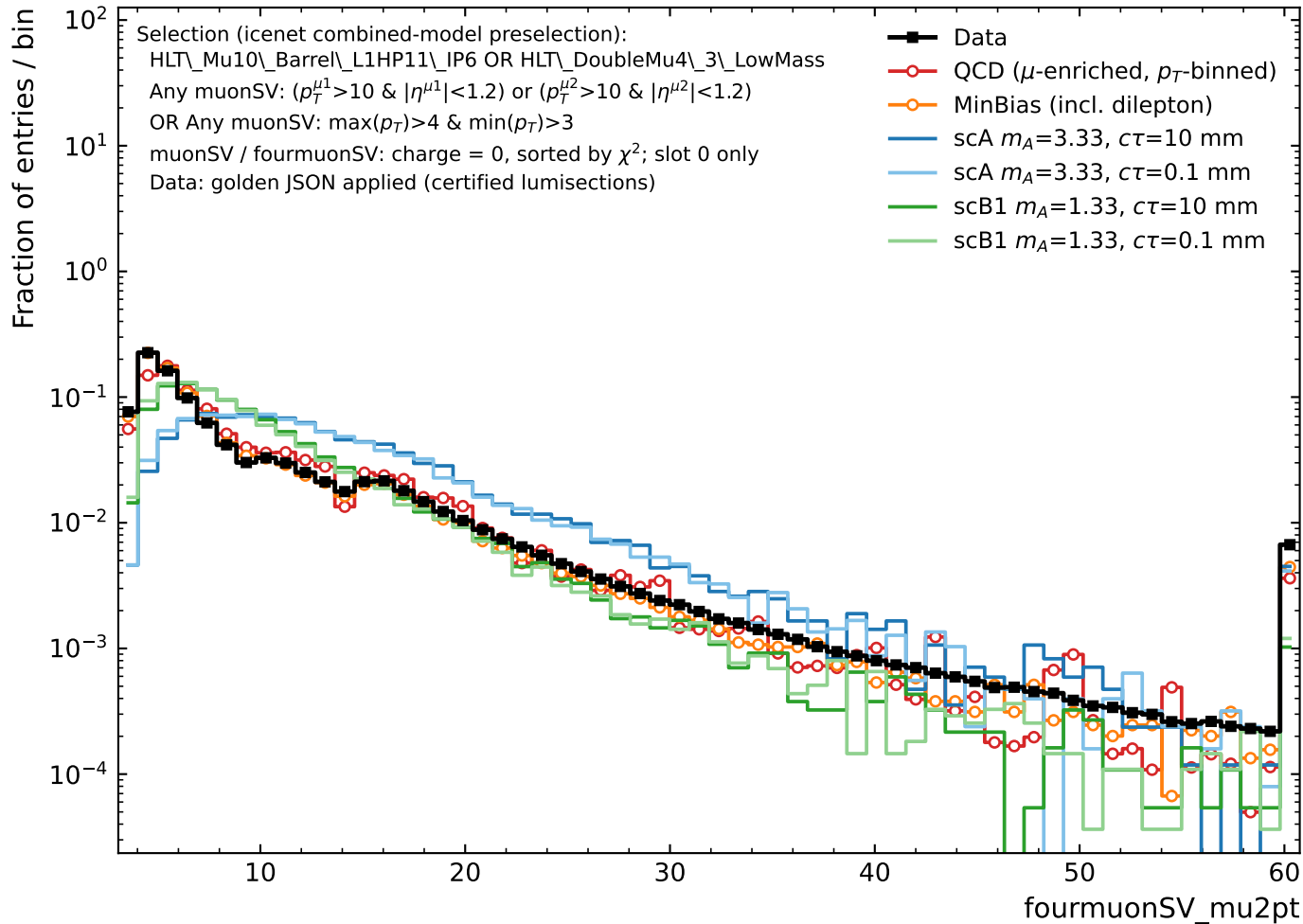
HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)



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HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

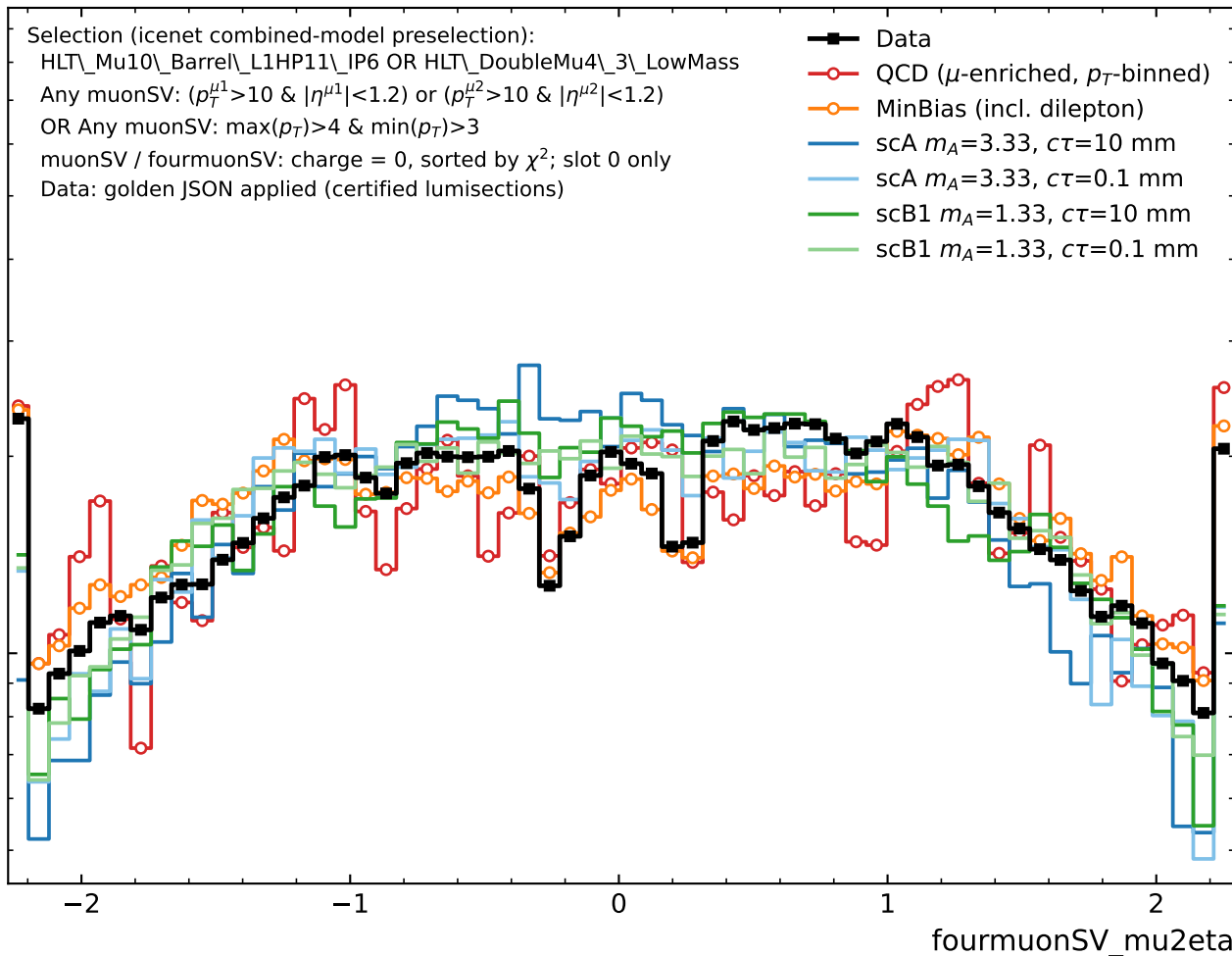
Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A = 3.33$, $c\tau = 10$ mm
- scA $m_A = 3.33$, $c\tau = 0.1$ mm
- scB1 $m_A = 1.33$, $c\tau = 10$ mm
- scB1 $m_A = 1.33$, $c\tau = 0.1$ mm

Fraction of entries / bin

10⁻²

fourmuonSV_mu2eta



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

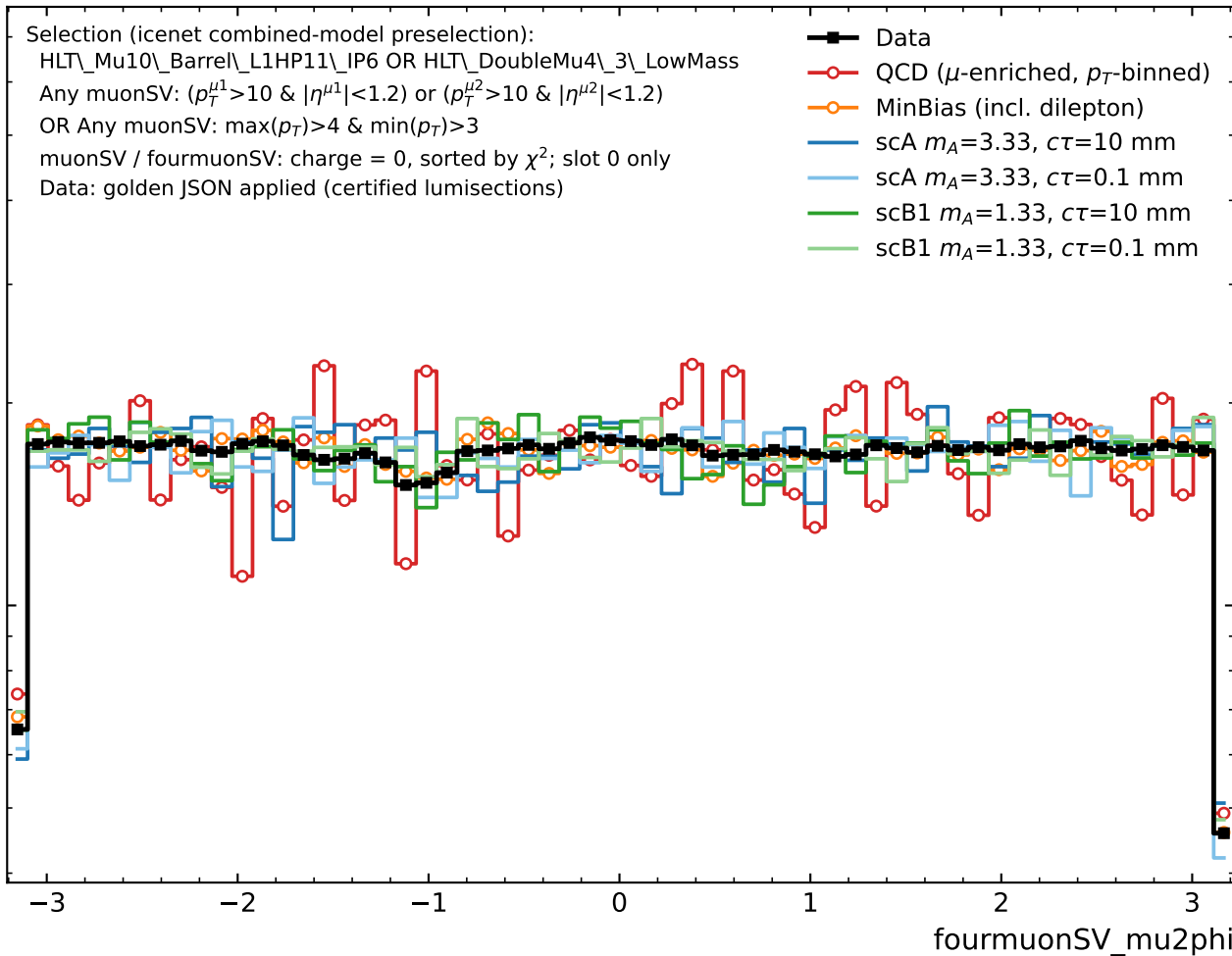
muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

Fraction of entries / bin

10^{-2}



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

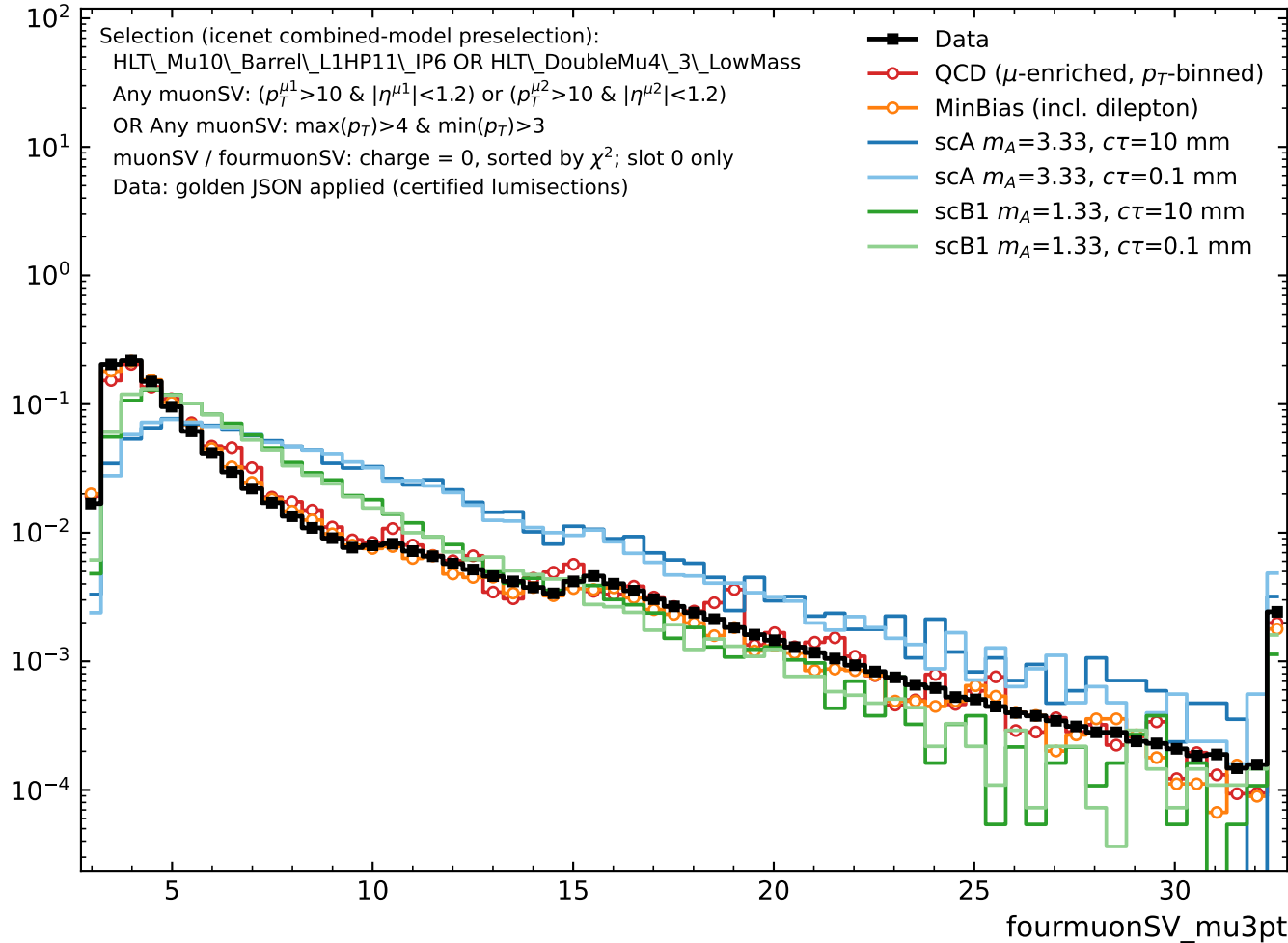
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

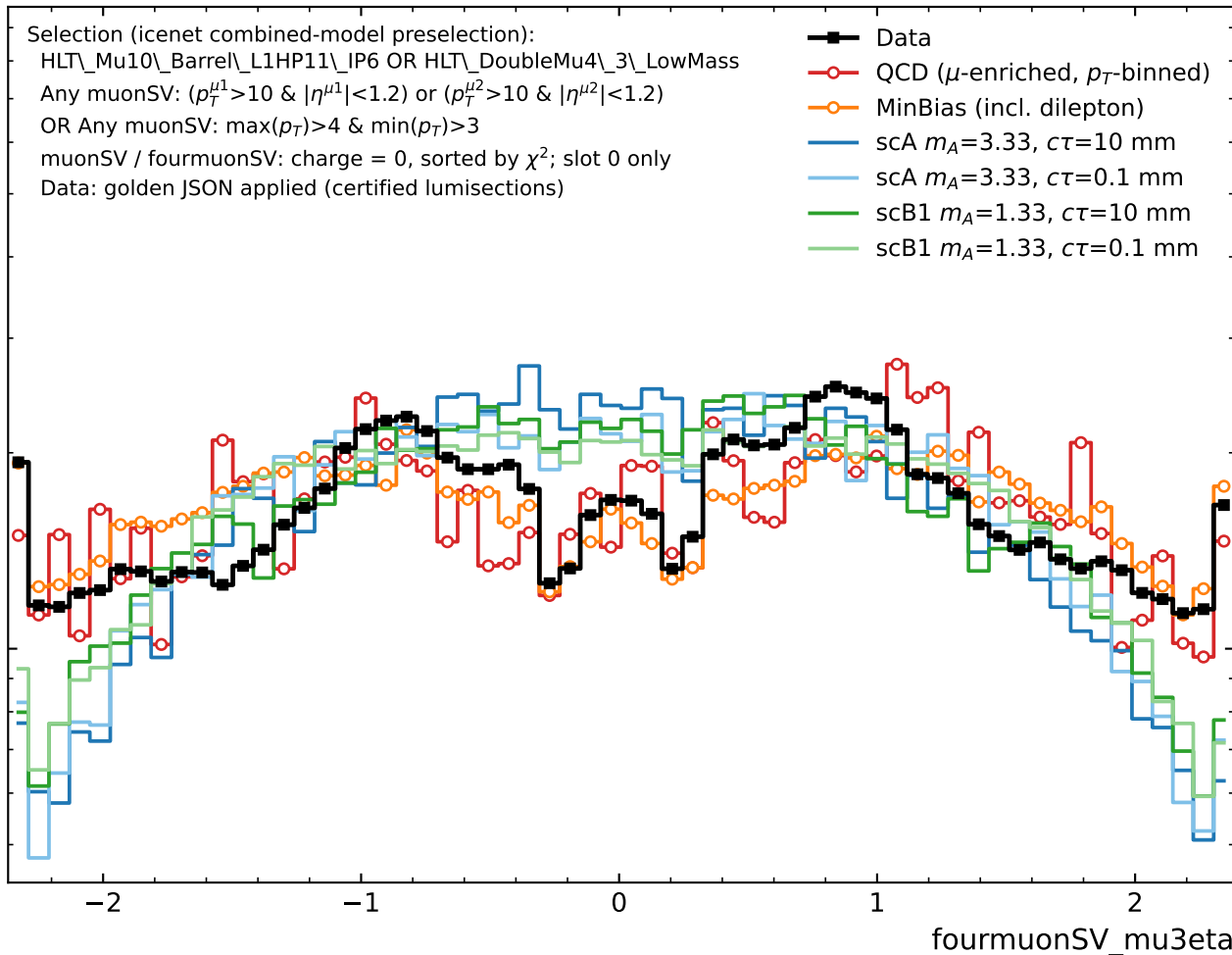
Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A = 3.33$, $c\tau = 10$ mm
- scA $m_A = 3.33$, $c\tau = 0.1$ mm
- scB1 $m_A = 1.33$, $c\tau = 10$ mm
- scB1 $m_A = 1.33$, $c\tau = 0.1$ mm

Fraction of entries / bin

10⁻²

fourmuonSV_mu3eta



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

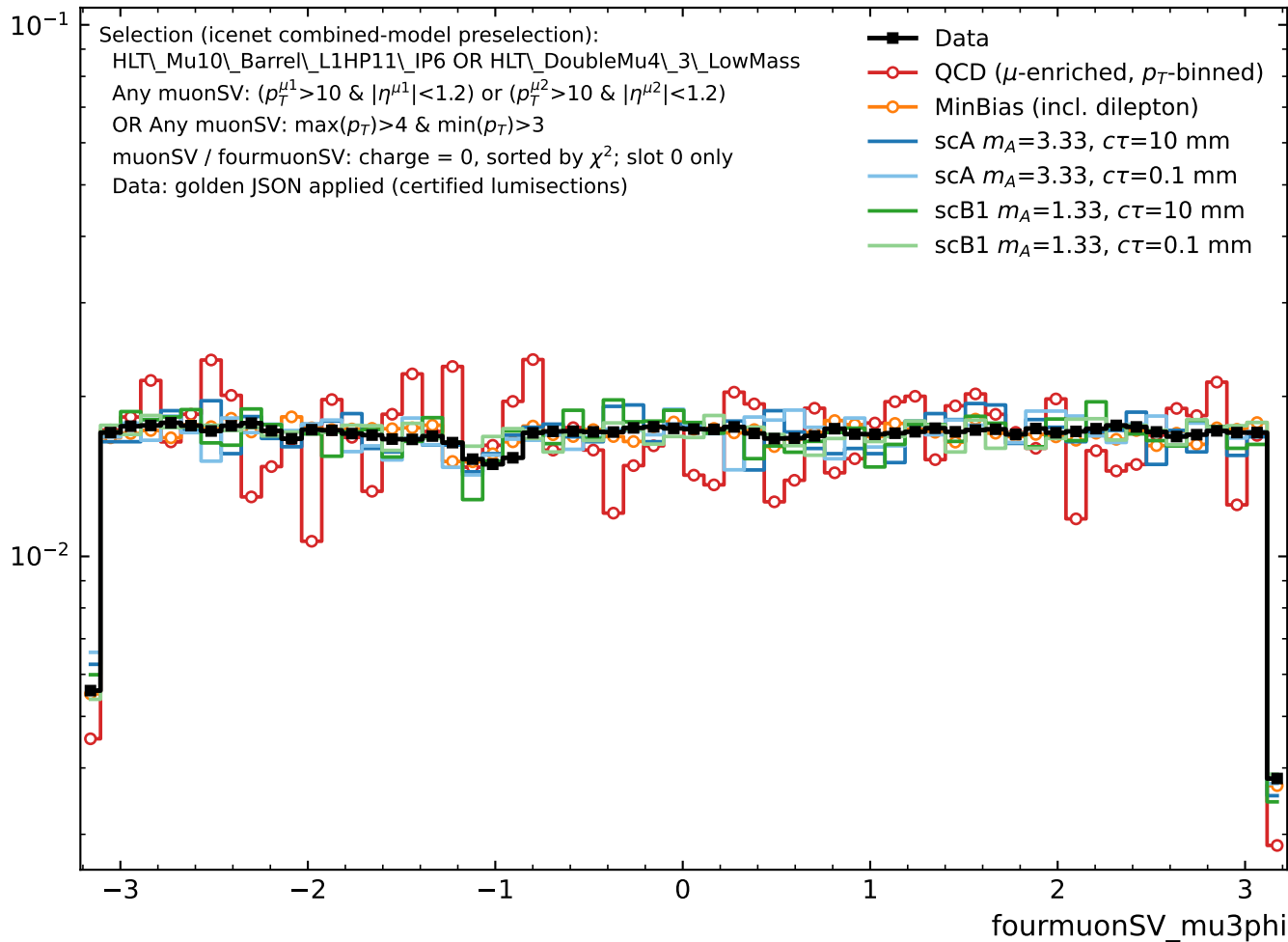
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

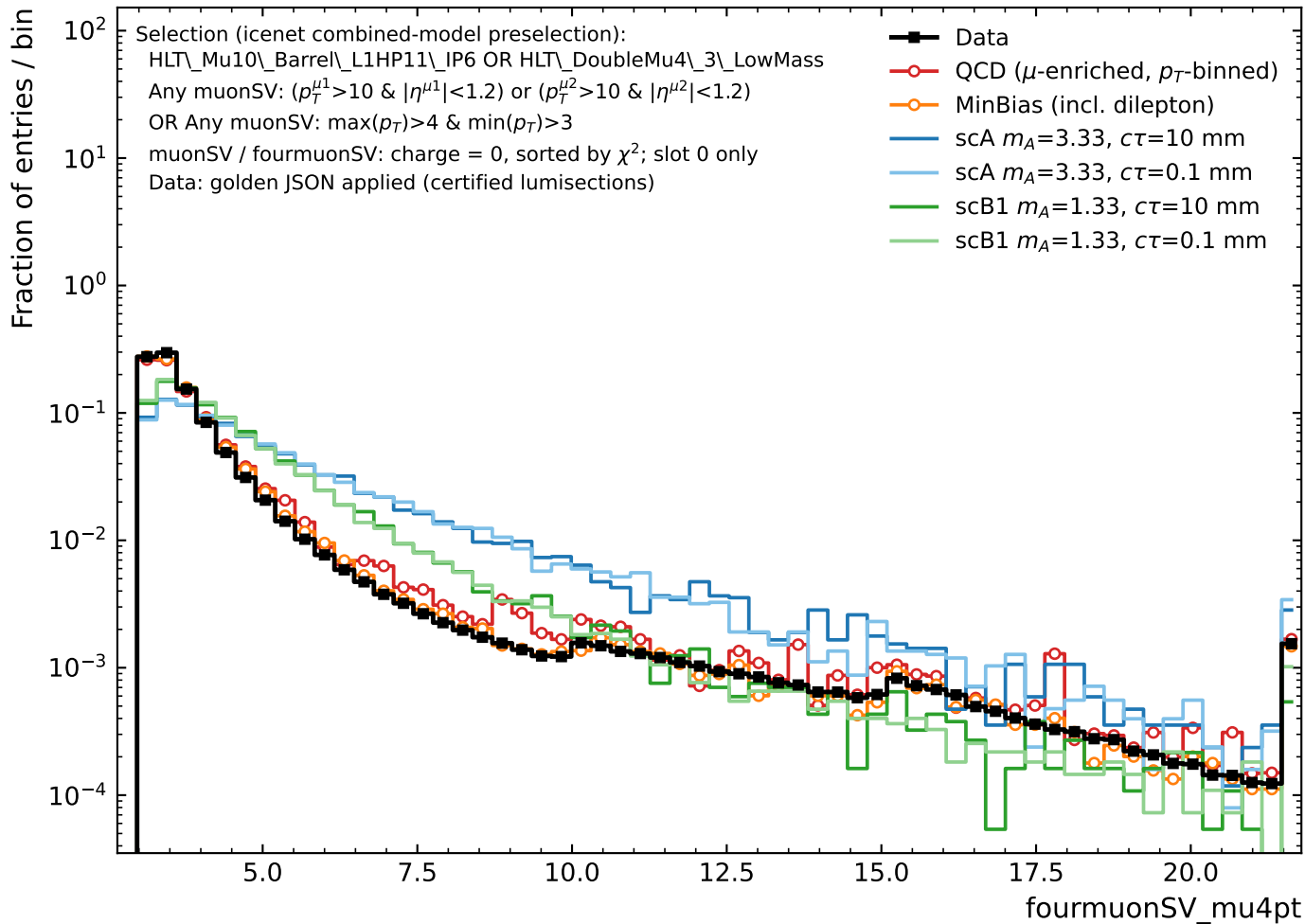
OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin





Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

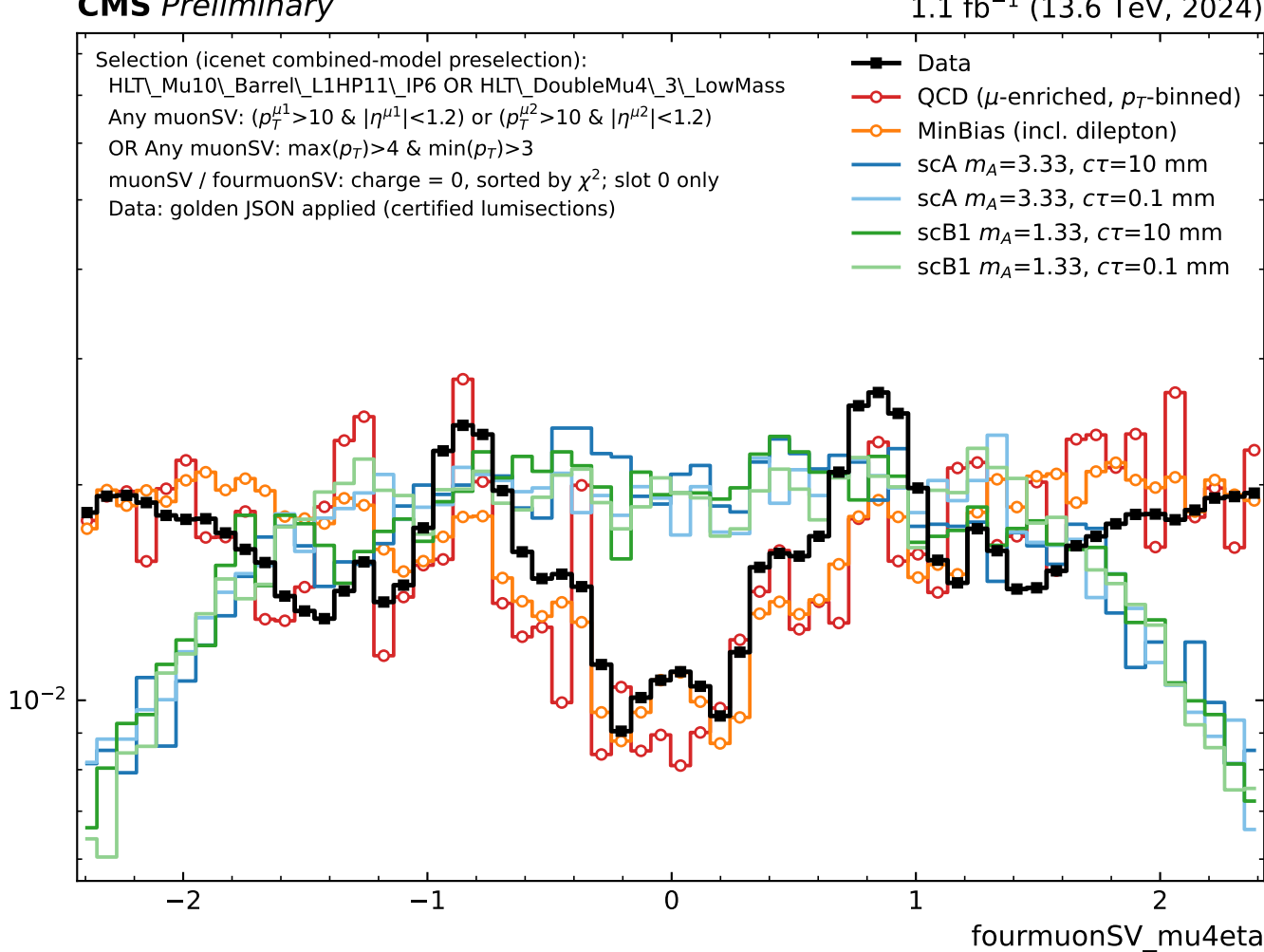
OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

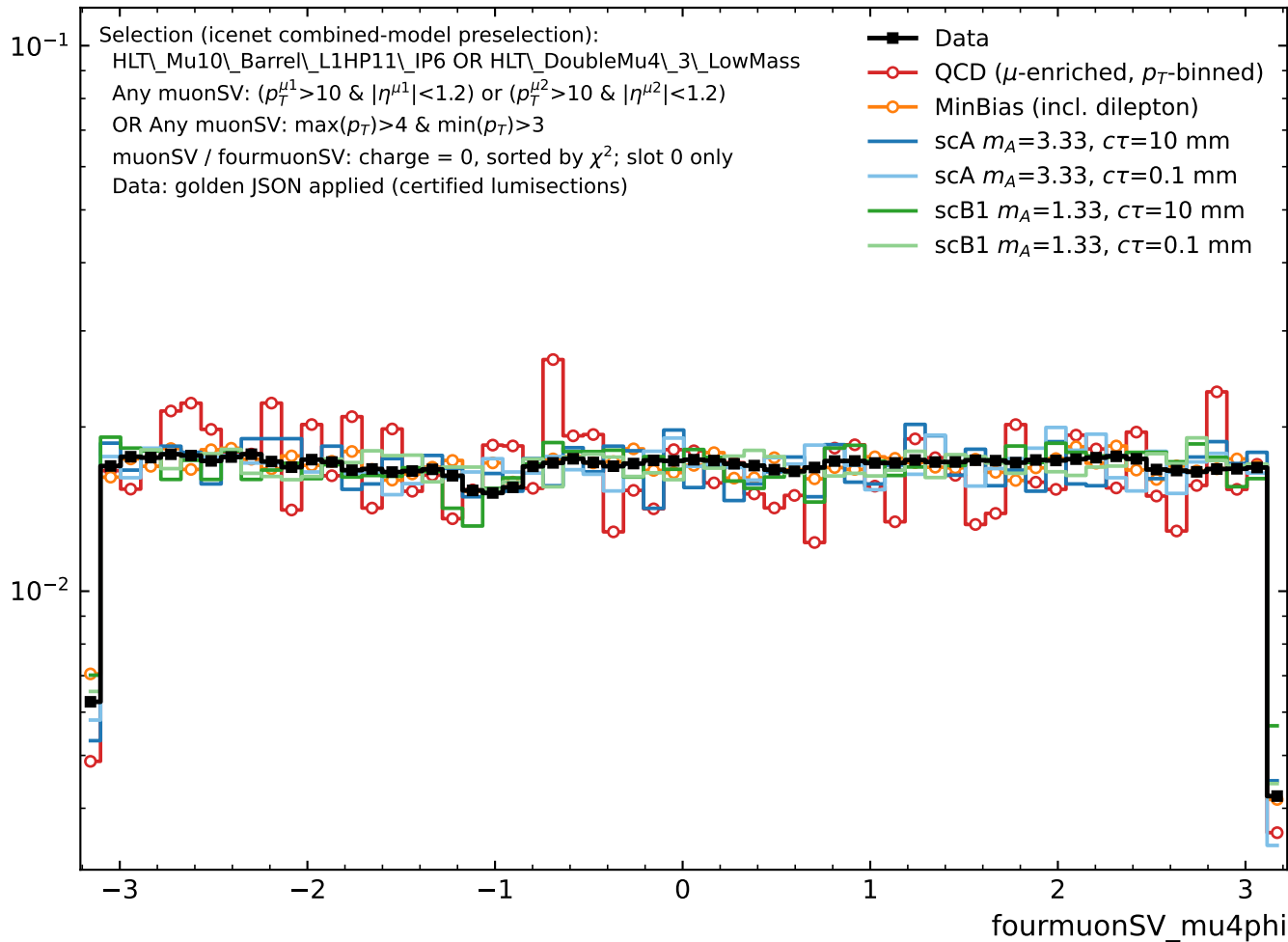
Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A = 3.33$, $c\tau = 10$ mm
- scA $m_A = 3.33$, $c\tau = 0.1$ mm
- scB1 $m_A = 1.33$, $c\tau = 10$ mm
- scB1 $m_A = 1.33$, $c\tau = 0.1$ mm

Fraction of entries / bin



Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

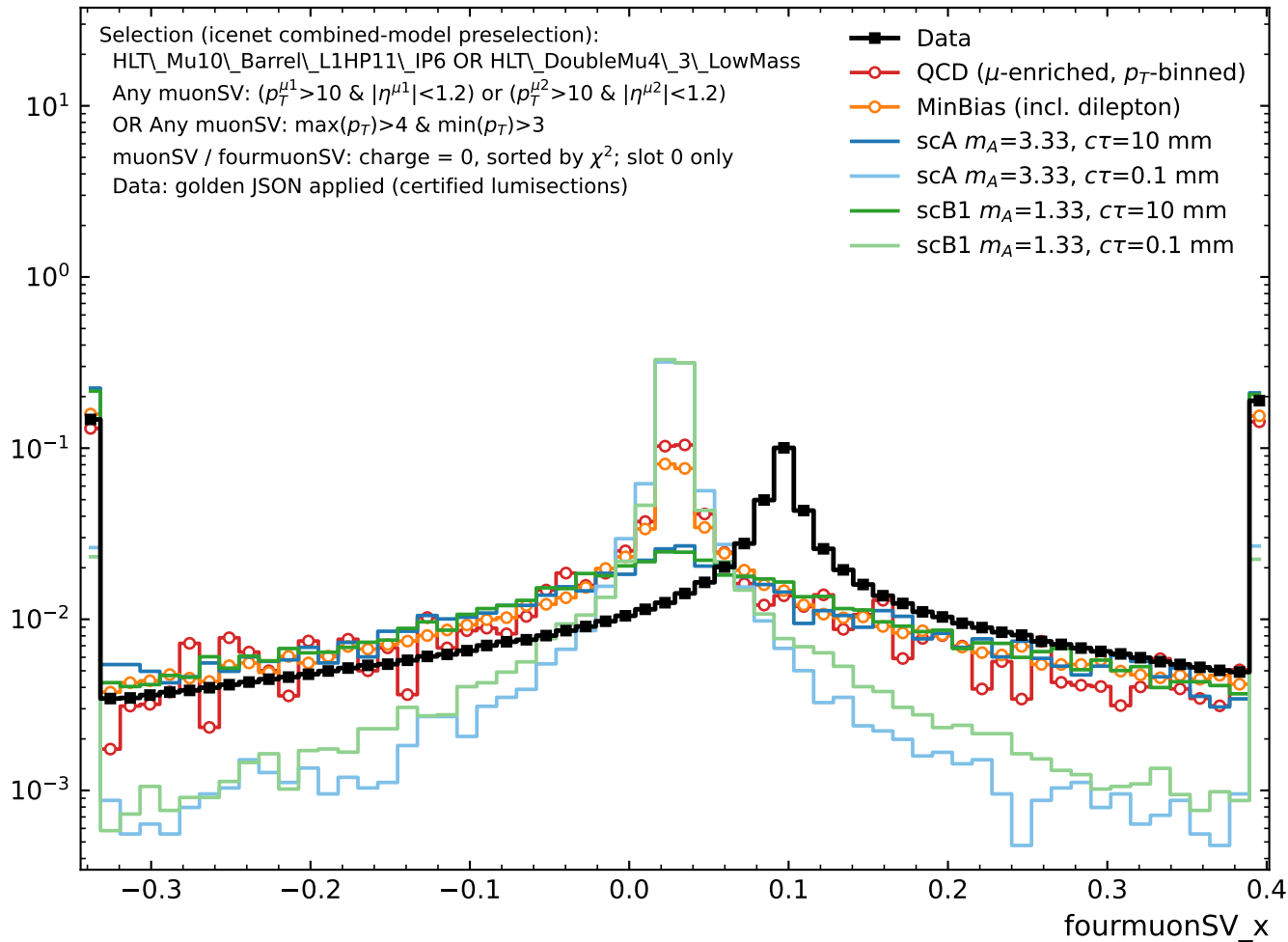
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

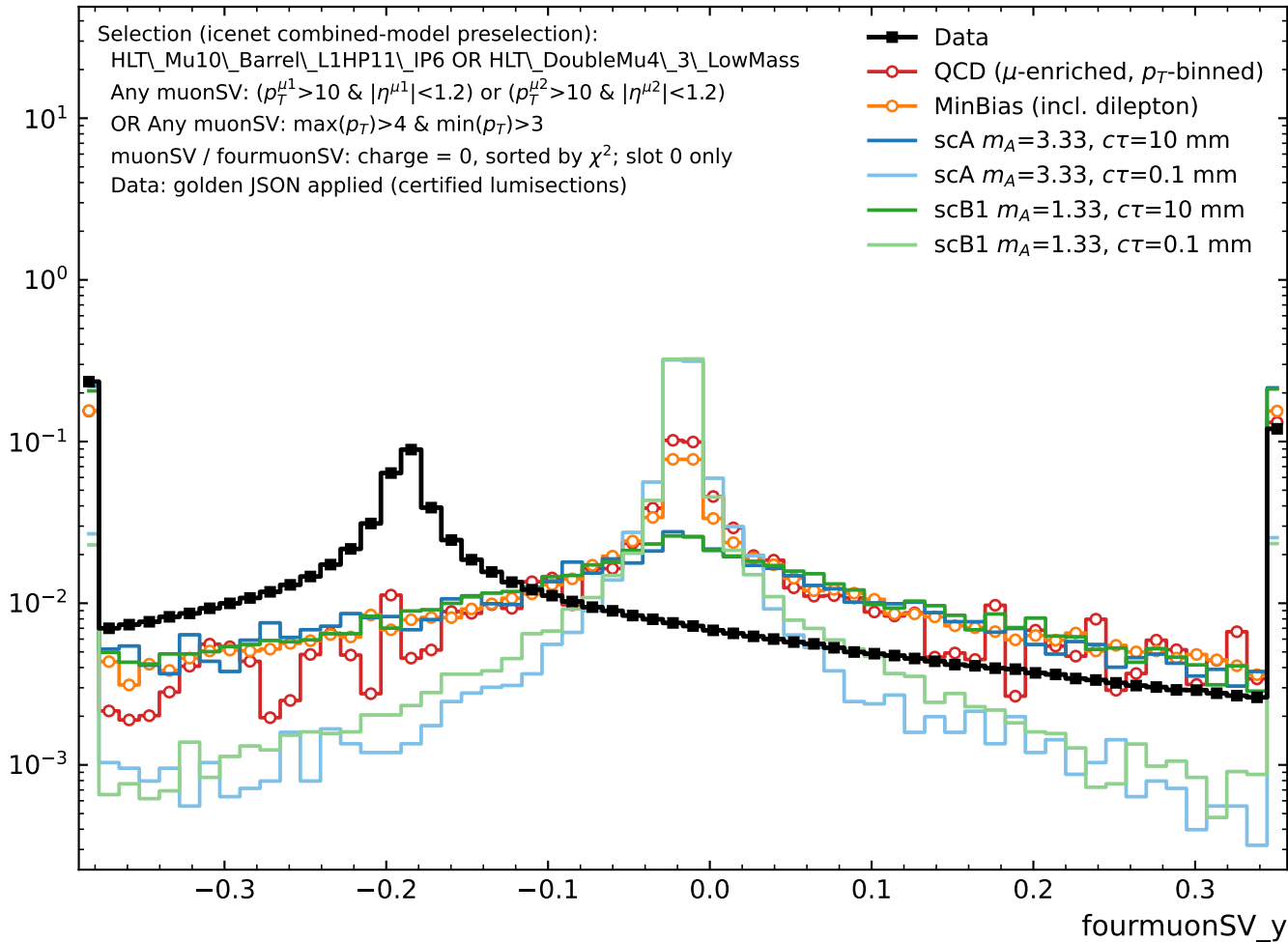
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

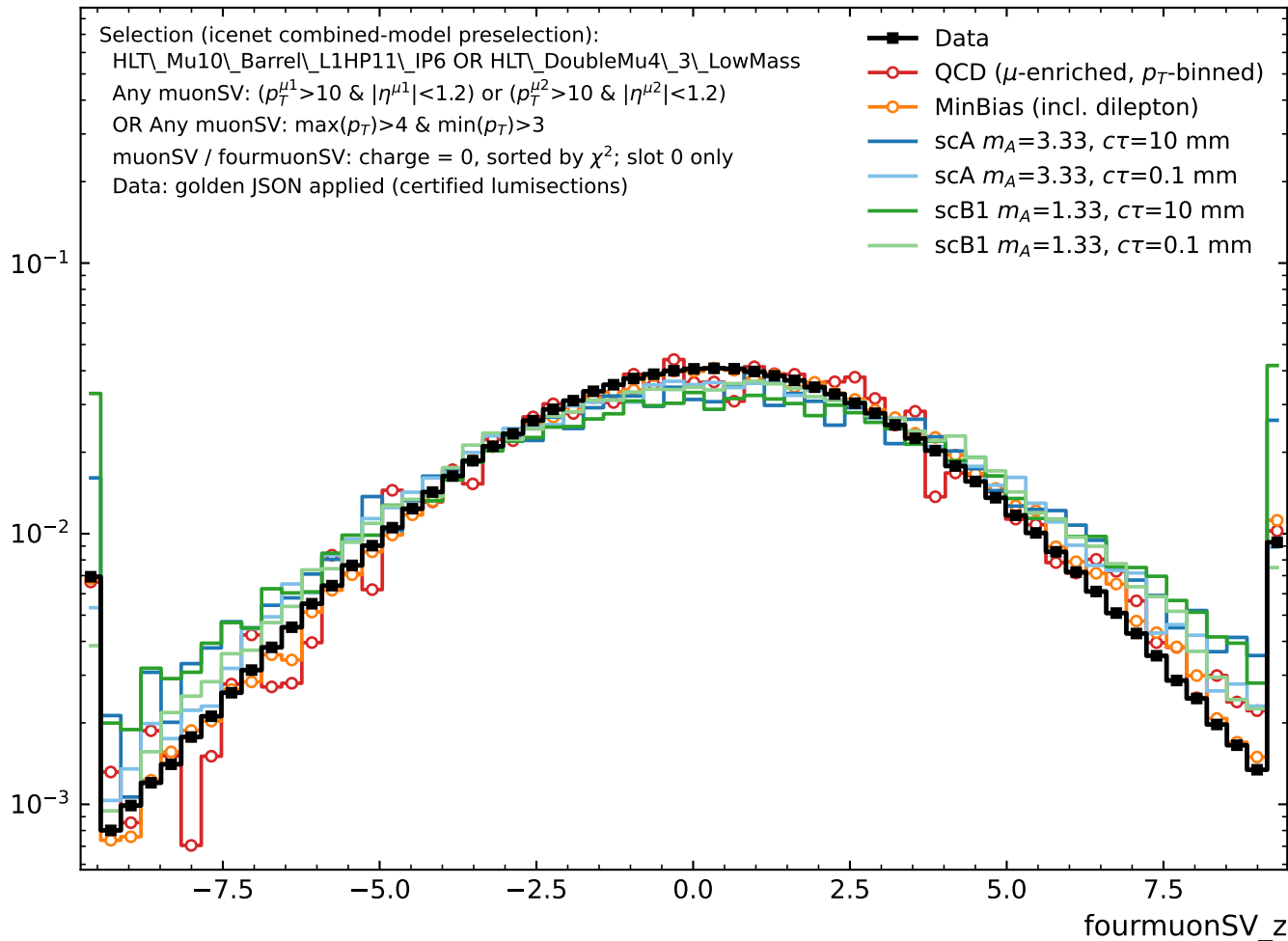
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

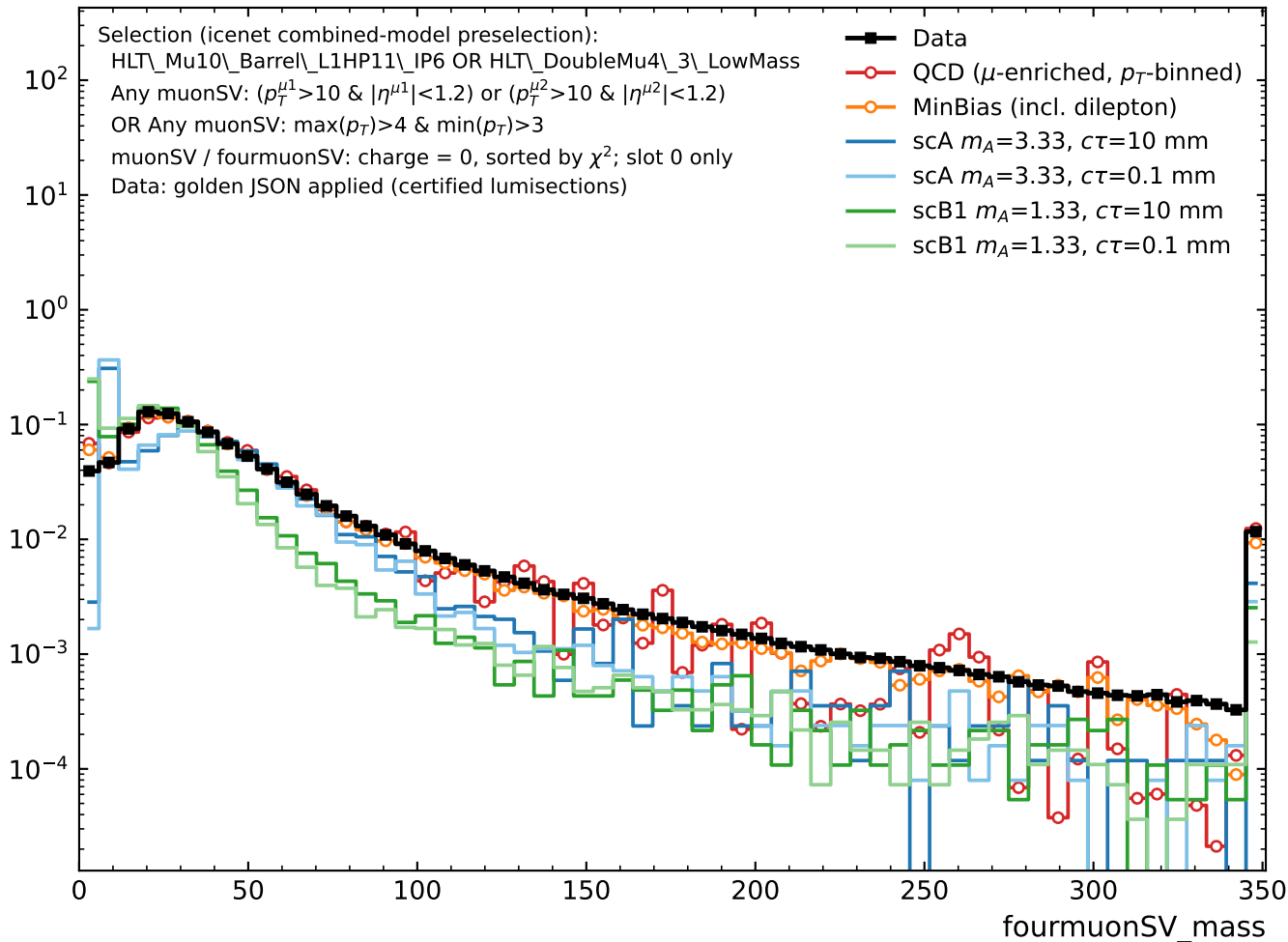
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

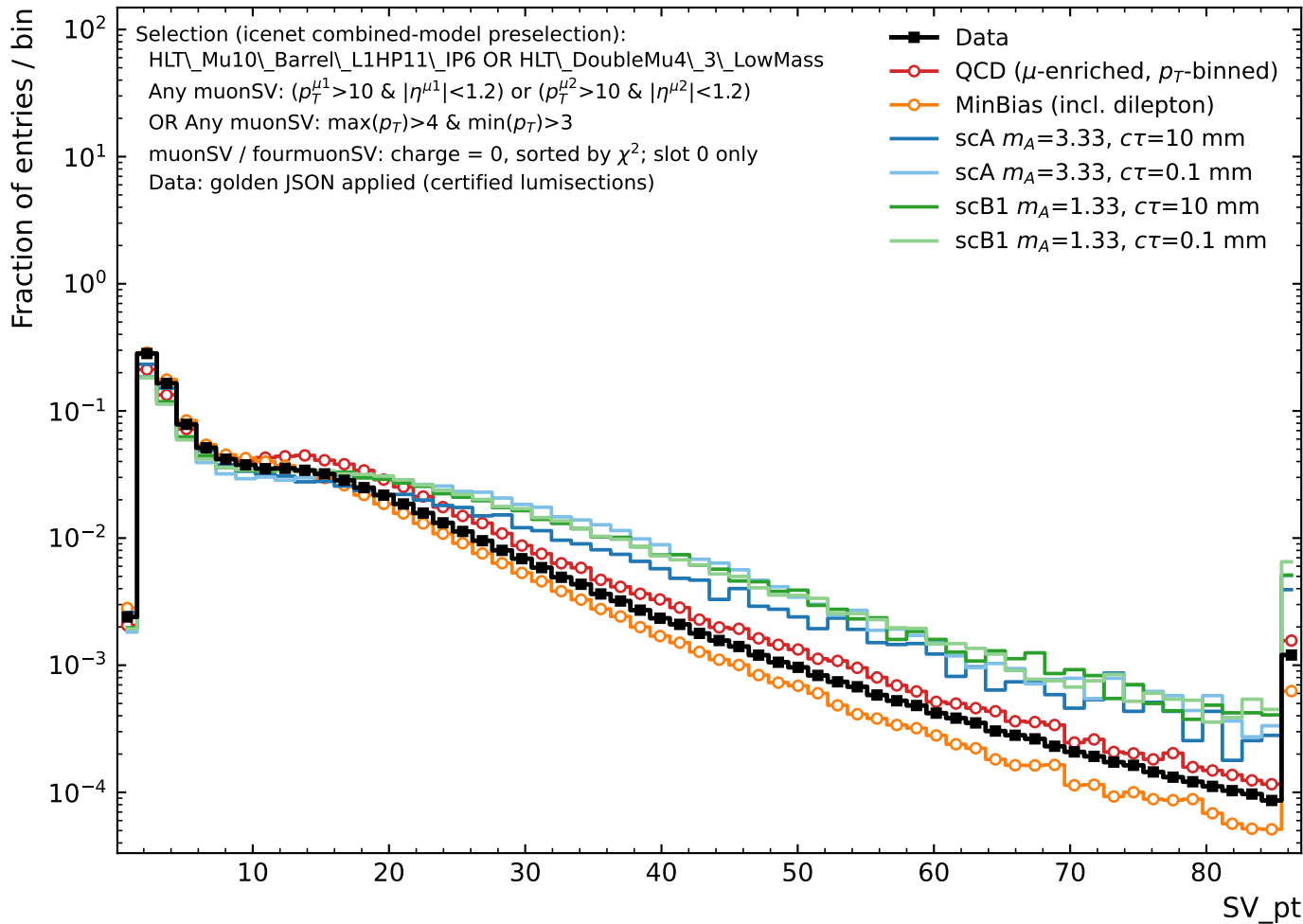
HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

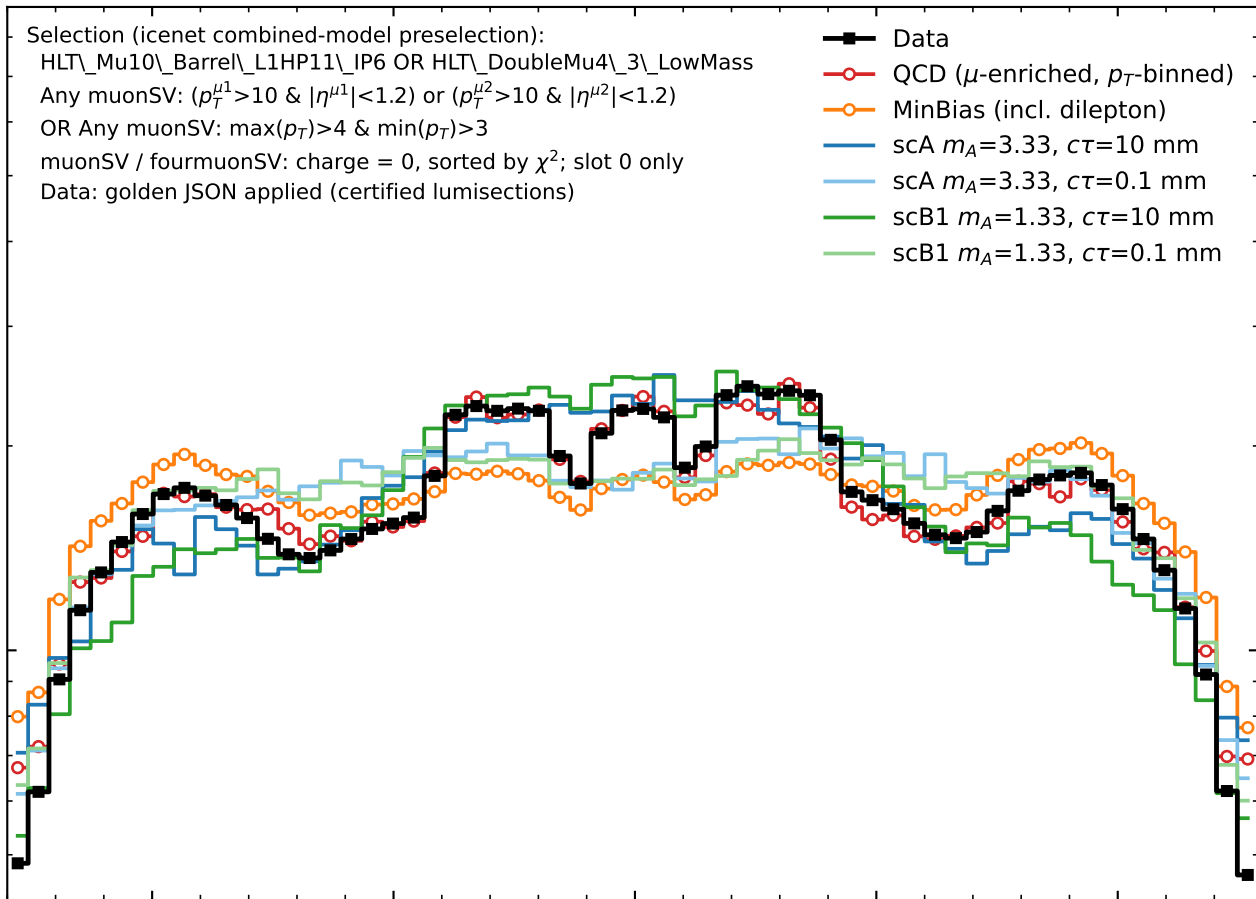
Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

Fraction of entries / bin

10⁻²

SV_eta



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

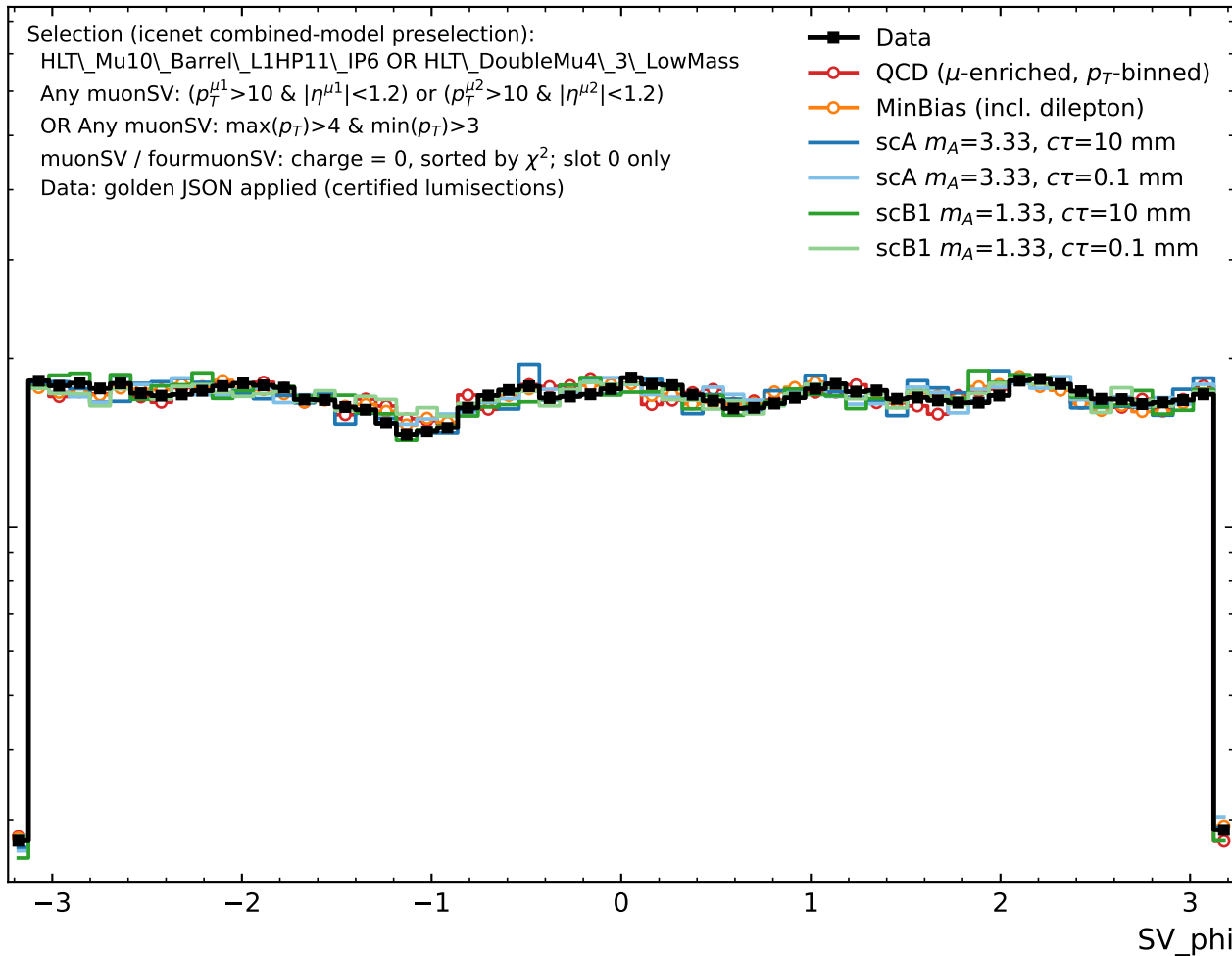
Data: golden JSON applied (certified lumisections)

- Data
- QCD (μ -enriched, p_T -binned)
- MinBias (incl. dilepton)
- scA $m_A=3.33$, $c\tau=10$ mm
- scA $m_A=3.33$, $c\tau=0.1$ mm
- scB1 $m_A=1.33$, $c\tau=10$ mm
- scB1 $m_A=1.33$, $c\tau=0.1$ mm

Fraction of entries / bin

10^{-2}

SV_phi



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

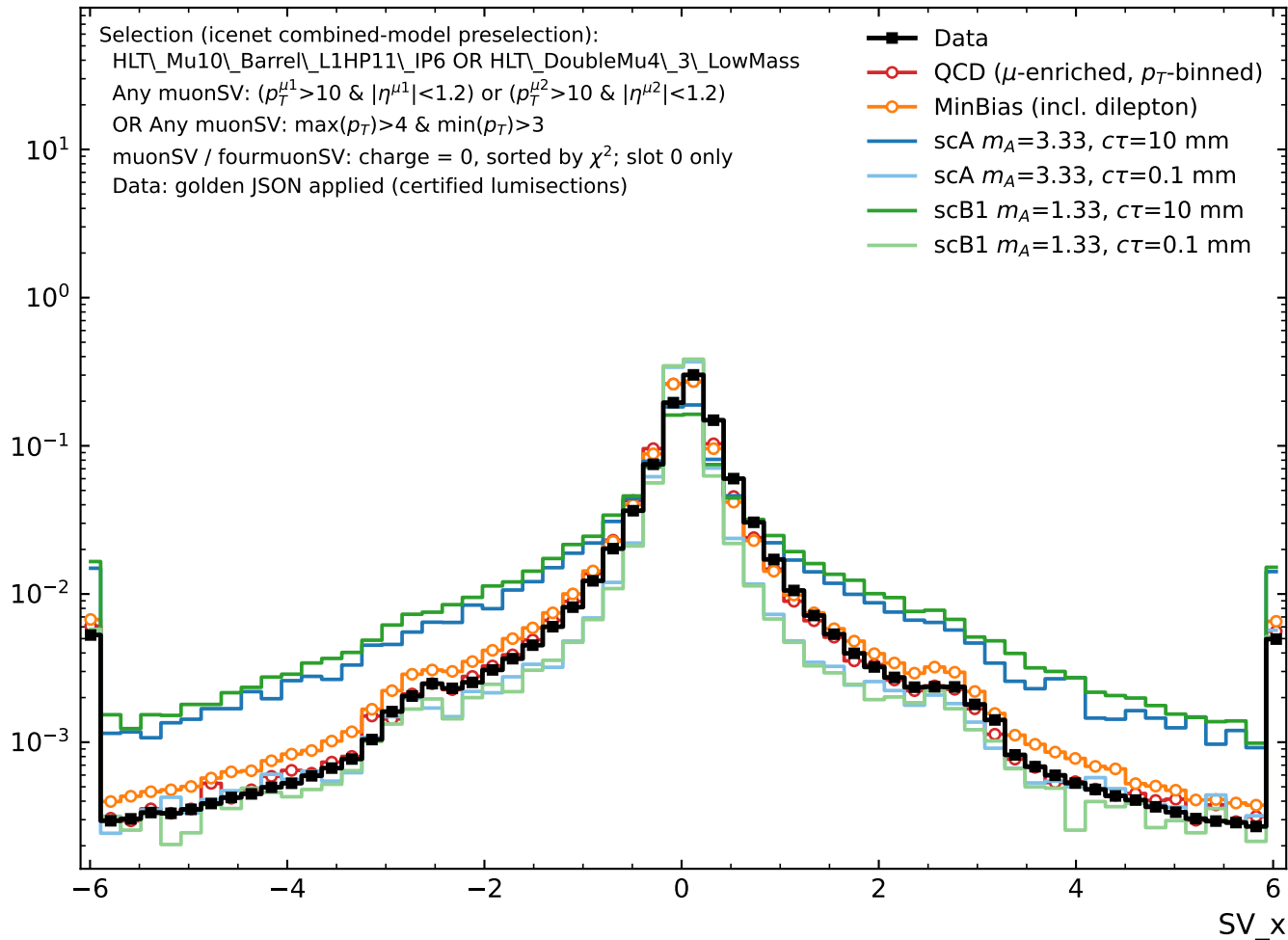
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

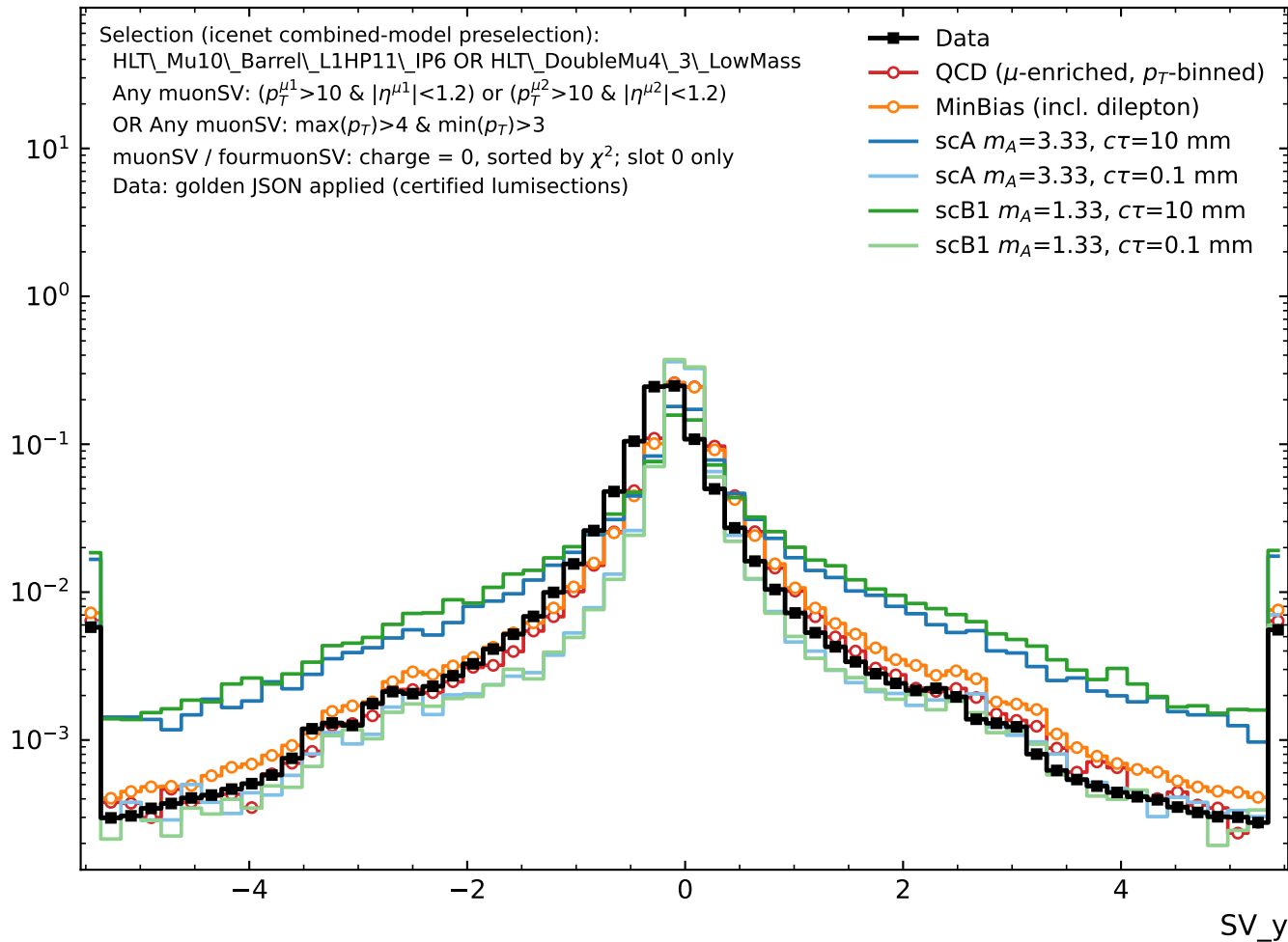
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

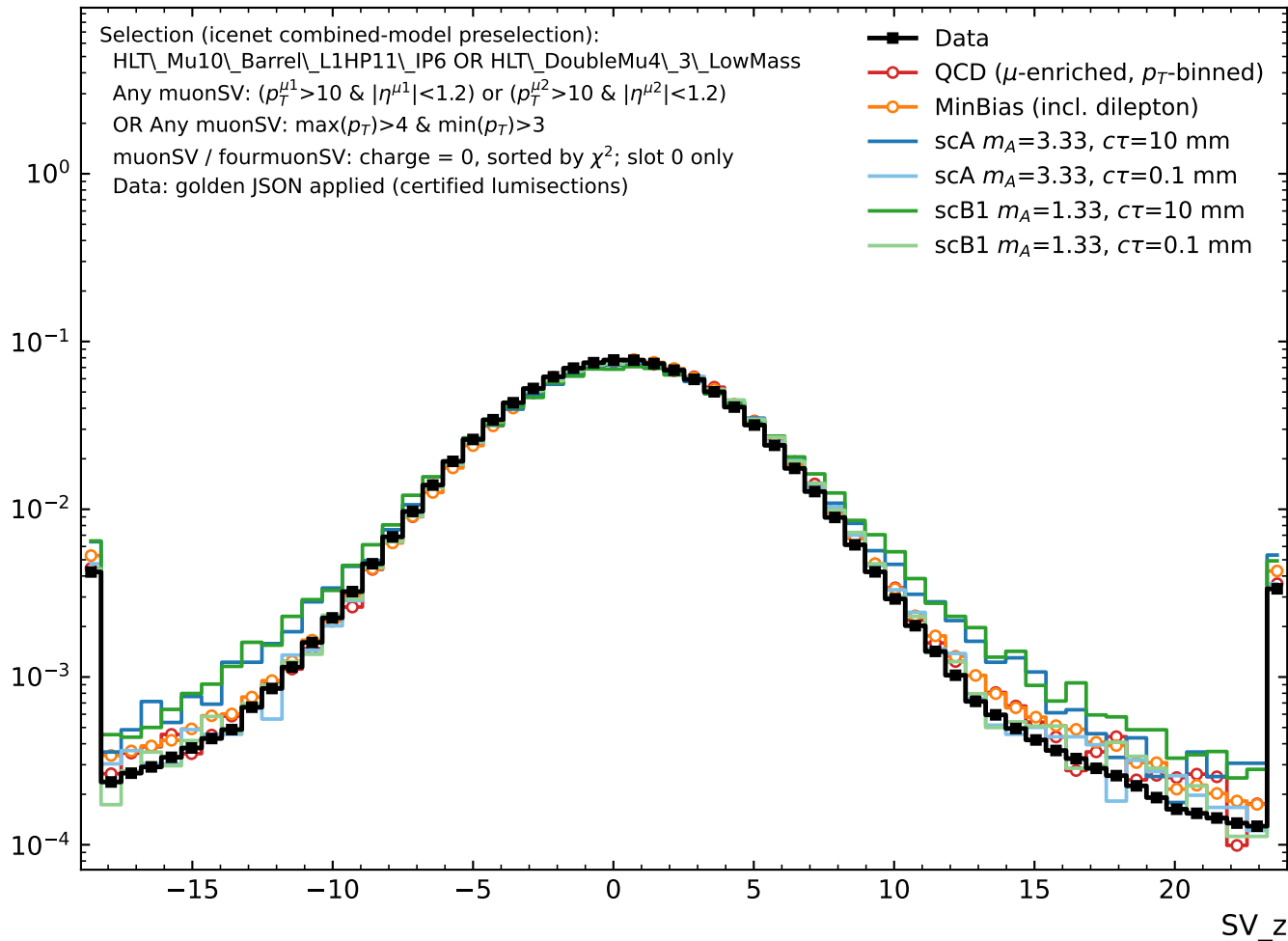
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

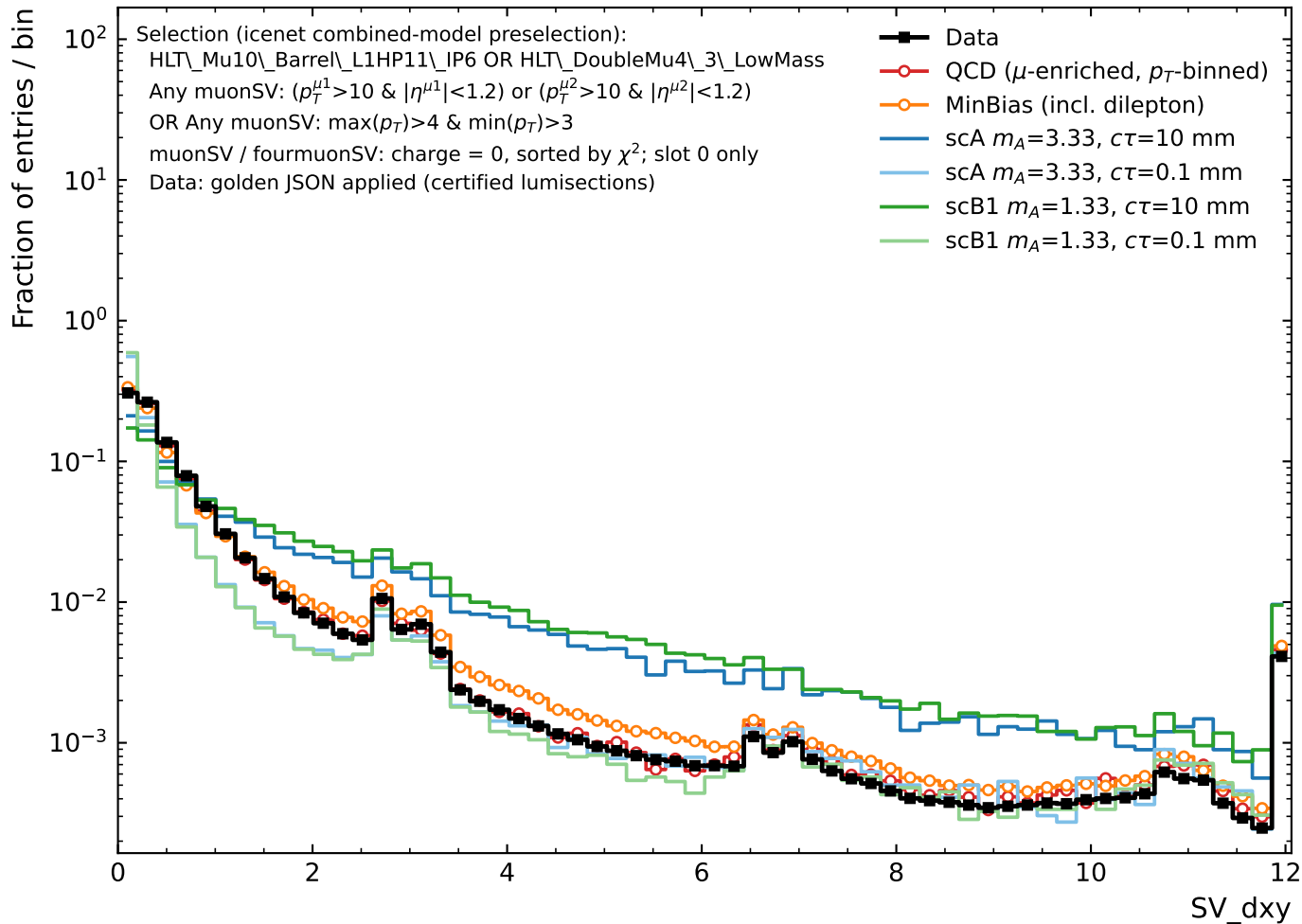
OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin





Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

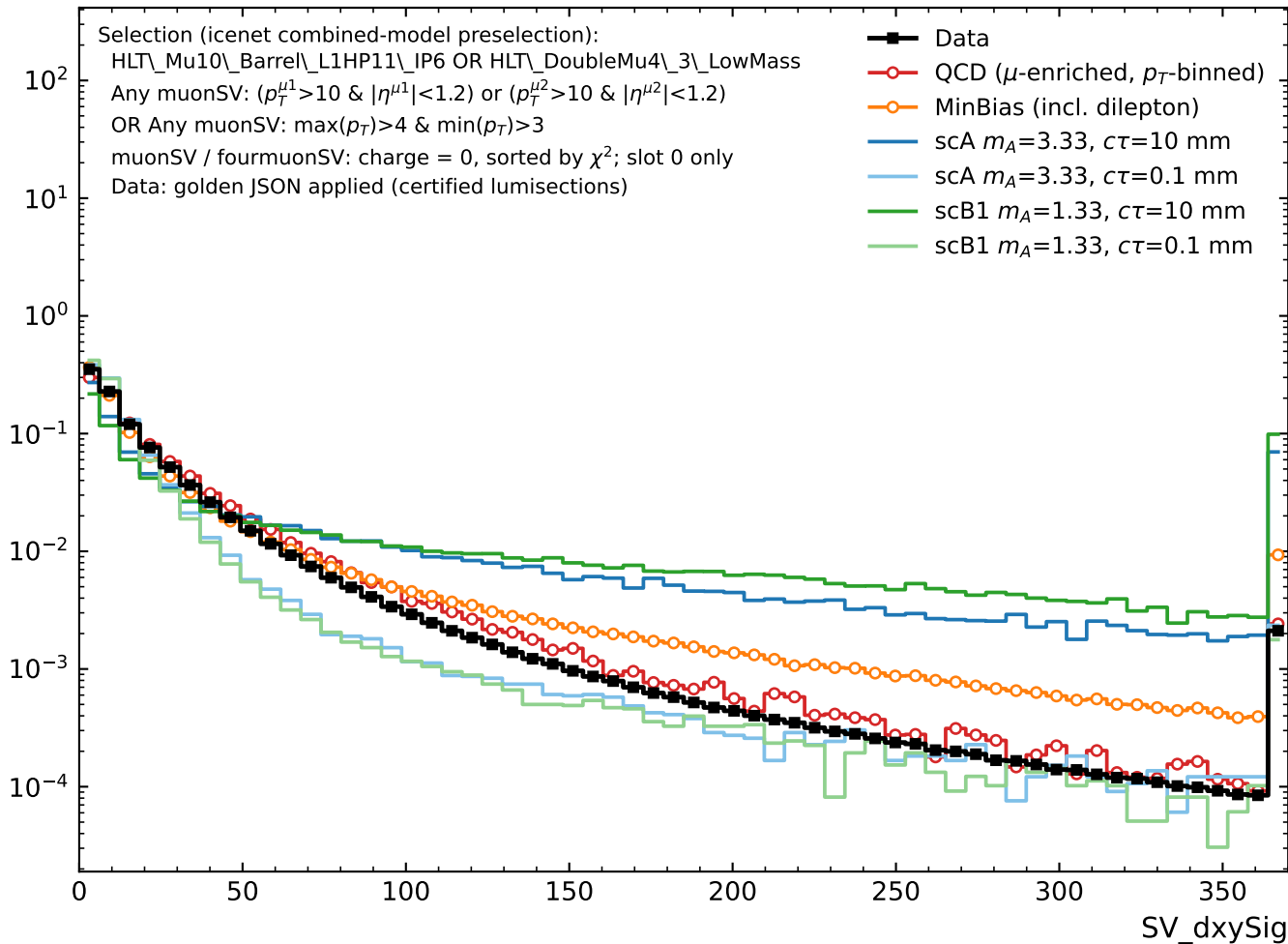
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

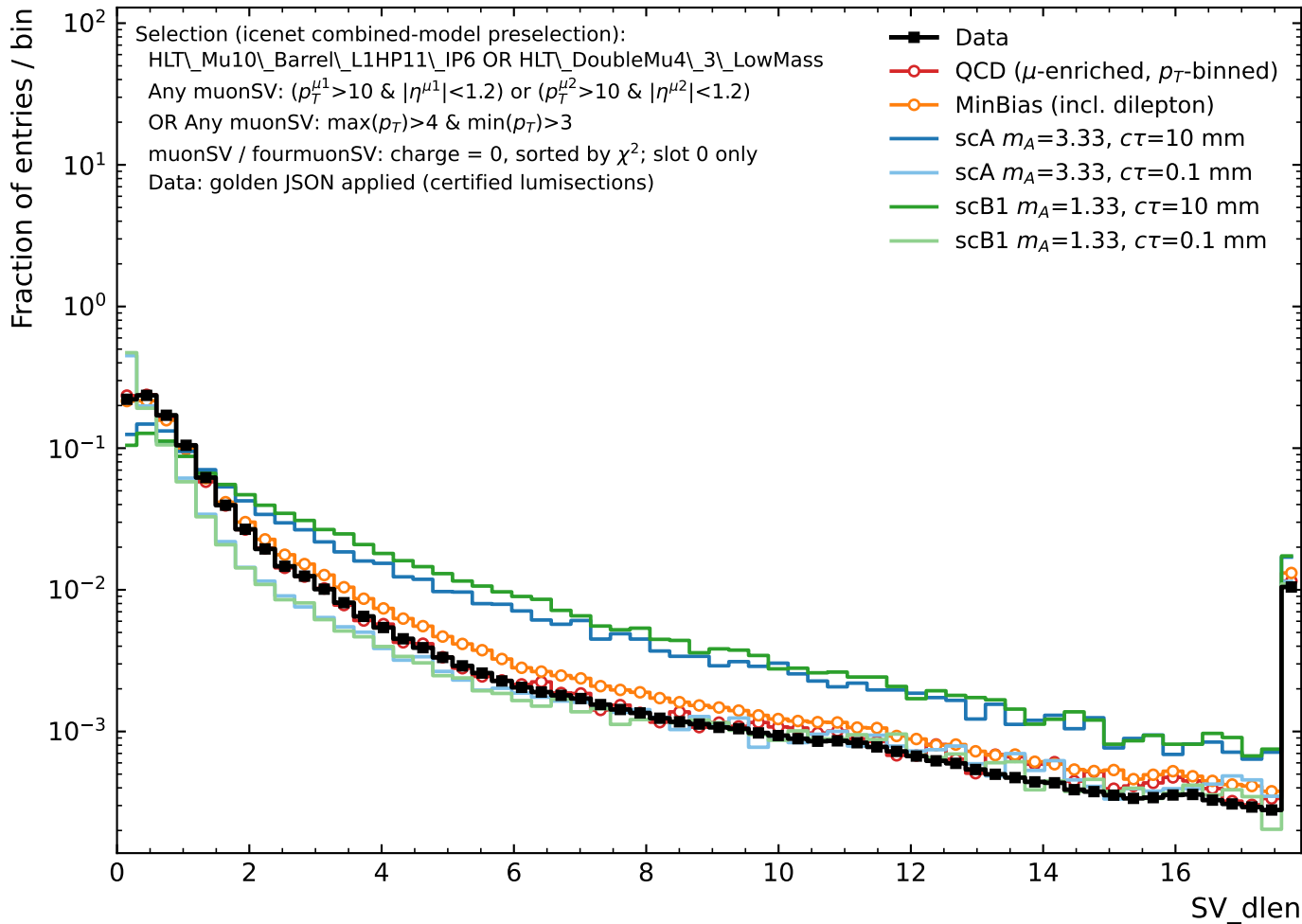
HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

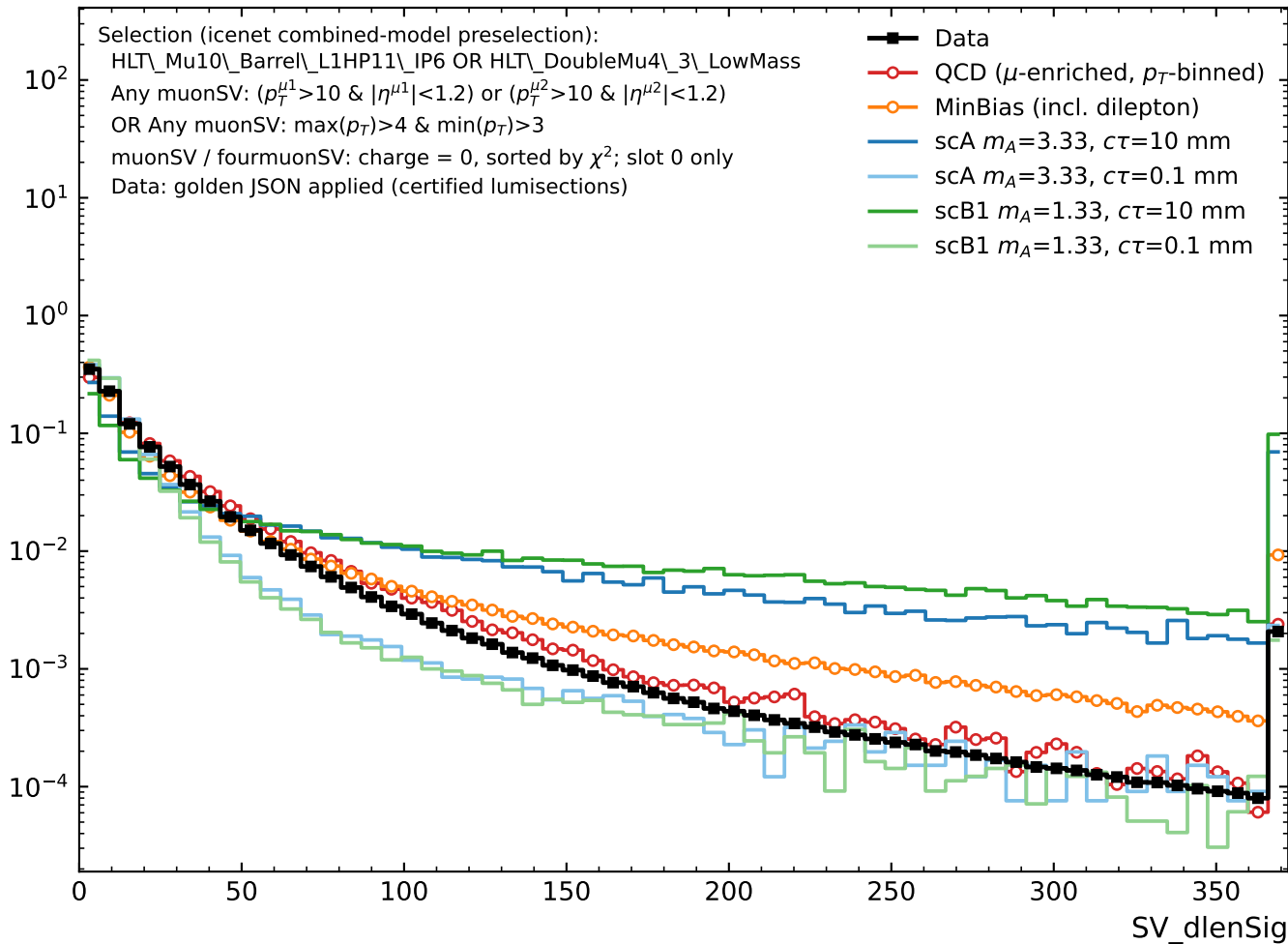
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

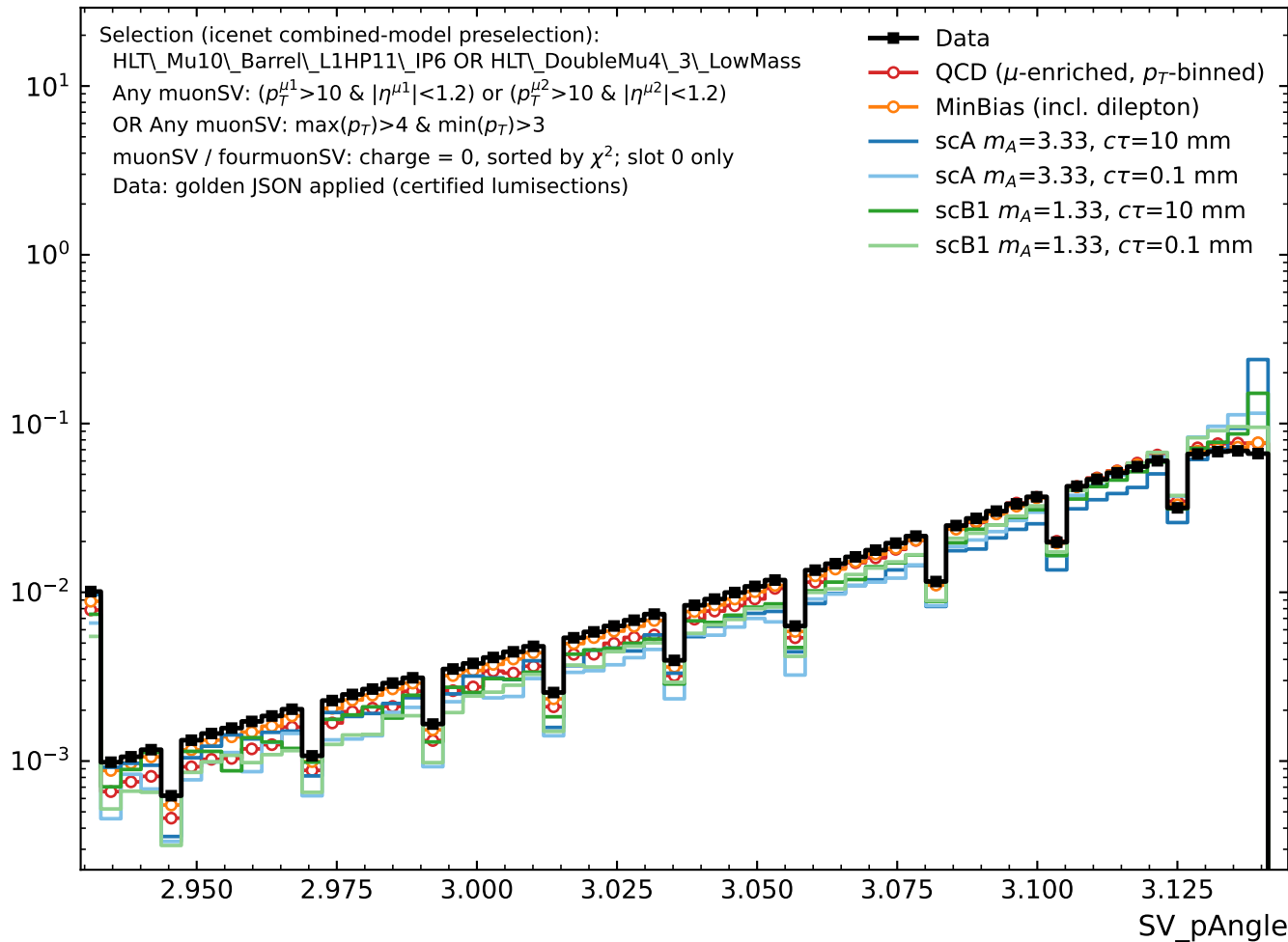
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

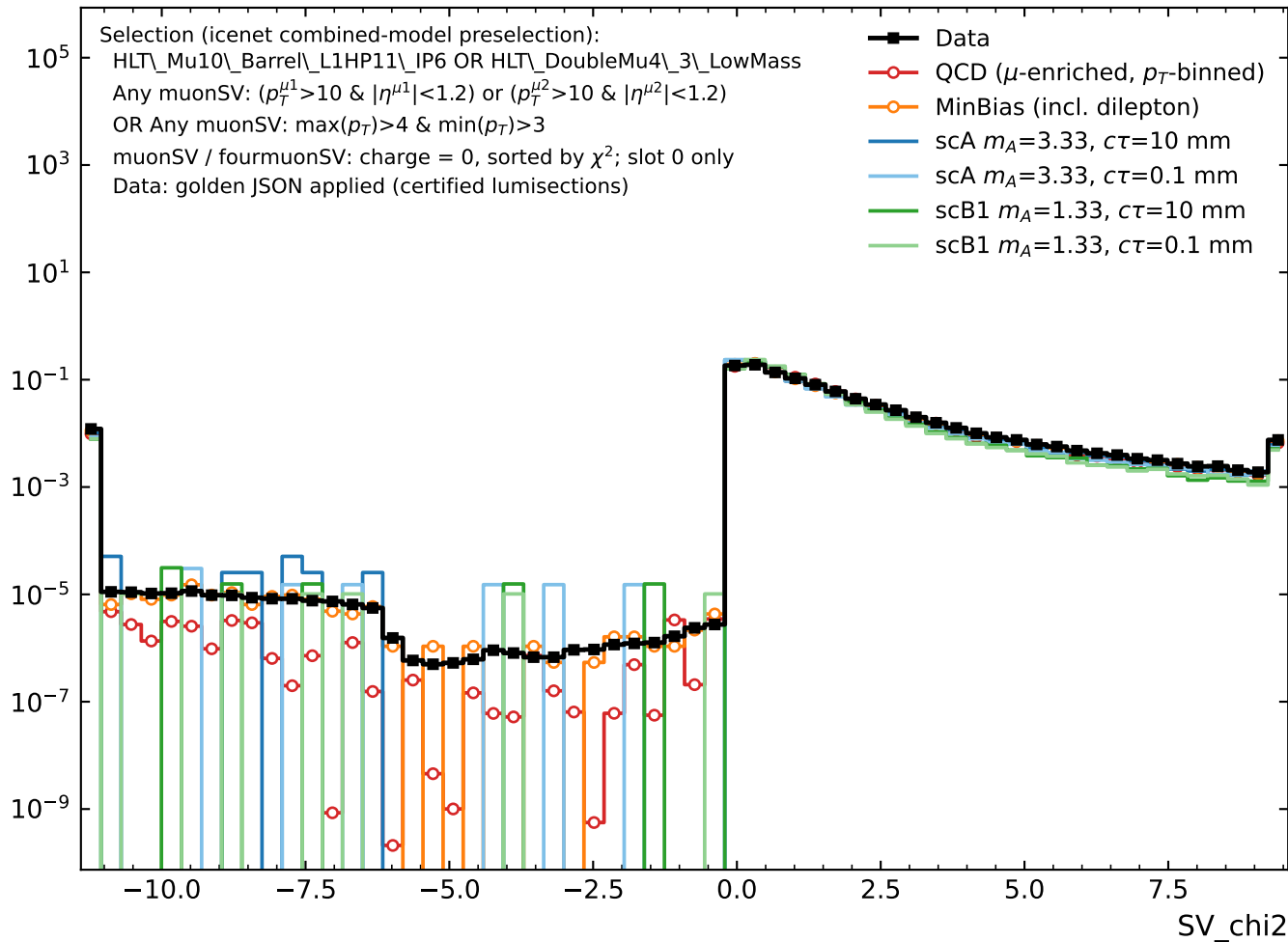
Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin



Selection (icenet combined-model preselection):

HLT_Mu10_Barrel_L1HP11_IP6 OR HLT_DoubleMu4_3_LowMass

Any muonSV: ($p_T^{\mu 1} > 10$ & $|\eta^{\mu 1}| < 1.2$) or ($p_T^{\mu 2} > 10$ & $|\eta^{\mu 2}| < 1.2$)

OR Any muonSV: $\max(p_T) > 4$ & $\min(p_T) > 3$

muonSV / fourmuonSV: charge = 0, sorted by χ^2 ; slot 0 only

Data: golden JSON applied (certified lumisections)

Fraction of entries / bin

